SCHWESERNOTES - BOOK 5

ALTERNATIVE INVESTMENTS

Reading 78: Alternative Investment Features, Methods, and Structures

LOS 78.a: Describe features and categories of alternative investments.

Alternative investments comprise various types of investments that do not fall under the heading of **traditional investments**, which refers to long-only investments in cash or publicly traded stocks and bonds.

Types of alternative investment structures include hedge funds, private equity funds, and various types of real estate investments. Alternative investments typically are actively managed and may include investments in commodities, infrastructure, and illiquid securities.

The perceived benefits of including alternative investments in portfolios are risk reduction from diversification (due to low correlations of alternative investments with traditional investments) and possible higher returns from holding illiquid securities, and from markets for some alternative investments possibly being less efficient than those for traditional investments.

Compared with traditional investments, alternative investments typically exhibit the following features:

- More specialized knowledge required of investment managers
- Relatively low correlations with returns of traditional investments
- Less liquidity of assets held
- Longer time horizons for investors
- Larger size of investment commitments

As a result of these unique features, alternative investments exhibit the following characteristics:

- Investment structures that facilitate direct investment by managers
- Information asymmetry between fund managers and investors, which funds typically address by means of incentive-based fee structures
- Difficulty in appraising performance, such as more problematic and less available historical returns and volatility data

Although correlations of returns on alternative investments with returns on traditional investments may be low on average, these correlations may increase significantly during periods of economic stress.

We will examine several types of alternative investments in detail in separate readings in this topic area. We may classify alternative investments into three broad categories of private capital, real assets, and hedge funds.

1. **Private capital** includes private equity and private debt:
    - As the name suggests, **private equity** funds invest in the equity of companies that are not publicly traded, or in the equity of publicly traded firms that the funds intend to take private. These firms are often in the mature or decline stages of their industry life cycle. **Leveraged buyout (LBO)** funds use borrowed money to purchase equity in established companies and comprise most

private equity investment funds. **Venture capital** funds invest in young, unproven companies at the start-up or early stages in their life cycles.
    - **Private debt** funds may make loans directly to companies, lend to early-stage firms (**venture debt**), or invest in the debt of firms that are struggling to make their debt payments or have entered bankruptcy (**distressed debt**).
2. **Real assets** include real estate, infrastructure, natural resources, and other assets such as digital assets:
    - **Real estate** Investments include residential or commercial properties, as well as real estate–backed debt. These investments are held in various structures, including full or leveraged ownership of individual properties, individual real estate–backed loans, private and publicly traded securities backed by pools of properties or mortgages, and limited partnerships.
    - **Natural resources** include commodities, farmland, and timberland. To gain exposure to **commodities**, investors can own physical commodities, commodity derivatives, or the equity of commodity-producing firms. Some funds seek exposure to the returns on various commodity indices, often by holding derivatives contracts (futures) that are expected to track a specific commodity index. **Farmland** can produce income from leasing the land out for farming or from raising crops or livestock for harvest and sale. **Timberland** investment involves purchasing forested land and harvesting trees to generate cash flows.
    - **Infrastructure** refers to long-lived assets that provide public services. These include economic infrastructure assets (e.g., roads, airports, and utility grids) and social infrastructure assets (e.g., schools and hospitals). While often financed and constructed by government entities, infrastructure investments have more recently been undertaken by **public-private partnerships**, with each holding a significant stake in the infrastructure assets. Various deal structures are employed, and the asset may revert to public ownership at some future date.
    - Other types of real assets include collectibles such as art, intangible assets such as patents, and **digital assets** such as cryptocurrencies.
3. **Hedge funds** are investment companies typically open only to qualified investors. These funds may use leverage, hold long and short positions, use derivatives, and invest in illiquid assets. Managers of hedge funds use many different strategies in attempting to generate investment gains. They do not necessarily hedge risk, as the name might imply.

LOS 78.b: Compare direct investment, co-investment, and fund investment methods for alternative investments.

Fund investing refers to investing in a pool of assets alongside other investors, using a fund manager who selects and manages a pool of investments using an agreed-upon strategy. In this case, the individual investors do not control the selection of assets for investment or their subsequent management and sale. The manager typically receives a percentage of the investable funds (management fee) as well as a percentage of the investment gains (incentive fee).

Compared to funds that invest in traditional asset classes, alternative investment funds typically require investors to commit larger amounts of capital for longer periods, provide less information on positions held and returns earned, and charge higher management fees. A fund's **term sheet** describes its investment policy, fee structure, and requirements for investors to participate.

With **co-investing**, an investor contributes to a pool of investment funds (as with fund investing) but also has the right to invest, directly alongside the fund manager, in some of the assets in which the manager invests. Compared to fund investing, co-investing can reduce overall fees while benefiting from the manager's expertise. Co-investing also can provide an investor with an opportunity to gain the skills and experience to pursue direct investing. For a fund manager, permitting co-investment may increase the availability of investment funds and expand the scope and diversification of the fund's investments.

Direct investing refers to an investor that purchases assets itself, rather than pooling its funds with others or using a specialized outside manager. Larger, more knowledgeable investors may purchase private companies or real estate directly. For example, a sovereign wealth fund may have its own specialized managers to invest in real estate, agricultural land, or companies in the venture stage.

Direct investing has advantages in that there are no fees to outside managers, and the investor has more control over investments choices. Disadvantages include the possibility of less diversification across investments, higher minimum investment amounts, and greater investor expertise required to evaluate deals and perform their own due diligence.

LOS 78.c: Describe investment ownership and compensation structures commonly used in alternative investments.

Alternative investments are often structured as **limited partnerships**. In a limited partnership, the **general partner (GP)** is the fund manager and makes all the investment decisions. The **limited partners (LPs)** are the investors, who own a partnership share proportional to their investment amounts. The LPs typically have no say in how the fund is managed and no liability beyond their investment in the partnership. The GP takes on the liabilities of the partnership, including the repayment of any partnership debt. Partnerships typically set a maximum number of LPs that may participate.

LPs commit to an investment amount, and in some cases, they only contribute a portion of that initially, providing the remaining funds over time as required by the GP (as fund investments are made). General partnerships are less regulated than publicly traded companies, and limited partnership shares are typically only available to **accredited investors**—those with sufficient wealth to bear significant risk and enough investment sophistication to understand the risks.

The rules and operational details that govern a partnership are contained in the **limited partnership agreement**. Special terms that apply to one limited partner but not to others can be stated in **side letters**. For example, an LP might negotiate an **excusal right** to withhold a capital contribution that the GP would otherwise require. Some limited partners may require that special terms offered to other LPs also be offered to them. This is known as a **most-favored-nation clause** in a side letter.

While most alternative investment limited partnership holdings are illiquid, a fund may be structured as a **master limited partnership (MLP)** that can be publicly traded. Master limited partnerships are most common in funds that specialize in natural resources or real estate.

Fee Structures

The total fees paid by investors in alternative investment funds often consist of a **management fee**, typically between 1% and 2% of the fund's assets, and a **performance fee** or **incentive fee** (sometimes referred to as **carried interest**).

The fund manager earns the management fee, regardless of investment performance. For hedge funds, the management fees are calculated as a percentage of assets under management (AUM), typically the net asset value of the fund's investments. For private equity funds, the management fee is calculated as a percentage of **committed capital**, not invested capital. Committed capital is typically not all invested immediately; rather, it is "drawn down" (invested) as securities are identified and added to the portfolio. Committed capital is usually drawn down over three to five years, but the drawdown period is at the discretion of the fund manager. Committed capital that has not yet been drawn down is referred to as **dry powder**. The reason for basing management fees on committed capital is that otherwise, the fund manager would have an incentive to invest capital quickly instead of selectively.

Performance fees (also referred to as incentive fees) are a portion of profits on fund investments. Most often, the partnership agreement will specify a **hurdle rate** (or **preferred return**) that must be met or exceeded before any performance fees are paid. Hurdle rates can be defined in two ways: either "hard" or "soft." If a **soft hurdle rate** is met, performance fees are a percentage of the total increase in the value of each partner's investment. With a **hard hurdle rate**, performance fees are based only on gains above the hurdle rate.

For example, consider a fund with a hurdle rate of 8% that has produced a return of 12% for the year. We will use a performance fee structure of 20% of gains. If the 8% is a soft hurdle rate, the performance fee will be 20% of the entire 12%, or 2.4%. If the 8% is a hard hurdle rate, the performance fee will be 20% of the gains above the hurdle rate (12% – 8% = 4%), which would be 0.8%.

Typically, performance fees are paid at the end of each year based on the increase in the value of fund investments, after management fees and other charges, which may include consulting and monitoring fees that are charged to individual portfolio companies.

A **catch-up clause** in a partnership agreement is based on a hurdle rate and is similar in its effect to a soft hurdle rate. Consider a fund with returns of 14%, a hurdle rate of 8%, and a 20% performance fee. A catch-up clause would result in the first 8% of gains going to the LPs and the next 2% going to the GP, allowing the GP to "catch up" to receiving 20% of the first 10% of gains. After the catchup, further gains are split 80/20 between the LPs and the GP.

Another feature that is often included is a **high-water mark**, which means no performance fee is paid on gains that only offset prior losses. Thus, performance fees are only paid to the extent that the current value of an investor's account is above the highest net-of-fees value previously recorded (at the end of a payment period). This feature ensures that investors will not be charged performance fees twice on the same gains in their portfolio values. Because investors invest in a fund at different times, they each may have a different high-water mark value.

A partnership's **waterfall** refers to the way in which payments are allocated to the GP and the LPs as profits and losses are realized on deals. With a **deal-by-deal waterfall** (or **American waterfall**), profits are distributed as each fund investment is sold and shared according to the partnership agreement. This favors the GP because performance fees are paid before 100% of the LPs' original investment plus the hurdle rate is returned to them. With a **whole-of-fund waterfall** (or **European waterfall**), the LPs receive all distributions until they have received 100% of their initial investment plus the hurdle rate (typically after all fund investments have been sold).

A **clawback provision** stipulates that if the GP accrues or receives incentive payments on gains that are subsequently reversed as the partnership exits deals, the LPs can recover previous (excess) incentive payments. With a deal-by-deal waterfall, successful deals might be exited initially, while losses are realized later. A clawback provision would allow the LPs to recover these performance fees to the extent that the subsequent losses negate prior gains on which performance fees had been paid.

LOS 78.a

Alternative investments comprise various types of investments that do not fall under the heading of traditional investments. Categories of alternative investments include the following:

- Hedge funds
- Private capital (private equity and private debt)
- Real assets (real estate, natural resources, and infrastructure)

Alternative investments have relatively low correlations with returns of traditional investments. Compared with traditional investments, alternative investments typically require more specialized knowledge of investment managers. Assets held tend to be less liquid, making alternative investments appropriate for investors who have long time horizons and can commit large amounts of capital.

LOS 78.b

Fund investing refers to pooling assets along with other investors, using a fund manager that selects and manages investments according to an agreed-upon strategy.

Co-investing refers to fund investing that includes the right to invest additional capital directly alongside the fund manager.

Direct investing refers to purchasing assets independently, rather than pooling funds with others or using a specialized outside manager.

LOS 78.c

Many alternative investments are structured as limited partnerships, in which the GP is the fund manager and the LPs are the investors. They are less regulated than publicly traded companies and are typically only available to accredited investors. LPs may commit to an investment amount and, in some cases, contribute only a portion of that initially, providing the remaining funds over time as required by the GP.

Fees in alternative investment funds often consist of a management fee and a performance fee. For hedge funds, management fees are a percentage of assets under management. For private capital, management fees are a percentage of committed capital rather than capital invested.

Performance fees are a portion of profits on fund investments. Typically, a hurdle rate must be exceeded before performance fees are paid. With a soft hurdle rate, performance fees are based on the total increase in the value of each partner's investment. With a hard hurdle rate, performance fees are based only on gains above the hurdle rate. A high-water mark is a provision that no performance fees are paid on gains that only offset prior losses.

A waterfall refers to the way payments are allocated to the GP and the LPs. With a deal-by-deal or American waterfall, profits are distributed as each fund investment is sold. With a whole-of-fund or European waterfall, the LPs receive all distributions until they have received 100% of their initial investment plus the hurdle rate.

With a clawback provision, if the GP receives incentive payments on gains that are subsequently reversed, the LPs can recover excess incentive payments.

LOS 79.a: Describe the performance appraisal of alternative investments.

Alternative investments are typically exposed to greater risks than unleveraged long-only traditional investments. These additional risks arise from the following:

- Timing of cash flows over an investment's life cycle
- Use of leverage by fund managers
- Valuation of investments that may or may not have observable market prices
- Complexity of fees, taxes, and accounting

Ideally, returns on alternative investments should be adjusted for these risks, although that may be difficult in practice. It is clear, however, that evaluating alternative investment returns (or expected returns) without considering these additional risks would be naïve and possibly misleading.

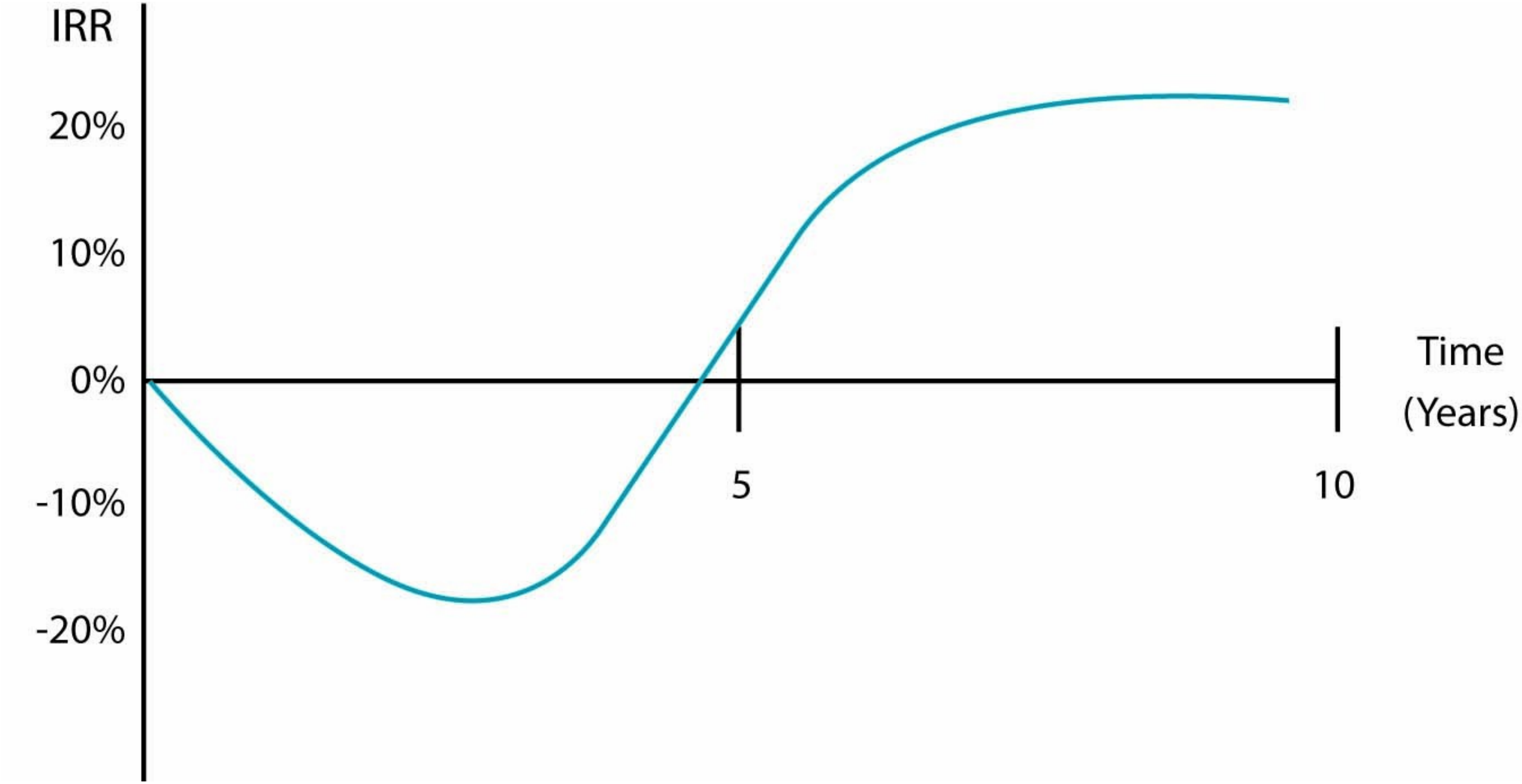
Timing of Cash Flows

Alternative investments often have a life cycle that exhibits three phases:

1. In the **capital commitment phase**, a fund's managers are identifying investments and making **capital calls** from the partners. Recall that limited partners commit a stated amount of capital that they will invest, but they do not necessarily deliver the entire amount immediately in cash. Managers make capital calls as they identify investments for which they require cash. Because of these cash outflows and the long-term nature of the typical investments, returns tend to be negative during the capital commitment phase.
2. During the **capital deployment phase**, the managers fund, and often involve themselves directly in, the firms or projects in which they invest. Returns typically remain negative in this phase, especially if the investments are in start-up companies or troubled firms that the managers are attempting to turn around.
3. If the fund's investments succeed and begin to generate income and cash flows, the fund enters a **capital distribution phase** during which its returns turn positive and accelerate.

A **J-curve effect** (illustrated in **J-Curve Example**) reflects the norm of negative returns in the capital commitment phase, followed by increasing returns in the capital deployment phase and maximum returns in the capital distribution phase. Returns may reach a plateau toward the end of a fund's life as the managers exit any remaining investments.

J-Curve Example



Given the variability of cash flows over a fund's life and the importance of management decisions in the timing and magnitude of after-tax cash flows, an IRR over the life of a fund is the most appropriate measure of after-tax investment performance. (Recall from Quantitative Methods that a money-weighted rate of return, which is an IRR, is appropriate when a manager controls the timing of cash inflows and outflows.) A drawback to using IRRs is that they rest on assumptions about the cost of capital for cash outflows and the reinvestment rate for cash inflows.

A simpler measure of investment success is the **multiple of invested capital** (or **money multiple**)—the ratio of total capital returned plus the value of any remaining assets, to the total capital paid in over the life of the investment. Because this measure does not consider the timing of cash inflows and outflows, which can affect annual returns on invested capital significantly, it can be considered somewhat naïve.

Use of Leverage

Some alternative investments, particularly hedge funds, use borrowing to magnify their gains (at the risk of magnifying losses). Hedge funds may arrange margin financing with **prime brokers** or employ leverage by means of derivatives.

To state the effect of leverage on returns, consider a fund that can invest the amount V_0 without leverage, and earn the rate of return r. The fund's unleveraged portfolio return (as a money amount) is simply this: $r \times V_0$. Now, let's say this fund can borrow the amount V_B at an interest rate of r_B, and earn r by investing the proceeds. The fund's leveraged portfolio return (again, as a money amount), after subtracting the interest cost, then becomes:

$$r \times (V_0 + V_B) - (r_B \times V_B)$$

Thus, stated as a rate of return on the initial portfolio value of V_0, the leveraged rate of return is as follows:

$$\text{leveraged return} = \frac{r(V_0 + V_B) - r_B V_B}{V_0}$$

One of the reasons that funds use leverage is that some strategies attempt to exploit relatively small pricing anomalies that might not produce meaningful results without leverage. A risk from using leverage is that a lender may issue margin calls if a fund's equity position decreases below a certain level. These can result in a fund having to realize losses by closing positions or liquidating investments at unfavorable prices. If the fund must sell a large position in a security, doing so may depress its price further. Another important risk of funds that depends on leverage is that lenders may limit their access to additional borrowing.

Valuation of Investments

As we have discussed, many alternative investments involve illiquid assets that do not trade frequently in transparent markets. While funds must recognize investments at fair value to comply with accounting standards, fair value might rest on assumptions of which an investor in a fund should be aware. A **fair value hierarchy** groups these assumptions into the following three levels:

- *Level 1.* The assets trade in active markets and have quoted prices readily available, such as exchange-traded securities.
- *Level 2*. The assets do not have readily available quoted prices, but they can be valued based on directly or indirectly observable inputs, such as many derivatives that can be priced using models.
- *Level 3.* The assets require unobservable inputs to establish a fair value, such as real estate or private equity investments, for which there have been few or no market transactions.

Particularly for Level 3 investments, the absence of market activity can result in valuations that remain near their initial cost for long periods. As a result, these values might not reflect the actual exit costs of the investments. Importantly, this relative lack of change in fair values can make reported returns for alternative investments appear higher, less risky, and less correlated with traditional investments than they really are.

Fee Structures

In the next LOS, we will examine some of the effects on investors' returns of different fee specifications, such as high-water marks, hard or soft hurdle rates, and waterfall structures. Keep in mind that fee structures are subject to negotiation (e.g., a limited partner might agree to a higher management fee in exchange for fewer restrictions on redemptions) and may differ depending on how early in a fund's life cycle an investor commits capital. Thus, different investors in the same fund might realize significantly different returns.

LOS 79.b: Calculate and interpret alternative investment returns both before and after fees.

We have seen how margin calls may require a leveraged fund to exit investments at unfavorable prices and unintended times. A similar risk arises from investor **redemptions**. The more negative a fund's returns, the more likely investors are to ask the manager to redeem their positions. For this reason, and because of the J-curve effect of negative returns in the early years, alternative investment funds (particularly hedge funds) typically take measures to restrict early redemptions.

A **lockup period** is the time after initial investment over which limited partners either cannot request redemptions or incur significant fees for redemptions. A **notice period** (typically between 30 and 90 days) is the amount of time a fund has to fulfill a redemption request. Notice periods allow time for managers to reduce positions in an orderly manner. Fund managers often incur significant transactions costs when they redeem shares. **Redemption fees** can offset these costs. Managers may also have the discretion to implement a **gate** that restricts redemptions for a temporary period.

Investors should be aware that other investors may receive terms that differ from those stated in the partnership agreement. Customized fee structures are contained in **side letters** with individual investors detailing how their terms differ from those in the standard offering documents.

Although "2 and 20" and "1 and 10" were, at one time, fairly standard fee structures for fund and fund-of-funds investments, these fee structures continue to be under competitive pressure. Investors making larger commitments can negotiate lower fees. There can also be a tradeoff between liquidity provisions and fees. Investors can negotiate for lower fees or better liquidity (shorter lockups and notice periods). Hurdle rates, hard versus soft hurdles, and catch-up provisions may also be subject to negotiation.

Early investors in a fund may also receive lower fees or better liquidity terms as an incentive to invest at the fund's inception. The investment interests of early investors who receive such relatively better terms are called **founders class shares**.

Annual investor fees can also be **either-or fees**, the maximum of the management fee or the incentive fee. Under such a structure, with a 1% management fee and a 30% incentive fee, investor fees each year would simply be the management fee unless the calculated incentive fee is higher. Such a structure may also stipulate that the 1% management fee be subtracted from the incentive fee in a subsequent year.

Biases in Alternative Investment Returns

Numerous alternative investment indexes exist to measure historical performance. However, they may not provide much meaningful information on the asset class because each fund's structure is unique, and the funds that exist at any given time can be in widely different phases of their life cycles. One way around this latter issue is to compare funds that originated in the same **vintage year**.

The effect of **survivorship bias** is greater for a hedge fund database than for other asset classes because by some estimates, more than 25% of hedge funds fail in the first three years of their existence. An index that does not include failed funds will overstate the returns and understate the risk of hedge funds as an asset class. These effects may be magnified by **backfill bias**, which occurs when managers only select their successful funds for inclusion in indexes.

Return Calculations for Alternative Investments

Before-fee returns on alternative investments are calculated the same way we calculate returns on any investment. The calculation of holding period returns and periodic rates of return, both with and without interim cash distributions, is described in Quantitative Methods and expanded on by application in other topic areas.

Calculating after-fee returns simply requires adjustment of the cash flows or values for the various fees involved, typically management and performance fees. For a simple case in which management fees are a fixed percentage of end-of-period assets and performance fees are a fixed percentage of total return with no hurdle rate, we can state the general partner's total fees in money terms as follows:

$$\text{total fees} = mV_1 + \max\,[0, p(V_1 - V_0)]$$

where:

m = management fee

p = performance fee

V_0 = beginning-of-period assets

V_1 = end-of-period assets

This is the rate of return for an investor after fees:

$$r = \frac{V_1 - V_0 - \text{total fees}}{V_0}$$

Fee structure provisions, such as hurdle rates and high-water marks, make calculating total fees more complex than this simple formula. Other provisions may state whether the performance fee is net of the management fee or independent of it, or whether the management fee is based on beginning-of-period assets or end-of-period assets.

Professor's Note

Because none of these provisions are standard across alternative investment funds, exam questions that require calculations will have to specify all of those that apply. Do not assume any provision applies unless it is stated in the question.

Some examples will illustrate the application of various fees and the relevant terminology.

Example: Hedge fund fees

BJI Funds is a hedge fund with a value of $110 million at initiation. BJI Funds charges a 2% management fee based on assets under management at the beginning of the year and a 20% performance fee with a 5% soft hurdle rate, and uses a high-water mark. Performance fees are calculated on gains net of management fees. The year-end values before fees are as follows:

- Year 1: $100.2 million
- Year 2: $119.0 million

Calculate the total fees and the investor's after-fee return for both years.

Answer:

Year 1:

Management fee:

110.0 million × 2% = $2.2 million

Return net of management fees:

$$\frac{100.2-2.2}{110.0}-1=-10.9\%$$

There is no performance fee because the return after the management fee is less than the 5% hurdle rate.

Total fees:

$2.2 million

Ending value net of fees:

100.2 million – $2.2 million = $98.0 million

Year 1 after-fees return:

$$\frac{98.0}{110.0}-1=-10.9\%$$

Year 2:

Management fee:

98.0 million × 2% = $1.96 million

Year-end value net of management fee:

$119.0 – $1.96 = $117.04 million

The high-water mark is $110 million.

Year 2 value net of management fee, above high-water mark:

117.04 million – 110.0 million = $7.04 million

Year 2 return net of management fee, above high-water mark:

$$\frac{7.04}{110.0}=6.4\%$$

Due to the high-water mark, the performance fee is calculated based on gains in value above $110 million.

The performance fee is calculated on the entire gain above the high-water mark because 6.4% is greater than the soft hurdle rate. If the 5% was a hard hurdle rate, the performance fee would be calculated only on the gains more than 5% above the high-water mark.

Performance fee:

7.04 × 0.20 = $1.41 million

Total fees:

$1.96 million + $1.41 million = $3.37 million

Year 2 year-end value after fees:

119.0 – 3.37 = $115.63 million

Year 2 after-fee return:

$$\frac{115.63}{98.0} - 1 = 18.0\%$$

Example: Fund-of-funds

An investor makes a total investment of $60 million in a fund-of-funds that has a "1 and 10" fee structure, with management and performance fees calculated independently based on year-end values. Of the $60 million investment, $40 million is allocated to the Alpha fund and $20 million is allocated to the Beta fund. One year later, the value of the Alpha fund investment is $45 million, and the value of the Beta fund investment is $28 million, both net of fund fees. Calculate the investor's return for the year net of fees.

Answer:

At year-end, the gross value of the investor's investment is $45 + $28 = $73 million.

The fund-of-funds management fee is 1% of $73 million, which is $0.73 million.

The investor's gain for the year before fund-of-funds fees is $73 – $60 = $13 million.

The fund-of-funds manager's performance fee is 10% of $13 million, which is $1.3 million.

The year-end value of the investor's fund-of-funds investment is $73 – $0.73 – $1.3 = $70.97 million.

The investor's one-year return after fees is: $\frac{70.97}{60} - 1 = 18.3\%$

Note that the same investments made directly with the Alpha and Beta funds would have returned $\frac{73}{60} - 1 = 21.7\%$.

Example: Waterfall structure and clawback provision

A private equity fund invests $100 million in a venture company that is sold for $130 million. The fund also invests $100 million in an LBO that goes poorly and is liquidated for $80 million.

1. If the carried interest performance fee for the GP is 20% and there is no clawback provision, calculate the investor's return after performance fees, assuming the investment outcomes are realized in the same year under the following:
    1. An American-style (deal-by-deal) waterfall structure
    2. A European-style (whole-of-fund) waterfall structure
2. How would the answers be affected if the venture investment was sold in Year 1 and the LBO investment was sold in Year 2?
3. How would including a clawback provision affect investor returns calculated in Question 1?

Answer:

1. Under an American-style (deal-by-deal) waterfall structure, a performance fee of 20% × ($130 – $100) = $6 million would be paid on the venture investment. Because there is a loss on the LBO investment, no performance fee is paid.

   investor's return on investment: $\frac{130 + 80 - 6}{200} - 1 = 2\%$

   Under a European-style (whole-of-fund) waterfall structure, the gain for the period is 130 + 80 – 200 = $10 million, and the performance fee is 20% × 10 = $2 million.

   investor's return on investment: $\frac{130 + 80 - 2}{200} - 1 = 4\%$

2. The European-style waterfall structure would have the same overall return as the American-style structure, as the performance fee for the venture investment of $6 million would be paid in Year 1 and no performance fee would be received on the LBO investment.
3. With a clawback provision, after the LBO investment is sold, the performance fee of $6 million paid on the venture investment is more than 20% of the return on the total investment. The fee is 60% of the total (net) gain of $10 million. The investor could "claw back" $4 million of the $6 million paid as a performance fee on the venture investment, which would reduce the total performance fee to 20% of the $10 million gain.

Reading 79: Key Concepts

LOS 79.a

Alternative investments are typically exposed to risks that traditional investments are not, including timing of cash flows, use of leverage, valuation assumptions for illiquid investments, and complex fee structures and taxation.

Alternative investment returns typically exhibit a J-curve effect: negative during the capital commitment phase, less negative during the capital deployment phase, and positive during the capital distribution phase.

Because managers control cash inflows and outflows, an IRR is appropriate for measuring returns. The multiple of invested capital is a simpler measure that does not depend on the timing of cash flows:

$$\text{leveraged return} = \frac{r(V_0+V_B) - r_B V_B}{V_0}$$

where:

V_0 = initial assets

V_B = borrowed funds

r = rate of return on investments

r_B = borrowing interest rate

Assumptions underlying the fair value of assets can be described by a three-level hierarchy:

- *Level 1*. Quoted prices readily available.
- *Level 2*. Can be valued based on observable inputs.
- *Level 3*. Require unobservable inputs to establish a fair value.

Fair values of Level 3 assets might be updated infrequently due to a lack of market activity. This can bias return measures upward and risk measures downward.

LOS 79.b

Before-fee returns on alternative investments are calculated the same way we calculate fees on any investment. Calculating after-fee returns requires us to adjust the cash flows or values for the various fees involved (typically management and performance fees), and for specifications such as hurdle rates and high-water marks.

LOS 80.a: Explain features of private equity and its investment characteristics.

Private capital is the funding provided to companies that is not raised from the public markets. Private capital includes private equity and private debt.

Private equity is equity capital raised from sources other than the public markets. Private equity funds usually invest in private companies or public companies that they plan to take private (leveraged buyout funds) or in early in companies' lives (venture capital funds). The companies in which a private equity fund invests are called its **portfolio companies**.

Private Equity Investment Categories

A popular type of private equity fund is the **leveraged buyout (LBO)** fund, which acquires public companies with a large percentage of the purchase price financed by debt. LBOs are a way for a company to "go private" because after the transaction, the target company's stock is no longer publicly traded.

In an LBO, the private equity firm seeks to add value by improving or restructuring the portfolio company's operations to increase its sales, profits, and cash flows. The cash flows can then be used to service and pay down the debt taken to fund the acquisition.

Two of the various types of LBOs are **management buyouts (MBOs)**, in which the portfolio company's existing management team participates in the purchase, and **management buy-ins (MBIs)**, in which the private equity manager replaces the portfolio company's current management team with a new team.

Venture capital (VC) funds provide financing to companies in the early stages of their development. VC funds typically receive common equity interest in the portfolio companies, but they may also get convertible debt or convertible preferred stock. Convertible securities help to align the interests of the VC investors and the start-up firm, in that both want to increase the value of the firm, but give the VC investors a higher priority of claims in the event of liquidation.

VC investment involves a high level of risk, but the returns can be substantial. VC investors are actively involved in developing their portfolio companies, often sitting on the boards or filling key management roles.

We can classify VC investments by the portfolio company's stage of development:

1. The **formative stage** refers to investments made during a firm's earliest period and comprises three different phases:
    - **Pre-seed capital** or **angel investing** is the capital provided at the idea stage. The investment funds are used for business plans and assessing market potential. The amount of financing is usually small, coming from individuals ("angels") rather than VC funds.

    - **Seed-stage financing** or **seed capital** generally supports product development, marketing, and market research. This is the first stage at which VC funds usually invest.
    - **Early-stage financing** or **start-up stage financing** refers to investments made to fund operations in the lead-up to production and sales.
2. **Later-stage financing** or **expansion venture capital** comes after production and sales have begun. Investment funds provided at this stage are used to support initial growth, expansion, product improvement, or marketing. In this stage, the owners (typically the founders and managers of the company) often sell control of the company to VC investors.
3. **Mezzanine-stage financing** refers to capital provided to prepare the firm for an initial public offering (IPO). The term *mezzanine stage* is used to indicate the timing of the financing rather than the method. A similar term, *mezzanine financing*, refers to hybrids of equity and debt, such as convertible securities. Mezzanine-stage financing can use these, but more often, it consists of equity or short-term debt.

A private equity firm that engages in **minority equity investing** buys a less-than-controlling interest in public companies that are looking for capital with which to expand. One way it can make such investments is through a **private investment in public equity (PIPE)**, a private offering to institutional investors that allows a publicly traded firm to raise capital more quickly and cost effectively than a public offering, with fewer disclosures and lower transaction costs.

Private Equity Exit Strategies

Private equity firms typically add value to young companies, then sell them, with an average holding period of five years. Funds have several methods of exiting an investment in a portfolio company:

1. **Trade sale**. Sell a portion of the private company to a strategic buyer via direct sale or auction. An advantage of a trade sale is that the strategic investor typically pays a premium to realize synergies with an existing business. Advantages compared with an IPO include faster execution and lower transaction costs. Disadvantages of a trade sale include potential resistance from the portfolio company's management and employees, as well as a limited universe of potential buyers.
2. **Public listing**. Listing on a stock exchange can take place through an IPO, a direct listing, or a special purpose acquisition company:
    - IPOs are the most common method of public listing, using investment banks to underwrite the offering. Compared to other methods, IPOs typically realize a higher price for the portfolio company. They may also improve its visibility to investors and the public, which can be positive for the value of shares that the private equity fund retains. However, IPOs involve high transaction and compliance costs and might not always be received favorably by the investing public. Portfolio companies best suited for exit via IPO are large firms in growing industries that have stable finances and clear strategies.
    - A less frequently used alternative to an IPO is a **direct listing**, in which the stock of the company is floated on the public market directly without underwriters. This decreases the cost of the transaction, but does not raise new capital for the portfolio company.
    - A **special purpose acquisition company (SPAC)** is an entity (sometimes referred to as a blank check company) set up solely to raise capital that it will use to acquire an unspecified private company within a stated time period; otherwise, it must return the capital to investors. This method can be more flexible than an IPO and reduce the uncertainty about the valuation of the portfolio company, as well as provide access to investors who transact regularly with the sponsors. However, SPACs have numerous disadvantages, including dilutive effects from SPAC shares and warrants, a spread between the value of the SPAC and the value of the acquired company, deal risk in the acquisition, and **stockholder overhang** from SPAC shareholders selling shares after the acquisition is announced. Scrutiny from securities regulators is also increasing.
3. **Recapitalization**. The company issues debt to fund a dividend distribution to equity holders. This is not an exit, in that the fund still controls the company, but a recapitalization allows the private equity fund to extract money from the company to pay its investors.
4. **Secondary sale**. Sell a portfolio company to another private equity firm or a group of investors.
5. **Write-off/liquidation**. Reassess and take losses from an unsuccessful investment in a portfolio company.

Risk and Return From Private Equity Investments

Empirical evidence shows that returns on private equity funds have been higher on average than overall stock returns. However, the standard deviation of private equity returns has been higher than the standard deviation of equity index returns, suggesting higher risk, including illiquidity and leverage risks. Furthermore, private equity indexes typically rely on self-reporting and are subject to survivorship and backfill biases, both of which lead to overstated returns. Because portfolio companies are revalued infrequently, measures of volatility and correlation with other investments may be biased downward.

LOS 80.b: Explain features of private debt and its investment characteristics.

Private debt refers to various forms of debt provided by investors directly to private entities. Categories of private debt include the following:

1. **Direct lending** refers to loans made directly to a private company without an intermediary. The debt is typically senior and secured, with covenants to protect the lender. A **leveraged loan** is a loan made by a private debt fund using money borrowed from other sources. That is, the fund's portfolio of loans is leveraged to magnify returns.
2. **Venture debt** is funding that provides VC backing to start-up or early-stage firms that are not yet profitable. Venture debt is often convertible to stock or combined with warrants. Managers of young companies may favor venture debt because it allows them to maintain ownership and control.
3. **Mezzanine debt** is private debt that is subordinated to senior secured debt. Mezzanine debt may have special features, such as conversion rights or warrants, to compensate investors for additional risk.
4. **Distressed debt** is the debt of mature companies in financial trouble, such as bankruptcy or default. In many cases, the fund becomes active in restructuring the existing debt or making other changes that increase the value of the acquired debt. Some distressed debt investors specialize in identifying otherwise good companies with temporary cash flow problems, anticipating that the value of the company and its debt will recover. Others focus on turnaround situations, acquiring a company's debt with an intent to be active in managing and restructuring the company.
5. **Unitranche debt** combines different classes of debt (secured and unsecured) into a single loan with an interest rate that reflects the blend of debt classes. The resulting debt typically ranks between senior and subordinated debt.

Private debt investments typically provide a higher rate of return relative to traditional bonds to compensate investors for higher risk and a lack of liquidity. Investors may also benefit from increased portfolio diversification because private debt returns have relatively low correlations with other traditional investments.

The interest rate of private debt is usually set relative to a reference rate, such as the Secured Overnight Financing Rate (SOFR). Therefore, the rate changes when the reference rate fluctuates due to changes to the interest rate environment.

The potential for higher returns of private debt is associated with higher risk, including illiquidity and default risk. Private debt investing requires specialized knowledge about the structure of the debt, the borrower's life cycle phase, and the features of the underlying assets for secured lending. Senior private debt has less risk and steadier yields than private mezzanine debt, but mezzanine debt offers greater upside potential.

LOS 80.c: Describe the diversification benefits that private capital can provide.

Private capital investments can provide some diversification benefit to a portfolio of traditional investments. Empirical evidence indicates that correlations of private capital fund index returns with public market index returns range from 0.63 to 0.83.[1]

Each private equity fund has a **vintage year**, which is the year the fund made its first investment. The performance of a fund is greatly influenced by its vintage year and the phase of the business cycle in that year. Funds that begin investing during a business cycle expansion are likely to earn higher rates of return if they specialize in early-stage companies. Funds that begin investing during business cycle contractions are likely to earn higher rates of return if they specialize in distressed companies. Investors in private capital should diversify across vintage years.

Risk and return profiles vary among categories of private capital. Private equity has the highest risk and return, followed by mezzanine debt, unitranche debt, senior direct lending, senior real estate debt, and infrastructure debt.

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[1] Preqin, annualized quarterly return of Private Capital Quarterly Index rebased to December 31, 2007.

Reading 80: Key Concepts

LOS 80.a

Private equity funds usually invest in the equity of private companies or companies wanting to become private, often financing their assets with high levels of debt:

- Venture capital funds provide capital to companies early in their development. Stages of venture capital investing include the formative stage (composed of the angel investing, seed, and early stages), the later stage (expansion), and the mezzanine stage (prepare for IPO).
- Leveraged buyouts include management buyouts, in which the existing management team is involved in the purchase, and management buy-ins, in which an external management team replaces the existing management.
- Developmental capital or minority equity investing refers to providing capital for business growth or restructuring. The firms financed may be public or private. In the case of public companies, such financing is referred to as a private investment in public equity.

Methods for exiting investments in portfolio companies include trade sale (sell to a competitor or another strategic buyer), IPO (sell some or all shares to investors), recapitalization (issue portfolio company debt), secondary sale (sell to another private equity firm or other investors), or write-off/liquidation.

LOS 80.b

Private debt refers to lending to private entities. Private debt investments include direct lending, venture debt, mezzanine loans, distressed debt, and unitranche debt. Private debt investing requires specialized knowledge about the structure of the debt, the borrower's life cycle phase, and the features of the underlying assets.

LOS 80.c

Private capital investments can provide some diversification benefit to a portfolio of traditional investments because their correlations of returns with traditional investments are relatively low.

Investors in private capital should diversify across vintage years. Funds that begin investing during a business cycle expansion are likely to earn higher rates of return if they specialize in early-stage companies. Funds that begin investing during business cycle contractions are likely to earn higher rates of return if they specialize in distressed companies.

Reading 81: Real Estate and Infrastructure

LOS 81.a: Explain features and characteristics of real estate.

Real estate can provide income in the form of rent, as well as the potential for price appreciation. Investment in real estate can be made directly in properties or indirectly through limited partnerships and publicly traded securities. The types of real estate we will discuss here include single-family residential and commercial. The four largest subcategories of commercial real estate are office buildings, shopping, industrial/warehouse/distribution, and rental residential (single-family detached and multifamily apartment).

As shown in **Basic Forms of Real Estate Investments(1)**, real estate investment can be described in terms of a quadrant. In the first (vertical) dimension, investment is differentiated between public and private real estate.

- In the private market, ownership usually involves direct investment, such as purchasing property or lending money to a purchaser. Direct investments can be solely owned or indirectly owned through partnerships, where the general partner (GP) provides property management services and limited partners (LPs) are investors (e.g., pension plans).
- The public market does not involve direct investment. Rather, ownership involves securities that serve as claims on the underlying assets. Public real estate investment includes ownership of real estate investment trust (REIT) shares, equity in a real estate company, exchange-traded funds (ETFs), residential mortgage-backed securities (MBSs), commercial mortgage-backed securities (CMBSs), mortgage REITs, and mortgage ETFs.

Basic Forms of Real Estate Investments[1]

	Debt	Equity
Private	Mortgage debt Construction loans Mezzanine debt	**Direct ownership** Sole ownership Joint ventures Limited partnerships **Indirect ownership** Real estate funds Private REITs
Public	MBS/CMBS/CMOs Covered bonds Mortgage REITs Mortgage ETFs	**Publicly traded shares** Construction Operating Development Public REITs UCITS/mutual funds/ETFs

The second (horizontal) dimension describes whether a real estate investment is in debt or equity.

- An equity investor has an ownership interest in real estate or securities of an entity that owns real estate. Equity investors control decisions such as borrowing money, property management, and the exit strategy.
- A debt investor is a lender that owns a mortgage or mortgage-backed securities. A mortgage is collateralized (secured) by the underlying real estate. The lender has a higher priority claim than an equity investor in the event of default. Because the lender must be repaid first, the value of an equity investor's interest is equal to the value of the property less the value of outstanding debt.

Each of the basic forms has its own risk, expected returns, regulations, legal issues, and market structure. Private real estate investments are usually larger than public investments because real estate is indivisible and illiquid. With public real estate investments, the investors can divide ownership without having to divide the property. As a result, public real estate investments are liquid and enable investors to diversify by participating in more properties.

Private real estate investment requires property management expertise on the part of the owner. In the case of publicly traded securities or private limited partnerships, the real estate is professionally managed.

Direct Real Estate Investment

Direct investment involves purchasing and financing a real estate investment. Benefits of direct investment include the following:

- *Control*. The owner can decide on what to purchase, how to finance it, what improvements to make, to which segment of tenants to market the property, and when to sell.
- *Diversification*. Real estate returns are less than perfectly correlated with the returns of stocks and bonds. Thus, adding private real estate investment to a portfolio can reduce risk relative to the expected portfolio return.
- *Tax benefits*. Real estate can provide deductions for noncash depreciation (even as properties typically appreciate) as well as interest expense.

Drawbacks of direct real estate investment include the following:

- Illiquidity and price opacity
- Complexity of managing property
- Need for specialized knowledge about current market conditions
- High initial investment/capital needed
- Concentration risk if a portfolio has one or few properties

Indirect Real Estate Investment

Indirect investments in real estate can be through limited partnerships, joint ventures, or publicly traded securities. REITs are a popular form of investment for income-producing real estate as they are exempt from double taxation. REITs are exchange traded, and therefore (unlike mutual funds) they do not face redemption/ liquidation risk. REITs have managers with the specialized knowledge necessary to invest in properties successfully. REITs report GAAP metrics such as earnings per share (EPS). Some REITs also provide non-GAAP metrics such as net asset value.

Types of REITs include equity REITs, which invest in real estate directly or via partnerships; mortgage REITs, which lend money for real estate or invest in MBSs/ CMBSs; or hybrid REITS that invest in both real estate and mortgages.

REITs that pursue **core real estate strategies** invest in high-quality commercial and residential properties that deliver stable returns. These REITs typically have openend structures with indefinite lives. Like open-end mutual funds, this structure gives investors the opportunity to invest or redeem at any time.

REITs with riskier investment strategies are most often structured as closed-end funds with finite lives. These strategies include:

- **Core-plus real estate strategies**, which accept a bit more risk than core strategies by undertaking modest development and redevelopment.
- **Value-add real estate strategies**, which undertake development and redevelopment on a somewhat larger scale than core-plus strategies.
- **Opportunistic real estate strategies**, which pursue large-scale redevelopment and repurposing of assets, invest in distressed properties, or speculate on upturns in real estate markets.

LOS 81.b: Explain the investment characteristics of real estate investments.

Among real estate investments, first mortgages or investment-grade CMBSs are the least risky and are similar to investing in bonds. Core strategies are the next-least risky and also have bond-like characteristics, in that they receive stable rental income from many lessors (i.e., a diversified income pool). Value-add and opportunistic strategies are the most risky and are more equity-like in their risk-and-return characteristics.

Because REITs invest in many properties, they provide diversification benefits in addition to liquidity of a publicly traded security. However, the correlation of REIT returns with equity returns is higher than that for direct investment, and correlations increase during steep market downturns. Even so, real estate investments do improve the risk-return profile of a portfolio relative to one comprising only traditional asset classes.

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[1]Reproduced from Level I CFA Curriculum learning module, "Real Estate and Infrastructure," Exhibit 2, with permission from CFA Institute.

LOS 81.c: Explain features and characteristics of infrastructure.

Infrastructure investments include transportation assets (e.g., roads, airports, ports, railways), utility assets (e.g., gas distribution facilities, electric generation and distribution facilities, and waste disposal and treatment facilities), information and communication technology (e.g., telecom towers and cable systems), and social infrastructure (e.g., prisons, schools, and health care facilities).

One way to invest in infrastructure is to construct the assets and either sell or lease them to the government or operate them directly. Alternatively, an investor might buy existing assets from a government to lease back or operate. Infrastructure investments can also be made by a **public-private partnership**.

Cash flows generated from infrastructure investments include **availability payments** for making the infrastructure available, **usage-based payments** such as highway tolls, and **take-or-pay arrangements** that require the buyer to pay minimum purchase price for an agreed-upon volume.

Investments in infrastructure assets that are already constructed are referred to as **brownfield investments**, and investments in infrastructure assets that are to be constructed are referred to as **greenfield investments**. An example of a brownfield investment is the privatization of public assets, which may require additional improvements or involve a sale-leaseback arrangement whereby the asset is purchased from and leased back to the government. Brownfield investments in fully operational facilities are called **secondary-stage investments**. Greenfield investments typically follow the build-operate-transfer (BOT) life cycle.

Infrastructure assets typically have a long life and are quite large in cost and scale, so direct investment in them has low liquidity. However, more liquid investments backed by infrastructure assets are available through ETFs, mutual funds, private equity funds, or master limited partnerships (MLPs). Publicly traded vehicles for investing in infrastructure are a small part of the overall universe of infrastructure investments and are relatively concentrated in a few categories of assets. Debt issued to fund infrastructure can be privately placed or publicly traded.

LOS 81.d: Explain the investment characteristics of infrastructure investments.

In general, investing in brownfield investments provides stable cash flows and relatively high current yields, but offers little potential for growth. Secondary-stage brownfield investments (e.g., existing toll roads and hospitals) are the least risky and offer the lowest return. Greenfield investments (e.g., new toll roads, renewable energy facilities) are subject to more uncertainty and may provide relatively lower yields in the near term, but offer greater growth potential. Infrastructure projects that rely on revenues from uncertain future demand tend to be the riskiest.

Because of long-term contracts and barriers to entry, cash flows from equity investment in infrastructure are stable and have low correlation with public equities. Infrastructure debt also tends to be safer and less affected by economic cycles. Greenfield investments in developing economies, while risky, have generated attractive returns over the long term as they benefit from increasing per-capita incomes and wealth.

Investing in infrastructure assets can provide diversification benefits, but investors should be aware that these investments are often subject to regulatory risk, risk from financial leverage, and the possibility that cash flows will be less than expected. Projects that involve building infrastructure assets have construction risk. When the assets are operated by a private owner, operational risk must also be considered.

Infrastructure investments tend to be most suitable for long-term institutional investors such as pension plans, life insurance companies, and sovereign wealth funds.

Reading 81: Key Concepts

LOS 81.a

Equity or debt investments in real estate can be direct or indirect. Advantages of direct (equity) investment include more operational control, higher tax savings, and diversification to reduce portfolio risk. Drawbacks of direct real estate investment include illiquidity, operational complexity, a need for specialized knowledge, high initial investment, and concentration risk.

Publicly traded securities such as REITs offer liquidity, professional management, exposure to a diversified pool of real estate, and exemption from double taxation. REITs following core strategies invest in high-quality commercial property. Core-plus strategies involve higher risk as they entail some redevelopment of the properties acquired. Value-add strategies take even more risk as they seek to undertake major redevelopment. Opportunistic strategies have the highest risk.

LOS 81.b

Real estate investments with bond-like risk-and-return characteristics include senior debt (the least risky) and core strategies. Real estate investments with equity-like characteristics include core-plus, value-add, and opportunistic strategies.

LOS 81.c

Infrastructure investments include transportation assets, utility assets, information and communication technology, and social infrastructure.

Brownfield investments are already constructed infrastructure facilities, while greenfield investments are yet to be constructed. Fully operational facilities that generate stable cash flows are called secondary-stage brownfield investments.

LOS 81.d

Brownfield investments are less risky than greenfield investments. Demand-based infrastructure projects tend to be the riskiest. Equity investment in infrastructure has a low correlation with public equities, while debt tends to be safer and less affected by economic cycles.

Reading 82: Natural Resources

Module 82.1: Farmland, Timberland, and Commodities

LOS 82.a: Explain features of raw land, timberland, and farmland and their investment characteristics.

Natural resources comprise raw land, land used for growing crops or timber, and commodities. Investment in natural resources can be a direct investment or via commingled funds such as exchange-traded funds, real estate investment trusts (REITs), limited partnerships, and limited liability companies (LLCs). Derivatives such as commodity futures and swaps are also commonly used for exposure to this asset class.

Raw land, farmland, and timberland are all illiquid investments, and their value is primarily driven by location. For timberland and farmland, proximity to transportation and markets increases value, as does access to water and quality of soil. While all three kinds of land can generate lease income and price appreciation, farmland and timberland also generate income from their output.

Farmland is primarily held by individuals, while institutions are most likely investors in raw land and timberland. Timberland lot sizes tend to be much larger than farmland. Timberland also requires specialized expertise. Investors who lack the expertise can invest through **timberland investment management organizations (TIMOs)**.

Farmland and timberland have fewer financing alternatives than residential and commercial real estate. Financing is often in the form of bank loans or direct private debt. Newer REITs in farmland allow retail investors to get a foothold in this category.

While most agricultural crops must be harvested within a short period, the choice of when to harvest timber is based on current prices and expected growth rates. Because commodity prices fluctuate, farmers often use short positions in commodity futures to hedge their expected harvest. Agricultural crops (including trees) consume carbon, and therefore, they are attractive to investors with an environmental, social, and governance (ESG) focus on climate change.

LOS 82.b: Describe features of commodities and their investment characteristics.

We can divide commodities into three major sectors: metals (including base or industrial metals and precious metals), agricultural products, and energy products. Commodity contracts are also classified based on grade (quality) and delivery location.

Governments may provide subsidies for some food crops for consumers or price support to farmers. They may also control access to extractable natural resources or even be directly involved in production. Climate change–related regulation may decrease demand for fossil fuels while increasing demand for minerals such as lithium, cobalt, and nickel.

While it is possible to invest directly in commodities such as grain and gold, derivatives such as futures are more commonly used to gain exposure to commodities. Because commodities are physical goods, they have costs for storage and transportation.

Futures, forwards, and options on futures are all available forms of commodity derivatives. Futures trade on exchanges and therefore have no counterparty risk.

Other methods of exposure to commodities include the following:

- Exchange-traded products (ETPs) include **exchange-traded funds (ETFs)** or **exchange-traded notes (ETNs)**. ETPs are suitable for investors who are limited to buying equity shares. ETFs can invest in commodities or commodity futures and can track prices or indexes.
- **Managed futures funds**, such as such as **commodity trading advisers (CTAs)**, are actively managed. Some managers concentrate on specific sectors (e.g., agricultural commodities), while others are more diversified. Managed futures funds can be structured as limited partnerships with fees like those of hedge funds and restrictions on the number, net worth, and liquidity of the investors. They can also be structured like mutual funds with shares that are publicly traded so that retail investors can benefit from professional management. This structure allows a lower minimum investment and offers greater liquidity than a limited partnership structure. **Separately managed accounts (SMAs)** are appropriate for larger investors who may require custom portfolios based on their individual preferences and needs.
- Specialized funds in specific commodity sectors can be organized under any of the structures we have discussed and focus on commodities such as oil and gas, grains, precious metals, or industrial metals.

Commodity Valuation

Wheat today and wheat six months from today are different products. Purchasing the commodity today will give the buyer the use of it if needed, while contracting for wheat to be delivered six months from today avoids storage costs (warehouse cost, insurance, and spoilage) and having cash tied up. An equation that considers these aspects is:

$$\text{futures price} \approx \text{spot price} \times (1 + \text{risk-free rate}) + \text{storage costs} - \text{convenience yield}$$

or

$$\text{futures price} = \text{spot price} + \text{net cost of carry}$$

where:

$$\text{net cost of carry} = \text{cost of capital} + \text{storage cost} - \text{convenience yield}$$

Convenience yield is the nonmonetary value of having a physical commodity for use over the period of a futures contract. If there is little or no convenience yield, the net cost of carry will be positive, and futures prices will be higher than spot prices, a situation termed **contango**. When the convenience yield is high, the net cost of carry will be negative, and futures prices will be less than spot prices, a situation referred to as **backwardation**. Contango decreases the return of long-only investors, while backwardation increases it.

LOS 82.c: Analyze sources of risk, return, and diversification among natural resource investments.

Commodity Prices and Investments

Spot prices for commodities are a function of supply and demand. Demand is affected by the value of the commodity to end users and by global economic conditions. Supply is affected by production, storage costs, and existing inventories.

For many commodities, supply is inelastic in the short run because of long lead times to alter production levels (e.g., drill oil wells, plant crops, or decide to plant less of them). As a result, commodity prices can be volatile when demand changes significantly over the economic cycle, or when there are supply shocks such as natural disasters.

Production of some commodities, especially agricultural commodities, can be significantly affected by the weather and plant disease, leading to high prices when production is low and low prices when production is high. Costs of extracting oil and minerals increase as more expensive methods or more remote areas are used.

To estimate future needs, commodity producers analyze economic events, government policy, and forecasts of future supply. Investors analyze inventory levels, forecasts of production, changes in government policy, and expectations of economic growth to forecast commodity prices.

In recent decades, returns and the volatility of returns have been higher for commodities than for global stocks or bonds. Timberland and farmland also have higher average returns, but with lower volatility than global stocks (but similar volatility to global bonds). As with other investments, speculators can earn high returns over short periods when their expectations about short-term commodity price movements are correct.

Historically, correlations of commodity returns with the returns of global equities and global bonds have been low. As a result, adding commodities to a portfolio of traditional assets can provide diversification benefits. Because commodity prices tend to move with inflation rates, holding commodities can act as a hedge of inflation risk. During periods of high (low) inflation, commodities outperform (underperform) both stocks and bonds. Commodity prices are more sensitive to geopolitical and weather-related risk factors.

Reading 82: Key Concepts

LOS 82.a

Returns on farmland and timberland are driven by changes in land prices and prices for the underlying output. Risks of investing in farmland and timberland include low liquidity, variable cash flows that depend on weather, and potential losses from natural disasters.

LOS 82.b

Natural resource investments include commodities, farmland, and timberland.

Exposure to commodity prices is most commonly achieved with derivatives. Other methods of exposure include commodity ETFs, managed futures funds, and specialized funds in specific commodity sectors.

futures price = spot price + net cost of carry

When the convenience yield is sufficiently high (low), net cost of carry is negative (positive), a situation known as backwardation (contango). Contango decreases the return of long-only investors, while backwardation increases it.

LOS 82.c

Returns on farmland and timberland include sales of the products raised on the land. Timberland returns also include price changes on the land, which depend on expected lumber prices and how much timber has been harvested. Farmland returns also depend on land price changes, changes in farm commodity prices, and the quality and quantity of the crops produced.

Commodity returns have low correlation with traditional asset classes and provide diversification benefits as well as a hedge against inflation.

Reading 83: Hedge Funds

Module 83.1: Hedge Funds

LOS 83.a: Explain investment features of hedge funds and contrast them with other asset classes.

Hedge funds are private pooled investment vehicles that are generally available only to qualified or accredited investors. While these funds were traditionally designed to use long and short exposures to generate positive returns in any market, they have evolved to pursue a wide variety of strategies. Unlike mutual funds, hedge funds tend to be lightly regulated, and their managers have great freedom in selecting investment strategies. The primary return drivers for hedge funds are market inefficiency and price volatility.

Hedge funds invest in traditional asset classes (debt and equity), use leverage, and use derivatives in pursuit of their strategies. Hedge funds are often evaluated on a total return or risk-adjusted return basis, rather than against a benchmark.

Hedge fund managers often invest their own money in their funds. Performance fees for hedge fund managers may be linked to a **high-water mark** requirement, whereby performance fees are payable only when the fund value exceeds its highest prior value.

Mutual funds, REITs, and ETFs are all publicly traded, while hedge funds are privately held. Unlike private equity funds, hedge funds tend to invest mostly in liquid asset classes and have a shorter time horizon. Private equity funds require a longer time horizon, while hedge funds have periodic redemptions.

Hedge Fund Categories

We can classify hedge funds by the strategies they pursue. Here we list some examples of each.

1. **Equity hedge fund strategies** seek to profit from long or short positions in publicly traded equities and derivatives with equities as their underlying assets. Short positions taken to reduce or remove overall market risk can be in a market index if a manager does not have a negative opinion on specific securities. Subcategories include the following:
    - **Fundamental long/short.** Use long positions in undervalued securities based on fundamental analysis while simultaneously having a short position in a portfolio of stocks or an index. This strategy seeks to capture alpha when a market correction occurs. Most managers have net long exposure (i.e., a long bias).
    - **Fundamental growth.** Use fundamental analysis to find high-growth companies. Identify and buy equities of companies that are expected to sustain relatively high rates of capital appreciation, and short equities of companies expected to have low or no revenue growth. Typically, these funds have a net long bias.
    - **Fundamental value.** Buy equity shares that are believed to be undervalued, and short equities believed to be overvalued, based on fundamental analysis. The performance of value stocks relative to growth stocks drives performance.
    - **Market neutral.** Use technical or fundamental analysis to select undervalued equities to be held long and to select overvalued equities to be sold short, in approximately equal amounts to profit from their relative price movements without exposure to market risk. Leverage may be used.

    - **Short bias.** Employ technical, quantitative, and fundamental analysis and take predominantly short positions in overvalued equities, possibly with smaller long positions but with negative market exposure overall. This is a contrarian strategy that focuses on the manager's skills at discerning flawed business strategies or accounting.
2. **Event-driven strategies** are typically based on a corporate restructuring or acquisition that creates profit opportunities for long or short positions in common equity, preferred equity, or debt of a specific corporation. Event-driven funds are typically long biased. Subcategories are as follows:
    - **Merger arbitrage.** Buy the shares of a firm being acquired and sell short the firm making the acquisition. Although the term *arbitrage* is used, such a strategy is not risk free because deal terms may change or an announced merger may not take place.
    - **Distressed/restructuring.** Buy the securities of firms in financial distress when analysis indicates that value will be increased by a successful restructuring, and possibly short overvalued securities at the same time.
    - **Activist shareholder.** Buy sufficient equity shares to influence a company's policies, with the goal of increasing company value (e.g., by restructuring, change in strategy or management, or return of capital to equity holders).
    - **Special situations.** Invest in the securities of firms that are issuing or repurchasing securities, spinning off divisions, selling assets, or distributing capital.
3. **Relative value strategies** involve buying a security and selling short a related security, with the goal of profiting when a perceived pricing discrepancy between the two is resolved. Subcategories include the following:
    - **Convertible arbitrage fixed income.** Exploit pricing discrepancies between convertible bonds and the common stock of the issuing companies and options on the common shares.
    - **Specific fixed income (ABS/MBS/high yield).** Exploit pricing and quality discrepancies.
    - **General fixed income.** Exploit pricing discrepancies among fixed-income securities of various issuers and types.
    - **Multistrategy.** Exploit pricing discrepancies among securities within and across asset classes and markets.
4. **Opportunistic strategies** focus on macro events and commodity trading. Often, these strategies are implemented using ETFs or derivatives in addition to individual securities. Subcategories include the following:
    - **Macro strategies** are based on global economic trends and events and may involve long or short positions in equities, fixed income, currencies, or commodities. These funds benefit from heightened volatility surrounding major events. Smoothing of economic shocks by central bank actions reduce the attractiveness of these strategies.
    - **Managed futures** funds may focus on trading commodity futures (these funds are known as commodity trading advisers, or CTAs) or incorporate financial futures. Commodity prices tend to behave differently than financial assets, in that high prices tend to decrease demand (which in turn decreases prices), while low prices reduce supply (which in turn increases prices).

Unique Characteristics of Hedge Fund Investing

Hedge funds differ from other private investment vehicles, such as mutual funds and REITs:

- They tend to be less regulated and have flexible mandates, allowing them freedom to deploy a wide range of strategies, use leverage, and choose from a larger universe of securities and derivatives.
- Hedge funds are higher cost, as they charge higher fees in the form of an incentive fee on top of generally high management fees.
- They have lower liquidity, including lockup periods and liquidity gates. A **lockup period** is the time after initial investment over which limited partners either cannot request redemptions or incur significant fees for redemptions (a soft lockup). A **notice period** is the amount of time a fund has to fulfill a redemption request made after the lockup period has passed. A **liquidity gate** is a partial restriction on redemptions (i.e., less than a full suspension of them).

Hedge fund managers often incur significant transactions costs when they redeem shares. Redemption fees can offset these costs. Notice periods and liquidity gates allow time for managers to reduce positions in an orderly manner. Redemptions often increase when hedge fund performance is poor over a period, and the costs of honoring redemptions may further decrease the value of the remaining partnership interests.

- There is reduced transparency about strategies followed and investments made by the fund to protect proprietary trading methods. This makes it difficult for investors and analysts to assign accurate values to a hedge fund's holdings.

LOS 83.b: Describe investment forms and vehicles used in hedge fund investments.

Hedge funds can be structured as commingled funds, whereby capital from many investors is pooled together, or as a **separately managed account (SMA)** for a single large investor.

Commingled funds often have a **master-feeder structure** that is designed to be tax efficient, enjoys economies of scale, and allows for funding from global investors. Under such a structure, there are two feeder funds: offshore (in a tax haven) and onshore. Both funds flow into a master fund, which makes the investments. The master-feeder structure bypasses regional regulatory requirements.

SMAs allow for a customized portfolio to meet an investor's risk/return objectives. One concern is that the manager has no stake in the fund, so the manager's interests are not as well aligned with those of the investor as they would be in a typical commingled fund. SMAs also require more operational oversight and thus are appropriate for larger or institutional investors. The benefit of lower negotiated fees in SMA structure is offset by the disadvantage of receiving allocations of only the fund manager's most liquid trades.

Hedge funds are typically structured as limited partnerships or limited liability corporations. The general partner is the fund manager and receives compensation based on the fund's performance. The contractual relationships between the GP and LPs are laid out in the fund documents: partnership agreement, **private placement memorandum**, or articles of incorporation. Typically, a fund is structured to have an indefinite life, though most funds do wind down as they are liquidated on a regular basis.

Recently, there has been market pressure to reduce hedge fund fees from the previous standard of 2% management fees and 20% performance fees. Some of the newer funds have 1% management fees plus 30% performance fees based on performance relative to a benchmark (as opposed to being based on total returns).

Indirect Investment Forms

A **fund-of-funds** is an investment company that invests in hedge funds. Fund-of-funds investing can give investors diversification among hedge fund strategies, offer a manager's expertise in selecting individual hedge funds, and provide smaller investors with access to hedge funds in which they may not be able to invest directly. These funds also have reduced lockup periods and greater exit liquidity.

Fund-of-funds managers charge an additional layer of fees beyond the fees charged by the individual hedge funds in the portfolio. Historically, these additional fees have been a 1% management fee and a 10% incentive fee. Because these fees to the fund-of-funds manager are on top of fees charged by the individual funds, they can significantly reduce investors' net returns.

LOS 83.c: Analyze sources of risk, return, and diversification among hedge fund investments.

Hedge fund returns come from three sources:

1. *Market beta*. This is the return attributable to the broad market index. Investors can get this from passive investments in index funds.
2. *Strategy beta*. This is the return attributable to specific sectors in which a fund has exposure.
3. *Alpha*. This is the additional return that is delivered by the manager through security selection.

Hedge fund managers use leverage to magnify the value added through strategy beta and alpha. High fees, however, act as a drag on performance.

Hedge fund performance as measured by indexes is often overstated. These indexes compile results based on voluntary reporting of performance data by participating fund managers. This likely biases returns and correlations with traditional investment returns. Because index reporting is voluntary, poor-performing funds are less likely to report.

- Index returns may exhibit **survivorship bias** because hedge funds might not be included until they have existed for a minimum time or reached a minimum size. In this case an index will not reflect the poor performance of funds that fail before reaching these minimums.
- **Selection bias** may result from index providers assigning funds to categories inconsistently or having different requirements for including a fund.
- **Backfill bias** refers to the effect on historical index returns of adding fund returns for prior years to index returns when a fund is added to an index. Like selection bias, this tends to overstate performance because funds with better historical returns are more likely to be added.

Historical data show that while a hedge fund's performance is highly dependent on the time period over which the performance is measured, hedge funds as an asset class provide diversification benefits relative to portfolios that invest in traditional asset classes. Hedge funds tend to be more correlated with equities than with fixed income.

Reading 83: Key Concepts

LOS 83.a

Hedge fund investments tend to have high fees, lower transparency, lower regulatory burden, and lower liquidity than mutual funds or ETFs.

Hedge funds generally use leverage, take both long and short positions, and use derivatives.

Equity hedge strategies take long and short positions in equities and equity derivatives. Strategies include market neutral, fundamental growth, fundamental value, fundamental long/short, and short bias.

Event-driven strategies include merger arbitrage, distressed/restructuring, activist, and special situations.

Relative value strategies seek profits from unusual pricing issues.

Opportunistic strategies include macro strategies and managed futures.

LOS 83.b

Hedge funds can be structured as commingled funds (master-feeder structure) or as a separately managed account.

A fund-of-funds invests in other hedge funds. Its advantages are that it can give investors diversification among hedge fund strategies, provides expertise in selecting hedge funds, and provides access to hedge funds that are otherwise unavailable. Its disadvantage is the additional layer of fees beyond the fees charged by the hedge funds in the portfolio.

LOS 83.c

Hedge fund returns comprise market beta, strategy beta, and alpha. Hedge funds often magnify strategy beta and alpha by using leverage.

Hedge fund performance is often overstated because of selection, backfill, and survivorship bias.

With proper due diligence, hedge funds can offer diversification benefits to a portfolio of traditional investments.

Reading 84: Introduction to Digital Assets

Module 84.1: Distributed Ledger Technology

LOS 84.a: Describe financial applications of distributed ledger technology.

Digital assets are assets that can be electronically created, stored, and transferred. This relatively new alternative asset class includes cryptocurrencies, tokens, and digital collectibles. A digital asset is secured and validated using **distributed ledger technology (DLT)**, also known as blockchain technology. Cryptocurrencies have their own blockchains, while crypto tokens are built on blockchains that already exist.

A **distributed ledger** is a database shared among market participants. It maintains a record of all transactions, which allows each participant to have an identical copy of the database. DLT benefits include accuracy, transparency, and security; rapid ownership transfer; and peer-to-peer (P2P) interactions for creating, trading, and monitoring digital assets. DLT disadvantages include data protection concerns, potential privacy violations, and the large amount of computational power needed for verifying transactions.

A DLT network consists of a digital ledger, a consensus mechanism, and a network of participants. A consensus mechanism establishes a common state of the ledger, which involves validating transactions and updating the ledger. These two steps generate records that are immutable (i.e., unchangeable) while providing transparency for all network participants. DLT uses cryptography to encrypt data, which prevents unauthorized parties from accessing data.

DLT can also implement **smart contracts**, where a computer program self-executes based on predetermined terms and conditions. Automating contingent claims and collateral transfers during default events are examples of smart contract applications.

A **blockchain** is a digital ledger that records information sequentially within blocks. These blocks are linked together (i.e., chained), and information is secured using cryptographic techniques. Blocks within the chain contain a group of transactions as well as a secure link (known as a hash) to the previous block. A newly created transaction is only added to a chain after it has been validated by authorized participants.

Consensus Protocols

Consensus protocols determine how blocks are chained together. They are structured to protect against market manipulation. The main types of consensus protocols are proof of work and proof of stake.

For the **proof of work (PoW) protocol**, when a transaction takes place, **miners** use computers to solve a cryptographic problem, which then verifies the transaction. Blocks are validated and locked into the blockchain thanks to powerful computers that use massive amounts of energy. This process rewards the miners with cryptocurrency. The resources required for mining impose substantial costs on any attempt to manipulate the blockchain's historical record. To do so would require one party to control most (i.e., 51%) of the network. For this reason, a blockchain is more likely to succeed when there is a large number of network participants. PoW is the most widely used consensus mechanism.

The **proof of stake (PoS) protocol** is an emerging consensus mechanism in which certain network participants pledge collateral (a process known as staking) to guarantee the validity of a block. These participants, who are known as **validators**, use their stake to signal to the network that a transaction is ready to be added to the blockchain. Additional validators will then verify that the transaction is, indeed, authentic. Validators also protect the network from potential malicious attacks by controlling most of the network's computational power. Validators are rewarded with a return on their staked collateral.

Permissionless and Permissioned Networks

The two forms of DLT networks are permissionless and permissioned.

In **permissionless networks**, transactions are visible to all users within the network, and any user can execute a transaction. In addition, all network functions can be performed by any network participant. Bitcoin is considered a permissionless network, as are many other types of cryptocurrency.

An advantage of a permissionless (i.e., open) network is that transactions are confirmed or denied through consensus mechanisms rather than by a centralized authority. Blockchain transactions cannot be manipulated once they have been added to the ledger, which creates an immutable record of all transactions. Transacting parties do not have to trust each other in permissionless networks.

In **permissioned networks**, users may be restricted from some network activities. Permissions (or controls) can modify the level of ledger accessibility. These levels range from adding transactions (for network participants) to viewing only the transaction history (for regulators). Because permissioned blockchains have stronger restrictions, they are more cost effective than open and decentralized permissionless networks.

Digital Asset Categories

DLT has many potential applications within the financial industry, including the creation of digital assets and the delivery of more efficient financial services. **Types of Digital Assets** outlines the main types of digital assets currently available.

Types of Digital Assets

Cryptocurrencies	Tokens

• Bitcoin
• Altcoins
  ◦ Other cryptocurrencies (e.g., Ether)
  ◦ Stablecoins (e.g., Tether)
  ◦ Meme coins (e.g., Dogecoin)
• Central bank digital currencies (CBDCs)
	• Nonfungible tokens (NFTs)
	• Security tokens
	  ◦ Initial coin offerings (ICOs)
	• Utility tokens
	• Governance tokens

A **cryptocurrency** is a digital currency issued privately with no backing from a central bank. By using this digital medium of exchange, parties can execute near-real-time transactions without the need for an intermediary. Cryptocurrencies commonly limit how many currency units can be issued. The objective of this supply limit is to maintain the unit as a store of value; however, many cryptocurrencies still experience extreme price volatility.

Various cryptocurrencies exist because their blockchains are designed and optimized differently. As the most widely traded cryptocurrency, Bitcoin has influenced the development of many other types of digital assets. Alternate cryptocurrencies, known as **altcoins**, are based on the technology used by Bitcoin. The most well-known altcoin is Ether, which is built on the Ethereum network. The Ethereum blockchain allows users to develop applications (i.e., smart contracts) to secure and validate transactions. Therefore, the value of Ethereum and other programmable blockchains extends beyond being only a store of value. To date, Bitcoin and Ether are the two most popular cryptocurrencies; as of July 2022, they made up more than 80% of the total cryptocurrency market.

Stablecoins offer stable digital currency values. These coins are linked to the value of another asset, such as the U.S. dollar, and secured by a basket of assets. The value of stablecoins is protected from price volatility through this reserve basket, which minimizes risk from potential transaction failures. Smart stablecoins or algorithmic stablecoins control their supply by using algorithms to mint additional assets when demand increases. Stablecoins cannot be exchanged for fiat currency, and they have no legal or regulatory backing. One of the most popular stablecoins is Tether, which is pegged to the U.S. dollar. A special case of stablecoins is an asset-backed token, which maintains price parity with a target asset, such as gold. We will discuss these tokens in more detail later in this reading.

> **Professor's Note**
>
> The TerraUSD stablecoin was pegged to the U.S. dollar "algorithmically" instead of being directly linked to the underlying collateral value. In May 2022, the 1:1 peg failed after significant investor withdrawals and its value collapsed, resulting in a run on several stablecoins.

In general, **meme coins** are launched for entertainment purposes. In some cases, their rapid rise in popularity has allowed early purchasers to sell these coins for a quick and significant profit. The most popular meme coin is Dogecoin, which is based on a popular internet meme. In May 2021, Dogecoin saw a dramatic increase in market value thanks to social media endorsements. However, a year later, its market value dropped sharply as its usefulness came into question.

As mentioned, governments do not currently back or regulate cryptocurrencies. Nevertheless, central banks are currently exploring the potential benefits of cryptocurrency equivalents as alternatives to physical currency. A **central bank digital currency (CBDC)** is essentially a digital version of a banknote or coin issued by a central bank.

When physical assets like real estate change hands, considerable work is required to verify ownership. **Tokenization** uses DLT to streamline this process by digitally tracking the historical record of ownership. A **nonfungible token (NFT)** is an example of tokenization in which a digital asset is linked to a certificate of authenticity. The key difference between NFTs and "fungible" tokens is that each NFT represents a distinct object. For example, digital artwork is a common application for NFTs.

A **security token** digitally tracks ownership rights in publicly traded securities. A blockchain can facilitate the custody, settlement, recordkeeping, and post-trade processing of security tokens. With this single ledger technology, participants can perform various transactions more easily and with more transparency. An example of a security token is an **initial coin offering (ICO)**, which is an unregulated process in which companies offer crypto tokens in exchange for money or other cryptocurrency. In an ICO, tokens are issued to investors for future purchases of products or services. An ICO is an alternative to regulated initial public offerings (IPOs). ICOs are potentially less expensive and take less time to raise capital compared to IPOs; however, most ICOs do not have attached voting rights. Many jurisdictions are considering regulating ICOs due to an increase in fraud connected to these offerings.

Utility tokens provide network services, such as service payments and network fees. In contrast to security tokens that may pay dividends, a utility token only compensates investors for network activities. **Governance tokens** are offered on permissionless networks and act as voting rights to determine how networks should operate. For example, a network can maintain stability and integrity by voting on solutions for a technical issue.

Module 84.2: Digital Asset Characteristics

LOS 84.b: Explain investment features of digital assets and contrast them with other asset classes.

Investing in digital assets is becoming more commonplace in the financial industry. In 2013, approximately 70 cryptocurrencies were recorded, but by early 2022, over 10,000 cryptocurrencies had been issued. Due to the high expected returns and diversification benefits of investing in cryptocurrencies, institutional investors are taking more interest in this alternative asset class. To accommodate the growing number of investors, financial service providers—including digital exchanges and custodians—have strengthened their infrastructures to support future institutional investment.

The nature of digital assets differs from traditional financial assets. Key investment differences between digital assets and other asset classes are as follows:

- *Inherent value differences*. Most digital assets are not backed by underlying assets or the cash flows they generate, such as interest or dividends. Because digital assets lack expected earnings, they have no fundamental value. In general, digital asset prices are determined by projected asset growth, which is based on scarcity (due to the limited currency supply) and the ability to transfer value in the future.
- *Transaction validation differences*. Traditional financial instruments are typically recorded on private ledgers maintained by central intermediaries. In contrast, digital assets are typically recorded on decentralized digital ledgers.
- *Medium of exchange differences*. Traditional financial assets are traded and priced in globally accepted fiat currencies. Financial systems rely on these fiat currencies as a medium of exchange. In contrast, digital assets are used as an alternative to fiat currencies and are mainly used for online transactions. However, the mainstream financial system is hesitant to accept cryptocurrencies as a medium of exchange. This is due to high transaction costs, cryptocurrencies not being legal tender in many jurisdictions, and cryptocurrencies being heavily restricted in several countries. CBDCs are currently being evaluated as tokenized forms of fiat currencies.
- *Regulatory differences*. Most jurisdictions have clear and established rules governing financial instruments. However, unlike traditional financial assets, digital assets lack well-developed standards. Regulations specific to digital assets are currently under development. Regulators in the United States consider some digital currencies to be commodities, while other regulators consider them as nonfinancial assets. In addition, the cryptocurrency exchanges used to trade digital currencies are typically unregulated. As a result, participants in these exchanges are not specifically prohibited from engaging in behaviors like market manipulation.

LOS 84.c: Describe investment forms and vehicles used in digital asset investments.

Investing in digital assets can be done directly on the blockchain, or indirectly through exchange-traded products and hedge funds. Direct ownership of cryptocurrencies requires the use of a "cryptocurrency wallet," which contains digital encryption keys for accessing and trading assets. The two categories of exchanges for trading cryptocurrencies are centralized exchanges and decentralized exchanges.

Centralized exchanges are privately held and offer trading platforms for price transparency and volume information. Despite being at odds with Bitcoin's decentralized nature, centralized exchanges are the most popular type of cryptocurrency exchange. Traders on these exchanges trade directly and electronically on private servers, which may introduce security vulnerabilities. Compromised exchange servers can cause trading to halt and valuable user information to leak, such as keys to cryptocurrency wallets. Depending on the jurisdiction, some of these exchanges are regulated as financial exchanges.

Similar in concept to Bitcoin, **decentralized exchanges** implement decentralized blockchain principles. A decentralized exchange does not have a centralized authority and it operates on a distributed framework. Multiple computers are used to service this type of exchange, so if one computer is attacked, the exchange will remain in operation. Therefore, attacks on decentralized exchanges are substantially more difficult. In addition, because no one person, group, or organization controls a decentralized exchange, regulations are difficult to enforce.

On either type of exchange, a lack of regulatory oversight raises investor protection concerns. Fraud and market manipulation are more likely to occur, such as **pump-and-dump schemes** involving questionable promotion of cryptocurrencies on social media to deliberately increase prices.

> **Professor's Note**
>
> The centralized exchange FTX became one of the largest crypto and stablecoin exchanges in 2021. However, falling cryptocurrency prices, increasing interest rates, and fraudulent management decisions led to extreme illiquidity issues for the exchange. In November 2022 these issues triggered substantial investor withdrawals, which ultimately led to the bankruptcy of FTX.

Direct Investment

Direct investment in cryptocurrencies occurs when a transaction is recorded on the blockchain. When a transaction takes place, it becomes validated and permanently stored on the blockchain. Examples of gaining direct exposure include purchasing tokens on a cryptocurrency exchange, trading an NFT, and investing in an ICO. Continuous trading is offered on most cryptocurrency exchanges.

Investing in cryptocurrencies directly has several risks. Fraud has become more prevalent with the rise in popularity of cryptocurrencies. This includes scam ICOs, pump-and-dump schemes, market manipulation, theft, and attempts to gain access to wallet credentials. Also, because cryptocurrencies are normally held in secure wallets that require unique passkeys to access, losing those passkeys makes the cryptocurrency in a wallet worthless. It is estimated that 20% of Bitcoin is in lost or abandoned wallets. In addition, many small cryptocurrencies are held primarily by a few large entities (known as whales) that may hold enough of a cryptocurrency to manipulate its price.

Indirect Investment

Indirect exposure to cryptocurrencies can be achieved through coin trusts, futures contracts, exchange-traded products, stocks, and hedge funds.

Cryptocurrency coin trusts offer shares in a trust that holds large amounts of cryptocurrency. This type of investment trades over the counter and is similar to a closed-end fund. Coin trusts increase transparency while eliminating the need for digital wallets and encryption keys. However, some trusts charge substantial fees and expenses.

Cryptocurrency futures contracts involve buying or selling a specific quantity of cryptocurrency at a predetermined future date. For example, the Chicago Mercantile Exchange trades Bitcoin futures based on spot Bitcoin prices from cryptocurrency exchanges. These contracts are typically cash settled and involve leverage. Because this market is newly developing, cryptocurrency futures are typically more volatile and less liquid than traditional futures markets.

Cryptocurrency exchange-traded products, such as exchange-traded funds, attempt to mimic the returns of digital assets. These products typically gain exposure to cryptocurrency price movements using cash or cryptocurrency derivatives.

Cryptocurrency stocks offer indirect exposure through their business connection to digital assets. Examples include public digital exchanges; payment providers that support cryptocurrencies; corporations that accept cryptocurrencies as payments, invest in cryptocurrencies, or mine cryptocurrencies; and companies that provide products or services for operating blockchain networks.

Cryptocurrency hedge funds employ strategies such as discretionary long, long/short, quantitative, and multistrategy to invest indirectly in cryptocurrencies. Some hedge funds also conduct Bitcoin mining activities to generate additional returns.

Digital Investment in Nondigital Assets

Asset-backed tokens represent digital ownership of physical assets or financial assets (i.e., nondigital assets). These tokenized assets are collateralized by the underlying asset, such as oil, gold, or equities. Asset-backed tokens can potentially increase liquidity of high-priced assets, such as real estate, by allowing for fractional ownership. A digital representation also allows for an immutable record of ownership, which improves transparency and reduces transaction costs. Regulators generally classify asset-backed tokens as securities.

The Ethereum network is often used to issue asset-backed tokens by allowing P2P interactions via smart contracts. Decentralized applications, referred to as dApps, enable transactions to be recorded on the blockchain without the need for a centralized system.

Decentralized finance (DeFi) grew out of this movement toward financial decentralization. DeFi seeks to develop sophisticated financial products and services using open-source financial applications. Essentially, DeFi is a marketplace of dApps that can offer a medium for exchange, a store of value, tokenization of assets, and immutable records of ownership. Currently, most dApps focus on digital asset speculation.

LOS 84.d: Analyze sources of risk, return, and diversification among digital asset investments.

Cryptocurrency values depend largely on asset appreciation, and a significant driver of prices is their limited supply. For instance, Bitcoin is limited to 21 million units. As a result, some investors refer to Bitcoin as digital gold. To date, Bitcoin has generated high returns and high volatility compared to traditional asset classes. For example, Bitcoin's price has risen from $0.05 when it was first made publicly available in mid-2010 to over $65,000 in November 2021, before falling below $20,000 in June 2022. **Monthly Log Returns for Bitcoin and Asset Class Benchmarks (January 2011–January 2022)(1)** illustrates the risk and return of Bitcoin compared to other major asset class indexes.

Monthly Log Returns for Bitcoin and Asset Class Benchmarks (January 2011–January 2022)[1]

	Bitcoin (In USD)	S&P 500 Index Total Return	MSCI World Index Return	Bloomberg Global Aggregate Index
Average	8.84%	1.13%	0.66%	0.16%

Standard deviation	0.32	0.04	0.04	0.01
Coefficient of variation	3.66	3.43	6.09	8.16

Other cryptocurrencies exhibit similar price volatility to Bitcoin. This volatility reflects the overall uncertainty associated with this asset class. Even though some of these assets are regulated as digital commodities in the United States and formal rules in the European Union are under development, investors remain wary due to recent fraud and crime. Consequently, cryptocurrencies have been severely restricted by several countries, including China, which banned them outright in 2021.

Historically, cryptocurrency returns have demonstrated low correlations with traditional asset class returns. Therefore, cryptocurrencies have the potential to provide portfolio diversification benefits. However, correlations may increase during periods of extreme market stress. In general, cryptocurrency prices are influenced by the unique aspects of this asset class, including market adoption, risk appetite, technological innovation, network impacts, speculation, and regulation. **Correlations Between Monthly Log Returns for Cryptocurrencies and Selected Asset Classes (January 2011–January 2022)(1)** displays a correlation matrix for Bitcoin and other major asset class indexes.

Correlations Between Monthly Log Returns for Cryptocurrencies and Selected Asset Classes (January 2011–January 2022)[2]

	Bitcoin in U.S. Dollars	S&P 500 Index Total Return	MSCI World Index Return	Bloomberg Global Aggregate Index
Bitcoin in U.S. dollars	1	—	—	—
S&P 500 Index total return	0.21	1	—	—
MSCI World Index return	0.22	0.97	1	—
Bloomberg Global Aggregate Index	0.14	0.25	0.33	1

[1] Reproduced from 2024 Level I CFA Curriculum, Volume 5, with permission from CFA Institute.

[2] Reproduced from 2024 Level I CFA Curriculum, Volume 5, with permission from CFA Institute.

Reading 84: Key Concepts

LOS 84.a

Digital assets are those that can be electronically created, stored, and transferred. This alternative asset class includes cryptocurrencies, tokens, and digital collectibles. A digital asset relies on distributed ledger technology (DLT).

A blockchain is a digital ledger that records information sequentially within blocks. Consensus protocols determine how blocks are chained together. The main types of consensus mechanisms are proof of work (miners use powerful computers to verify transactions) and proof of stake (validators pledge collateral to guarantee the validity of transactions).

In permissionless networks, transactions are visible to all users within the network, and any user can execute a transaction. In permissioned networks, some network activities may be modified or restricted for users.

A cryptocurrency is a digital currency issued privately with no backing from a central bank. Stablecoins offer stable digital currency values by linking to another asset (e.g., the U.S. dollar). Meme coins are launched for entertainment purposes. A central bank digital currency is a digital version of a bank note or coin issued by a central bank.

Tokenization uses blockchain technology to digitally track the historical record of asset ownership.

LOS 84.b

The key differences in investment features between digital assets and other asset classes are as follows:

- Most digital assets are not backed by underlying assets or cash flows, as traditional financial instruments are. Digital assets do not have expected earnings, so they have no fundamental value.
- Traditional financial assets are typically recorded on private ledgers maintained by central intermediaries. Digital assets are typically recorded on decentralized digital ledgers.
- Traditional financial assets are traded and priced in fiat currencies. Digital assets are an alternative to fiat currencies and are mainly used for online transactions.
- Most jurisdictions have clear and established rules that govern traditional financial instruments. Digital assets currently lack these well-developed standards.

LOS 84.c

Centralized exchanges allow participants to trade directly and electronically on private servers. Decentralized exchanges operate on a distributed framework where multiple computers service the network. For both exchanges, investor protection concerns arise due to a lack of regulatory oversight.

Direct investment in cryptocurrencies occurs when a transaction is recorded directly on the blockchain. Investing in cryptocurrencies directly has several risks, including scam ICOs, pump-and-dump schemes, market manipulation, theft, and attempts to gain access to cryptocurrency wallet credentials.

Indirect investment in cryptocurrencies can be achieved through coin trusts, futures contracts, exchange-traded products, stocks, and hedge funds.

Asset-backed tokens represent digital ownership of nondigital assets. The movement toward decentralization and using digital investment forms for nondigital assets has led to decentralized finance (DeFi) products and services.

LOS 84.d

Cryptocurrency values are driven by asset appreciation and the limited supply of currency units. To date, digital assets have generated high returns and high volatility compared to traditional asset classes. Because digital asset returns have demonstrated low correlations with traditional asset class returns, this alternative asset class has the potential to provide portfolio diversification benefits.

Reading 85: Portfolio Management: An Overview

Module 85.1: Portfolio Management Process

LOS 85.a: Describe the portfolio approach to investing.

The **portfolio perspective** refers to evaluating individual investments by their contribution to the risk and return of an investor's portfolio. The alternative to taking a portfolio perspective is to examine the risk and return of individual investments in isolation. An investor who holds all his wealth in a single stock because he believes it to be the best stock available is not taking the portfolio perspective—his portfolio is very risky compared to holding a diversified portfolio of stocks. Modern portfolio theory concludes that the extra risk from holding only a single security is not rewarded with higher expected investment returns. Conversely, diversification allows an investor to reduce portfolio risk without necessarily reducing the portfolio's expected return.

In the early 1950s, the research of Professor Harry Markowitz provided a framework for measuring the risk-reduction benefits of diversification. Using the standard deviation of returns as the measure of investment risk, he investigated how combining risky securities into a portfolio affected the portfolio's risk and expected return. One important conclusion of his model is that unless the returns of the risky assets are perfectly positively correlated, risk is reduced by diversifying across assets.

In the 1960s, professors Treynor, Sharpe, Mossin, and Lintner independently extended this work into what has become known as modern portfolio theory (MPT). MPT results in equilibrium expected returns for securities and portfolios that are a linear function of each security's or portfolio's market risk (the risk that cannot be reduced by diversification).

One measure of the benefits of diversification is the **diversification ratio**. It is calculated as the ratio of the risk of an equally weighted portfolio of *n* securities (measured by its standard deviation of returns) to the risk of a single security selected at random from the *n* securities. If the average standard deviation of returns for the *n* stocks is 25%, and the standard deviation of returns for an equally weighted portfolio of the *n* stocks is 18%, the diversification ratio is 18 / 25 = 0.72. If the standard deviation of returns for an equally weighted portfolio is 25%, there are no diversification benefits and the diversification ratio equals one. A *lower* diversification ratio indicates a *greater* risk-reduction benefit from diversification.

While the diversification ratio provides a quick measure of the potential benefits of diversification, an equal-weighted portfolio is not necessarily the portfolio that provides the greatest reduction in risk. Computer optimization can calculate the portfolio weights that will produce the lowest portfolio risk (standard deviation of returns) for a given group of securities.

Portfolio diversification works best when financial markets are operating normally; diversification provides less reduction of risk during market turmoil, such as the credit contagion of 2008. During periods of financial crisis, correlations tend to increase, which reduces the benefits of diversification.

LOS 85.b: Describe the steps in the portfolio management process.

There are three major steps in the portfolio management process:

Step 1: The **planning step** begins with an analysis of the investor's risk tolerance, return objectives, time horizon, tax exposure, liquidity needs, income needs, and any unique circumstances or investor preferences.

This analysis results in an **investment policy statement (IPS)** that details the investor's investment objectives and constraints. It should also specify an objective benchmark (such as an index return) against which the success of the portfolio management process will be measured. The IPS should be updated at least every few years and any time the investor's objectives or constraints change significantly.

Step 2: The **execution step** involves an analysis of the risk and return characteristics of various asset classes to determine how funds will be allocated to the various asset types. Often, in what is referred to as a *top-down* analysis, a portfolio manager will examine current economic conditions and forecasts of such macroeconomic variables as GDP growth, inflation, and interest rates, in order to identify the asset classes that are most attractive. The resulting portfolio is typically diversified across such asset classes as cash, fixed-income securities, publicly traded equities, hedge funds, private equity, and real estate, as well as commodities and other real assets.

Once the asset class allocations are determined, portfolio managers may attempt to identify the most attractive securities within the asset class. Security analysts use model valuations for securities to identify those that appear undervalued in what is termed *bottom-up* security analysis.

Step 3: The **feedback step** is the final step. Over time, investor circumstances will change, risk and return characteristics of asset classes will change, and the actual weights of the assets in the portfolio will change with asset prices. The portfolio manager must monitor these changes and **rebalance** the portfolio periodically in response, adjusting the allocations to the various asset classes back to their desired percentages. The manager must also measure portfolio performance and evaluate it relative to the return on the benchmark portfolio identified in the IPS.

LOS 85.c: Describe types of investors and distinctive characteristics and needs of each.

Individual investors save and invest for a variety of reasons, including purchasing a house or educating their children. In many countries, special accounts allow citizens to invest for retirement and to defer any taxes on investment income and gains until the funds are withdrawn. Defined contribution pension plans are popular vehicles for these investments. Pension plans are described later in this reading.

Many types of **institutions** have large investment portfolios. An **endowment** is a fund that is dedicated to providing financial support on an ongoing basis for a specific purpose. For example, in the United States, many universities have large endowment funds to support their programs. A **foundation** is a fund established for charitable purposes to support specific types of activities or to fund research related to a particular disease. A typical foundation's investment objective is to fund the activity or research on a continuing basis without decreasing the real (inflation adjusted) value of the portfolio assets. Foundations and endowments typically have long investment horizons, high risk tolerance, and, aside from their planned spending needs, little need for additional liquidity.

The investment objective of a **bank**, simply put, is to earn more on the bank's loans and investments than the bank pays for deposits of various types. Banks seek to keep risk low and need adequate liquidity to meet investor withdrawals as they occur.

Insurance companies invest customer premiums with the objective of funding customer claims as they occur. Life insurance companies have a relatively long-term investment horizon, while property and casualty (P&C) insurers have a shorter investment horizon because claims are expected to arise sooner than for life insurers.

Investment companies manage the pooled funds of many investors. **Mutual funds** manage these pooled funds in particular styles (e.g., index investing, growth investing, bond investing) and restrict their investments to particular subcategories of investments (e.g., large-firm stocks, energy stocks, speculative bonds) or particular regions (emerging market stocks, international bonds, Asian-firm stocks).

Sovereign wealth funds refer to pools of assets owned by a government. For example, the Abu Dhabi Investment Authority, a sovereign wealth fund in the United Arab Emirates funded by Abu Dhabi government surpluses, has approximately USD 700 billion in assets.[1]

Characteristics of Different Types of Investors provides a summary of the risk tolerance, investment horizon, liquidity needs, and income objectives for different types of investors.

Characteristics of Different Types of Investors

Investor	Risk Tolerance	Investment Horizon	Liquidity Needs	Income Needs
Individuals	Depends on individual	Depends on individual	Depends on individual	Depends on individual
Banks	Low	Short	High	Pay interest
Endowments	High	Long	Low	Spending level
Insurance	Low	Long—life Short—P&C	High	Low
Mutual funds	Depends on fund	Depends on fund	High	Depends on fund
Defined benefit pensions	High	Long	Low	Depends on age

LOS 85.d: Describe defined contribution and defined benefit pension plans.

A **defined contribution pension plan** is a retirement plan in which the firm contributes a sum each period to the employee's retirement account. The firm's contribution can be based on any number of factors, including years of service, the employee's age, compensation, profitability, or even a percentage of the employee's contribution. In any event, the firm makes no promise to the employee regarding the future value of the plan assets. The investment decisions are left to the employee, who assumes all of the investment risk.

In a **defined benefit pension plan**, the firm promises to make periodic payments to employees after retirement. The benefit is usually based on the employee's years of service and the employee's compensation at, or near, retirement. For example, an employee might earn a retirement benefit of 2% of her final salary for each year of service. Consequently, an employee with 20 years of service and a final salary of $100,000, would receive $40,000 ($100,000 final salary × 2% × 20 years of service) each year upon retirement until death. Because the employee's future benefit is defined, the employer assumes the investment risk. The employer makes contributions to a fund established to provide the promised future benefits. Poor investment performance will increase the amount of required employer contributions to the fund.

[1]Source: SWF Institute (https://www.swfinstitute.org/).

LOS 85.e: Describe aspects of the asset management industry.

The asset management industry comprises firms that manage investments for clients. Asset management firms include both independent managers and divisions of larger financial services companies. They are referred to as **buy-side firms**, in contrast with **sell-side firms** such as broker-dealers and investment banks.

Full-service asset managers are those that offer a variety of investment styles and asset classes. **Specialist asset managers** may focus on a particular investment style or a particular asset class. A **multi-boutique firm** is a holding company that includes a number of different specialist asset managers.

A key distinction is between firms that use active management and those that use passive management. **Active management** attempts to outperform a chosen benchmark through manager skill, for example by using fundamental or technical analysis. **Passive management** attempts to replicate the performance of a chosen benchmark index. This may include traditional broad market index tracking or a **smart beta** approach that focuses on exposure to a particular market risk factor.

Passive management represents about one-fifth of assets under management. Its share of industry revenue is even smaller because fees for passive management are lower than fees for active management.

Asset management firms may also be classified as traditional or alternative, based on the asset classes they manage. Traditional asset managers focus on equities and fixed-income securities. Alternative asset managers focus on asset classes such as private equity, hedge funds, real estate, or commodities. Profit margins tend to be higher for the alternative asset classes. As a result, many traditional asset managers have been moving into this area, somewhat blurring the distinction between these types of firms.

Some trends in the asset management industry are worth noting:

- The market share for passive management has been growing over time. This is due in part to the lower fees passive managers charge investors, and in part to questions about whether active managers are actually able to add value over time on a risk-adjusted basis, especially in developed markets that are believed to be relatively efficient.
- The amount of data available to asset managers has grown exponentially in recent years. This has encouraged them to invest in information technology and third-party services to process these data, attempting to capitalize on information quickly to make investment decisions.
- **Robo-advisors** are a technology that can offer investors advice and recommendations based on their investment requirements and constraints, using a computer algorithm. These advisors increasingly appeal to younger investors and those with smaller portfolios than have typically been served by asset management firms. They have also lowered the barriers to entry into the asset management industry for firms such as insurance companies.

LOS 85.f: Describe mutual funds and compare them with other pooled investment products.

Mutual funds are one form of **pooled investments** (i.e., a single portfolio that contains investment funds from multiple investors). Each investor owns shares representing ownership of a portion of the overall portfolio. The total net value of the assets in the fund (pool) divided by the number of such shares issued is referred to as the **net asset value (NAV)** of each share.

With an **open-end fund**, investors can buy newly issued shares at the NAV. Newly invested cash is invested by the mutual fund managers in additional portfolio securities. Investors can **redeem** their shares (sell them back to the fund) at NAV as well. All mutual funds charge a fee for the ongoing management of the portfolio assets, which is expressed as a percentage of the net asset value of the fund. **No-load funds** do not charge additional fees for purchasing shares (up-front fees) or for redeeming shares (redemption fees). **Load funds** charge either up-front fees, redemption fees, or both.

Closed-end funds are professionally managed pools of investor money that do not take new investments into the fund or redeem investor shares. The shares of a closed-end fund trade like equity shares (on exchanges or over-the-counter). As with open-end funds, the portfolio management firm charges ongoing management fees.

Types of Mutual Funds

Money market funds invest in short-term debt securities and provide interest income with very low risk of changes in share value. Fund NAVs are typically set to one currency unit, but there have been instances over recent years in which the NAV of some funds declined when the securities they held dropped dramatically in value. Funds are differentiated by the types of money market securities they purchase and their average maturities.

Bond mutual funds invest in fixed-income securities. They are differentiated by bond maturities, credit ratings, issuers, and types. Examples include government bond funds, tax-exempt bond funds, high-yield (lower rated corporate) bond funds, and global bond funds.

A great variety of **stock mutual funds** are available to investors. **Index funds** are **passively managed**; that is, the portfolio is constructed to match the performance of a particular index, such as the Standard & Poor's 500 Index. **Actively managed** funds refer to funds where the management selects individual securities with the goal of producing returns greater than those of their benchmark indexes. Annual management fees are higher for actively managed funds, and actively managed funds have higher turnover of portfolio securities (the percentage of investments that are changed during the year). This leads to greater tax liabilities compared to passively managed index funds.

Other Forms of Pooled Investments

Exchange-traded funds (ETFs) are similar to closed-end funds in that purchases and sales are made in the market rather than with the fund itself. There are important differences, however. While closed-end funds are often actively managed, ETFs are most often invested to match a particular index (passively managed). With closed-end funds, the market price of shares can differ significantly from their NAV due to imbalances between investor supply and demand for shares at any point in time. Special redemption provisions for ETFs are designed to keep their market prices very close to their NAVs.

ETFs can be sold short, purchased on margin, and traded at intraday prices, whereas open-end funds are typically sold and redeemed only daily, based on the share NAV calculated with closing asset prices. Investors in ETFs must pay brokerage commissions when they trade, and there is a spread between the bid price at which market makers will buy shares and the ask price at which market makers will sell shares. With most ETFs, investors receive any dividend income on portfolio stocks in cash, while open-end funds offer the alternative of reinvesting dividends in additional fund shares. One final difference is that ETFs may produce less capital gains liability compared to open-end index funds. This is because investor sales of ETF shares do not require the fund to sell any securities. If an open-end fund has significant redemptions that cause it to sell appreciated portfolio shares, shareholders incur a capital gains tax liability.

A **separately managed account** is a portfolio that is owned by a single investor and managed according to that investor's needs and preferences. No shares are issued, as the single investor owns the entire account.

Hedge funds are pools of investor funds that are not regulated to the extent that mutual funds are. Hedge funds are limited in the number of investors who can invest in the fund and are often sold only to qualified investors who have a minimum amount of overall portfolio wealth. Minimum investments can be quite high, often between $250,000 and $1 million.

Private equity and **venture capital** funds invest in portfolios of companies, often with the intention to sell them later in public offerings. Managers of funds may take active roles in managing the companies in which they invest.

Professor's Note

Hedge funds, private equity, and venture capital are addressed in the Alternative Investments topic area.

Reading 85: Key Concepts

LOS 85.a

A diversified portfolio produces reduced risk for a given level of expected return, compared to investing in an individual security. Modern portfolio theory concludes that investors who do not take a portfolio perspective bear risk that is not rewarded with greater expected return.

LOS 85.b

The three steps in the portfolio management process are:

1. **Planning:** Determine client needs and circumstances, including the client's return objectives, risk tolerance, constraints, and preferences. Create, and then periodically review and update, an investment policy statement (IPS) that spells out these needs and circumstances.
2. **Execution:** Construct the client portfolio by determining suitable allocations to various asset classes based on the IPS and on expectations about macroeconomic variables such as inflation, interest rates, and GDP growth (top-down analysis). Identify attractively priced securities within an asset class for client portfolios based on valuation estimates from security analysts (bottom-up analysis).
3. **Feedback:** Monitor and rebalance the portfolio to adjust asset class allocations and securities holdings in response to market performance. Measure and report performance relative to the performance benchmark specified in the IPS.

LOS 85.c

Types of investment management clients and their characteristics:

Investor Type	Risk Tolerance	Investment Horizon	Liquidity Needs	Income Needs
Individuals	Depends on individual	Depends on individual	Depends on individual	Depends on individual
Banks	Low	Short	High	Pay interest
Endowments	High	Long	Low	Spending level
Insurance	Low	Long—life Short—P&C	High	Low
Mutual funds	Depends on fund	Depends on fund	High	Depends on fund

Defined benefit pension	High	Long	Low	Depends on age

LOS 85.d

In a defined contribution plan, the employer contributes a certain sum each period to the employee's retirement account. The employer makes no promise regarding the future value of the plan assets; thus, the employee assumes all of the investment risk.

In a defined benefit plan, the employer promises to make periodic payments to the employee after retirement. Because the employee's future benefit is defined, the employer assumes the investment risk.

LOS 85.e

The asset management industry comprises buy-side firms that manage investments for clients. Asset management firms include both independent managers and divisions of larger financial services companies and may be full-service or specialist firms offering investments in traditional or alternative asset classes.

Active management attempts to outperform a chosen benchmark through manager skill. Passive management attempts to replicate the performance of a chosen benchmark index. Most assets under management are actively managed, but the market share for passive management has been increasing.

LOS 85.f

Mutual funds combine funds from many investors into a single portfolio that is invested in a specified class of securities or to match a specific index. Many varieties exist, including money market funds, bond funds, stock funds, and balanced (hybrid) funds. Open-ended shares can be bought or sold at the net asset value. Closed-ended funds have a fixed number of shares that trade at a price determined by the market.

Exchange-traded funds are similar to mutual funds, but investors can buy and sell ETF shares in the same way as shares of stock. Management fees are generally low, though trading ETFs results in brokerage costs.

Separately managed accounts are portfolios managed for individual investors who have substantial assets. In return for an annual fee based on assets, the investor receives personalized investment advice.

Hedge funds are available only to accredited investors and are exempt from most reporting requirements. Many different hedge fund strategies exist. A typical annual fee structure is 20% of excess performance plus 2% of assets under management.

Buyout funds involve taking a company private by buying all available shares, usually funded by issuing debt. The company is then restructured to increase cash flow. Investors typically exit the investment within three to five years.

Venture capital funds are similar to buyout funds, except that the companies purchased are in the start-up phase. Venture capital funds, like buyout funds, also provide advice and expertise to the start-ups.

Reading 86: Basics of Portfolio Planning and Construction

LOS 86.a: Describe the reasons for a written investment policy statement (IPS).

An investment manager is very unlikely to produce a good result for a client without understanding that client's needs, circumstances, and constraints.

A written **investment policy statement** will typically begin with the investor's goals in terms of risk and return. These should be determined jointly, as the goals of high returns and low risk (while quite popular) are likely to be mutually exclusive in practice. Investor expectations in terms of returns must be compatible with investor's tolerance for risk (uncertainty about portfolio performance).

LOS 86.b: Describe the major components of an IPS.

The major components of an IPS typically address the following:

- *Description of Client* circumstances, situation, and investment objectives.
- *Statement of the Purpose* of the IPS.
- *Statement of Duties and Responsibilities* of investment manager, custodian of assets, and the client.
- *Procedures* to update IPS and to respond to various possible situations.
- *Investment Objectives* derived from communications with the client.
- *Investment Constraints* that must be considered in the plan.
- *Investment Guidelines* such as how the policy will be executed, asset types permitted, and leverage to be used.
- *Evaluation of Performance*, the benchmark portfolio for evaluating investment performance, and other information on evaluation of investment results.
- *Appendices* containing information on strategic (baseline) asset allocation and permitted deviations from policy portfolio allocations, as well as how and when the portfolio allocations should be rebalanced.

In any case, the IPS will, at a minimum, contain a clear statement of client circumstances and constraints, an investment strategy based on these, and some benchmark against which to evaluate the account performance.

LOS 86.c: Describe risk and return objectives and how they may be developed for a client.

The **risk objectives** in an IPS may take several forms. An **absolute risk objective** might be to "have no decrease in portfolio value during any 12-month period" or to "not decrease in value by more than 2% at any point over any 12-month period." Low absolute percentage risk objectives such as these may result in portfolios made up of securities that offer guaranteed returns (e.g., U.S. Treasury bills).

Absolute risk objectives can also be stated in terms of the probability of specific portfolio results, either percentage losses or dollar losses, rather than strict limits on portfolio results. Examples are as follows:

- "No greater than a 5% probability of returns below –5% in any 12-month period."
- "No greater than a 4% probability of a loss of more than $20,000 over any 12-month period."

An absolute return objective may be stated in nominal terms, such as "an overall return of at least 6% per annum," or in real returns, such as "a return of 3% more than the annual inflation rate each year."

Relative risk objectives relate to a specific benchmark and can also be strict, such as, "Returns will not be less than 12-month euro LIBOR over any 12-month period," or stated in terms of probability, such as, "No greater than a 5% probability of returns more than 4% below the return on the MSCI World Index over any 12-month period."

Return objectives can be relative to a benchmark portfolio return, such as, "Exceed the return on the S&P 500 Index by 2% per annum." For a bank, the return objective may be relative to the bank's cost of funds (deposit rate). While it is possible for an institution to use returns on peer portfolios, such as an endowment with a stated objective to be in the top quartile of endowment fund returns, peer performance benchmarks suffer from not being *investable* portfolios. There is no way to match this investment return by portfolio construction before the fact.

In any event, the account manager must make sure that the stated risk and return objectives are compatible, given the reality of expected investment results and uncertainty over time.

LOS 86.d: Explain the difference between the willingness and the ability (capacity) to take risk in analyzing an investor's financial risk tolerance.

An investor's **ability to bear risk** depends on financial circumstances. Longer investment horizons (20 years rather than 2 years), greater assets versus liabilities (more wealth), more insurance against unexpected occurrences, and a secure job all suggest a greater ability to bear investment risk in terms of uncertainty about periodic investment performance.

An investor's **willingness to bear risk** is based primarily on the investor's attitudes and beliefs about investments (various asset types). The assessment of an investor's attitude about risk is quite subjective and is sometimes done with a short questionnaire that attempts to categorize the investor's risk aversion or risk tolerance.

When the adviser's assessments of an investor's ability and willingness to take investment risk are compatible, there is no real problem selecting an appropriate level of investment risk. If the investor's willingness to take on investment risk is high but the investor's ability to take on risk is low, the low ability to take on investment risk will prevail in the adviser's assessment.

In situations where ability is high but willingness is low, the adviser may attempt to educate the investor about investment risk and correct any misconceptions that may be contributing to the investor's low stated willingness to take on investment risk. However, the adviser's job is not to change the investor's personality characteristics that contribute to a low willingness to take on investment risk. The approach will most likely be to conform to the lower of the investor's ability or willingness to bear risk, as constructing a portfolio with a level of risk that the client is clearly uncomfortable with will not likely lead to a good outcome in the investor's view.

LOS 86.e: Describe the investment constraints of liquidity, time horizon, tax concerns, legal and regulatory factors, and unique circumstances and their implications for the choice of portfolio assets.

Professor's Note

When some of us were studying for the CFA exams over 20 years ago, we memorized R-R-T-T-L-L-U as a checklist for addressing the important points of portfolio construction, and it still works today. Then, as now, the important points to cover in an IPS were Risk, Return, Time horizon, Tax situation, Liquidity, Legal restrictions, and the Unique constraints of a specific investor.

Investment constraints include the investor's liquidity needs, time horizon, tax considerations, legal and regulatory constraints, and unique needs and preferences.

Liquidity: Liquidity refers to the ability to turn investment assets into spendable cash in a short period of time without having to make significant price concessions to do so. Investor needs for money to pay tuition, to pay for a parent's assisted living expenses, or to fund other possible spending needs may all require that some liquid assets be held. As we noted in an earlier reading discussing property and casualty insurance companies, claims arrive unpredictably to some extent and therefore their portfolios must hold a significant proportion of liquid (or maturing) securities in order to be prepared to honor these claims. Illiquid investments in hedge funds and private equity funds, which typically are not traded and have restrictions on redemptions, are not suitable for an investor who may unexpectedly need access to the funds.

Time horizon: In general, the longer an investor's time horizon, the more risk and less liquidity the investor can accept in the portfolio. While the expected returns on a broad equities portfolio may not be too risky for an investor with a 20-year investment horizon, they likely are too risky for an investor who must fund a large purchase at the end of this year. For such an investor, government securities or a bank certificate of deposit may be the most appropriate investments because of their low risk and high liquidity at the time when the funds will be needed.

Tax situation: Besides an individual's overall tax rate, the tax treatment of various types of investment accounts is also a consideration in portfolio construction. For a fully taxable account, investors subject to higher tax rates may prefer tax-free bonds (U.S.) to taxable bonds or prefer equities that are expected to produce capital gains, which are often taxed at a lower rate than other types of income. A focus on expected after-tax returns over time in relation to risk should correctly account for differences in tax treatments as well as investors' overall tax rates.

Some types of investment accounts, such as retirement accounts, may be tax exempt or tax deferred. Investors with such accounts may choose to put securities that generate fully taxed income, such as corporate bond interest, in accounts that are tax deferred, while seeking long-term capital gains, tax-exempt interest income, and dividend income (in jurisdictions where dividends receive preferential tax treatment) in their personal accounts, which have no tax deferral benefit.

Legal and regulatory: In addition to financial market regulations that apply to all investors, more specific legal and regulatory constraints may apply to particular investors. Trust, corporate, and qualified investment accounts may all be restricted by law from investing in particular types of securities and assets. There may also be restrictions on percentage allocations to specific types of investments in such accounts. Corporate officers and directors face legal restrictions on trading in the securities of their firms that the account manager should be aware of.

Unique circumstances: Each investor, whether individual or institutional, may have specific preferences or restrictions on which securities and assets may be purchased for the account. Some of these may be nonfinancial considerations, which are commonly categorized as **responsible investing**. Ethical preferences, such as prohibiting investment in securities issued by tobacco or firearms producers, are not uncommon. Restrictions on investments in companies or countries where human rights abuses are suspected or documented would also fall into this category. Religious preferences may preclude investment in securities that make explicit interest payments.

Unique investor preferences may also be based on diversification needs when the investor's income depends heavily on the prospects for one company or industry. An investor who has founded or runs a company may not want any investment in securities issued by a competitor to that company.

LOS 86.f: Explain the specification of asset classes in relation to asset allocation.

After having determined the investor objectives and constraints through the exercise of creating an IPS, a **strategic asset allocation** is developed which specifies the percentage allocations to the included asset classes. In choosing which asset classes to consider when developing the strategic asset allocation for the account, the correlations of returns *within* an asset class should be relatively high, indicating that the assets within the class are similar in their investment performance. On the other hand, it is low correlations of returns *between* asset classes that leads to risk reduction through portfolio diversification.

Historically, only the broad categories of equities, bonds, cash, and real estate were considered. More recently, a group of several investable asset classes, referred to collectively as alternative investments, has gained more prominence. Alternative investment asset classes include hedge funds of various types, private equity funds, managed or passively constructed commodity funds, artwork, and intellectual property rights.

We can further divide equities by whether the issuing companies are domestic or foreign, large or small, or whether they are traded in emerging or developed markets. An example of specifying asset classes is world equities. A U.S. investor may want to divide world equities into different regions.

With bonds, we can divide the overall universe of bonds into asset classes based on maturities or on criteria such as whether they are foreign or domestic, government or corporate, or investment grade or speculative (high yield). Overall, the asset classes considered should approximate the universe of permissible investments specified in the IPS.

Once the universe of asset classes has been specified, the investment manager will collect data on the returns, standard deviation of returns, and correlations of returns with those of other asset classes for each asset class.

Strategic Asset Allocation illustrates the strategic asset allocation for a pension fund.

Strategic Asset Allocation

The Vermont Pension Investment Committee manages more than $4 billion in retirement assets for various teachers and state and municipal employees in that state. VPIC's investment policy specifies the following strategic asset allocation:

Asset Class	Target
Growth assets	
Passive global equities	24%
Active global equities	5%
Large cap U.S. equities	4%
Small-/mid-cap U.S. equities	3%
Non–U.S. developed market equities	5%
International small-cap equities	2%

Private equity	10%
Core plus fixed income	6%
Emerging market debt	4%
Private debt	5%
Non-core real estate	3%
Total growth assets	**71%**
Downturn hedging assets	
Core fixed income	14%
Short-term quality credit	5%
Total downturn hedging	**19%**
Inflation hedging assets	
Core real estate	5%
U.S. TIPS	3%
Infrastructure/farmland	2%
Total inflation hedging	**10%**

Source: State of Vermont, Office of the State Treasurer. Target allocation as of March 25, 2019. www.vermonttreasurer.gov/content/pension.

LOS 86.g: Describe the principles of portfolio construction and the role of asset allocation in relation to the IPS.

Once the portfolio manager has identified the investable asset classes for the portfolio and the risk, return, and correlation characteristics of each asset class, an *efficient frontier*, analogous to one constructed from individual securities, can be constructed using a computer program. By combining the return and risk objectives from the IPS with the actual risk and return properties of the many portfolios along the efficient frontier, the manager can identify that portfolio which best meets the risk and return requirements of the investor. The asset allocation for the efficient portfolio selected is then the strategic asset allocation for the portfolio.

So far, we have not concerned ourselves with deviations from strategic asset allocations or with selection of individual securities within individual asset classes. These activities are referred to as active (versus passive) portfolio management strategies. A manager who varies from strategic asset allocation weights in order to take advantage of perceived short-term opportunities is adding **tactical asset allocation** to the portfolio strategy. **Security selection** refers to deviations from index weights on individual securities within an asset class. For example, a portfolio manager might overweight energy stocks and underweight financial stocks, relative to the index weights for U.S. large-cap equities as an asset class. For some asset classes, such as hedge funds, individual real estate properties, and artwork, investable indexes are not available. For these asset classes, selection of individual assets is required by the nature of the asset class.

While each of these active strategies may produce higher returns, they each also increase the risk of the portfolio compared to a passive portfolio of asset class indexes. A practice known as **risk budgeting** sets an overall risk limit for the portfolio and budgets (allocates) a portion of the permitted risk to the systematic risk of the strategic asset allocation, the risk from tactical asset allocation, and the risk from security selection.

Active portfolio management has two specific issues to consider.

1. An investor may have multiple managers actively managing to the same benchmark for the same asset class (or may have significant benchmark overlap). In this case, one manager may overweight an index stock while another may underweight the same stock. Taken together, there is no net active management risk, although each manager has reported active management risk. Overall, the risk budget is underutilized as there is less net active management than gross active management.
2. When all managers are actively managing portfolios relative to an index, trading may be excessive overall. This extra trading could have negative tax consequences, specifically potentially higher capital gains taxes, compared to an overall efficient tax strategy.

One way to address these issues is to use a **core-satellite approach**. The core-satellite approach invests the majority, or core, portion of the portfolio in passively managed indexes and invests a smaller, or satellite, portion in active strategies. This approach reduces the likelihood of excessive trading and offsetting active positions.

Clearly, the success of security selection will depend on the manager's skill and the opportunities (mispricings or inefficiencies) within a particular asset class. Similarly, the success of tactical asset allocation will depend both on the existence of short-term opportunities in specific asset classes and on the manager's ability to identify them.

LOS 86.h: Describe how environmental, social, and governance (ESG) considerations may be integrated into portfolio planning and construction.

Approaches to ESG investing include:

- **Negative screening**, excluding specific companies or industries based on ESG factors.
- **Positive screening**, investing in companies that have positive ESG practices.
- **Thematic investing**, selecting sectors or companies to promote specific ESG-related goals.
- **Impact investing**, selecting investments both to provide a return and to promote positive ESG practices.
- **Engagement/active ownership**, using share ownership as a platform to promote improved ESG practices at a company.
- **ESG integration**, considering ESG factors throughout the asset allocation and security selection process.

If a portfolio's investment universe is constrained by negative screening, measuring its performance against a broad market index is unlikely to be appropriate. Indexes excluding companies or industries that investors with ESG concerns commonly avoid are available.

While many investors use positive screening, thematic, or impact investing approaches, choices of which specific companies to invest in and which ESG factors to focus on differ among investors. Thus, portfolios and performance benchmarks must be customized under these approaches, and they may require investment managers who specialize in these style of investing.

For investment managers with clients who wish to engage in active ownership, it is important to clarify whether the clients intend to vote their shares themselves or direct the managers to vote the shares according to specified ESG factors.

Regardless of the approach chosen, investors should be aware that imposing constraints based on ESG factors will likely affect portfolio performance. How these constraints will affect portfolio performance in practice is uncertain. Both limiting the universe of investment choices and incurring the costs involved in considering ESG factors may decrease returns. On the other hand, investing in companies with good corporate governance practices and avoiding those that face ESG-related risks may increase portfolio returns.

Reading 86: Key Concepts

LOS 86.a

A written investment policy statement(IPS), the first step in the portfolio management process, is a plan for achieving investment success. An IPS forces investment discipline and ensures that goals are realistic by requiring investors to articulate their circumstances, objectives, and constraints.

LOS 86.b

Many IPS include the following sections:

- Introduction: Describes the client.
- Statement of Purpose: The intentions of the IPS.
- Statement of Duties and Responsibilities: Of the client, the asset custodian, and the investment managers.
- Procedures: Related to keeping the IPS updated and responding to unforeseen events.
- Investment Objectives: The client's investment needs, specified in terms of required return and risk tolerance.
- Investment Constraints: Factors that may hinder the ability to meet investment objectives; typically categorized as time horizon, taxes, liquidity, legal and regulatory, and unique needs.
- Investment Guidelines: For example, whether leverage, derivatives, or specific kinds of assets are allowed.
- Evaluation and Review: Related to feedback on investment results.
- Appendices: May specify the portfolio's strategic asset allocation (policy portfolio) or the portfolio's rebalancing policy.

LOS 86.c

Risk objectives are specifications for portfolio risk that are developed to embody a client's risk tolerance. Risk objectives can be either absolute (e.g., no losses greater than 10% in any year) or relative (e.g., annual return will be within 2% of FTSE return).

Return objectives are typically based on an investor's desire to meet a future financial goal, such as a particular level of income in retirement. Return objectives can be absolute (e.g., 9% annual return) or relative (e.g., outperform the S&P 500 by 2% per year).

The achievability of an investor's return expectations may be hindered by the investor's risk objectives.

LOS 86.d

Willingness to take financial risk is related to an investor's psychological factors, such as personality type and level of financial knowledge.

Ability or capacity to take risk depends on financial factors, such as wealth relative to liabilities, income stability, and time horizon.

A client's overall risk tolerance depends on both his ability to take risk and his willingness to take risk. A willingness greater than ability, or vice versa, is typically resolved by choosing the more conservative of the two and counseling the client.

LOS 86.e

Investment constraints include:

- Liquidity—The need to draw cash from the portfolio for anticipated or unexpected future spending needs. High liquidity needs often translate to a high portfolio allocation to bonds or cash.
- Time horizon—Often the period over which assets are accumulated and before withdrawals begin. Risky or illiquid investments may be inappropriate for an investor with a short time horizon.
- Tax considerations—Concerns the tax treatments of the investor's various accounts, the relative tax treatment of capital gains and income, and the investor's marginal tax bracket.
- Legal and regulatory—Constraints such as government restrictions on portfolio contents or laws against insider trading.
- Unique circumstances—Restrictions due to investor preferences (religious, ethical, etc.) or other factors not already considered.

LOS 86.f

An asset class is a group of securities with similar risk and performance characteristics. Examples of major asset classes include equity, fixed income, cash, and real estate. Portfolio managers also use more narrowly defined asset classes, such as large-cap U.S. equities or speculative international bonds, and alternative asset classes, such as commodities or investments in hedge funds.

LOS 86.g

Strategic asset allocation is a set of percentage allocations to various asset classes that is designed to meet the investor's objectives. The strategic asset allocation is developed by combining the objectives and constraints in the IPS with the performance expectations of the various asset classes. The strategic asset allocation provides the basic structure of a portfolio.

Tactical asset allocation refers to an allocation that deviates from the baseline (strategic) allocation in order to profit from a forecast of shorter-term opportunities in specific asset classes.

LOS 86.h

Imposing portfolio constraints based on ESG factors may affect performance. Limiting the universe of investment choices may decrease returns, but good corporate governance and low ESG-related risks may increase returns.

If a portfolio's investment universe is constrained by negative screening, its performance should be measured against a benchmark that excludes companies with negative ESG factors.

Positive screening, best-in-class, or thematic investing approaches typically require portfolio construction to be customized for the investor's choices of which ESG factors to focus on.

For active ownership, it is important to clarify whether investors intend to vote their shares themselves or direct managers to vote the shares according to specified ESG factors.

Reading 87: The Behavioral Biases of Individuals

Module 87.1: Cognitive Errors vs. Emotional Biases

Traditional finance assumes that individuals act as perfectly rational economic beings who objectively consider all relevant information to make rational decisions, and that this process results in efficient markets. Research results have called these assumptions into question. Kahneman and Tversky pioneered this work in the early 1970s, initially setting logic tests where individuals' intuitive answers were predictably flawed. This supported the idea that human decision-making has systematic biases that lead to irrational decisions.

This work was extended by Kahneman, Tversky, and others, who have suggested that a better understanding of these biases by clients (and by the professionals who work with those clients) should produce securities prices and returns over time that better match the informationally efficient markets of traditional finance theory.

LOS 87.a: Compare and contrast cognitive errors and emotional biases.

Individuals, when facing complex decision-making, often lack the time or ability to derive the optimal course of action prescribed by traditional finance. Cognitive limitations and emotional responses introduce bias into the decision-making process, leading to decisions that are biased (not perfectly rational). Behavioral finance asserts that certain biases, which are not simply random errors, are widespread and therefore predictable.

Cognitive errors are due primarily to faulty reasoning or irrationality. They can arise from not understanding statistical analysis, information processing errors, illogical reasoning, or memory errors. Such errors can possibly be reduced by increased awareness, better training, or more information.

Emotional biases are not related to conscious thought. Rather, they stem from feelings, impulses, or intuition. As such, they are difficult to overcome and may have to be accommodated.

Despite the distinction in grouping biases as either cognitive or emotional, a bias may have elements of both cognition and emotion. When trying to overcome or mitigate biases that are both emotional and cognitive, success is more likely by focusing on the cognitive issues.

LOS 87.b: Discuss commonly recognized behavioral biases and their implications for financial decision making.

Cognitive errors can be divided into *belief perseverance* biases that reflect an irrational reluctance to change prior conclusions and decisions, and *processing errors* where the information analysis is flawed.

Cognitive Errors: Belief Perseverance

Psychologists use the term **cognitive dissonance** to refer to a situation where an individual holds conflicting beliefs or receives information that causes a current belief to be questioned. Cognitive dissonance causes stress that individuals seek to reduce. They may do so by letting go of prior beliefs in favor of the conflicting belief. On the other hand, they might discount the conflicting information or viewpoints by questioning their truth, source, applicability, or significance. To the extent that it is easier to do the latter than the former, bias in favor of currently held beliefs is the result.

- **Conservatism bias** occurs when market participants rationally form an initial view but then fail to change that view as new information becomes available. That is, they overweight their prior probabilities and do not adjust them appropriately as new information becomes available. Individuals displaying this bias tend to maintain prior forecasts and securities allocations, ignoring or failing to recognize the significance of new information. Individuals may react slowly to new data or ignore information that is complex to process.

  Example: Conservatism bias

  John Molinari allocates assets based on his observation that over the last 80 years, recessions occurred in 20% of those years. When a coworker informs Molinari that the country's central bank has announced a policy change to a tightening of monetary conditions, Molinari does not adjust his recommended asset allocations. Does this reflect conservatism bias?

  Answer:

  Molinari should consider that the conditional probability of a recession, given that the central bank is tightening, may differ from the unconditional probability of a recession that he previously estimated. He is showing conservatism bias by not considering the impact of this new information.

  Conservatism bias may result in market participants holding investments too long because they are unwilling or slow to update a view or forecast. They may be avoiding the mental effort or stress of updating prior beliefs by not considering the implications of new information.

- **Confirmation bias** occurs when market participants focus on or seek information that supports prior beliefs, while avoiding or diminishing the importance of conflicting information or viewpoints. They may distort new information in a way that remains consistent with their prior beliefs.

  For example, after buying a car from a given manufacturer, the buyer would exhibit confirmation bias by reading articles about how great cars from that manufacturer are but avoiding news about problems with that particular brand. Because the buyer already purchased the car, information suggesting that it was a bad decision is unwelcome.

  Consequences and implications of confirmation bias may include market participants who:

  - Consider positive information but ignore negative information.
  - Set up a decision process or data screen incorrectly to support a preferred belief.
  - Become overconfident about the correctness of a presently held belief.

  Market participants can reduce confirmation bias by seeking out contrary views and information—for example, by reading analyses and viewpoints that disagree with a presently held belief, rather than only reading those that reinforce the belief.

- **Representativeness bias** occurs when certain characteristics are used to put an investment in a category and the individual concludes that it will have the characteristics of investments in that category. Individuals systematically make the error of believing that two things that are similar in some respects are more similar in other respects than they actually are.

  Two forms of representativeness bias are base-rate neglect and sample-size neglect.

- Base-rate neglect refers to analyzing an individual member of a population without adequately considering the probability of a characteristic in that population (the base rate). Consider this example of base-rate neglect: a group was asked to identify the most likely occupation of a man who was characterized as somewhat shy as a salesperson or a librarian. Most participants chose librarian, thinking that most librarians would tend to be more shy on average than salespeople, who tend to be outgoing. Their mistake was in not considering that there are relatively few male librarians and a great number of male salespeople. Even though a greater percentage of librarians may be characterized as somewhat shy, the absolute number of salespeople who could be characterized as somewhat shy is significantly greater.

- **Sample-size neglect** refers to making a classification based on a small and potentially unrealistic data sample. The error is believing the population reflects the characteristics of the small sample.

  For example, a fund manager may show strong performance over a three-year time horizon. This may lead investors to assume this is evidence of superior skill. However, examination of longer sample periods shows a lack of persistence; there will be some three-year "winners" even when annual returns results are actually random. The evidence of the lack of persistence of managers' relative performance over time does not support investors' conclusion that this manager will continue to outperform.

Example: Representativeness bias

XYZ company has long been recognized as a growth stock, delivering superior earnings growth and stock price appreciation. While earnings have continued to grow, last year's revenue has not, and neither has the stock price. Under the following two conditions, would an analyst be more likely to buy or sell the stock?

1. The analyst suffers from base-rate and sample-size neglect.
2. The analyst treats the growth classification as representative.

Answer:

If the analyst exhibits sample-size neglect and base-rate neglect biases, the analyst will ignore XYZ's long record as a growth stock, focus on the short-term disappointing results, and may recommend selling the stock without considering the long-term possibility it will revert to growth behavior.

However, if the analyst over-relies on the initial growth classification, the analyst may assume that the stock will return to growth and recommend buying it, without properly considering the reasons for its recent results or their longer-term implications.

Representativeness bias may lead market participants to attach too much importance to a few characteristics based on a small sample size or make decisions based on simple rules and classifications rather than conducting a more-thorough and complex analysis.

- **Illusion of control bias** exists when market participants believe they can control or affect outcomes when they cannot. It is often associated with emotional biases: *illusion of knowledge* (belief you know things you do not know), *self-attribution* (belief you personally caused something to happen), and *overconfidence* (an unwarranted belief that your beliefs will prove to be correct)

  An example of control bias can be seen in humans' attempts to control the weather using ritual ceremonies. This allowed them to believe they had some control of the environment, when in fact it is highly unlikely that a ceremony can influence the weather.

Illusion of control may cause market participants to overweight securities for which they believe they have control over outcomes, such as a company they work for or are otherwise associated with. This can result in their portfolios being inadequately diversified.

- **Hindsight bias** is a selective memory of past events, actions, or what was knowable in the past, resulting in an individual's tendency to see things as more predictable than they really are. People tend to remember their correct predictions and forget their incorrect ones. They also overestimate what could have been known. This behavior results from individuals being able to observe outcomes that did occur but not the outcomes that did not materialize. Hindsight bias is sometimes referred to as the *I-knew-it-all-along phenomenon*.

  Examples of behavior that reflects hindsight bias include the following:

  - Individuals distort their earlier predictions when looking back. This is the tendency to believe that we knew the outcome of an uncertain event all along.
  - Individuals tend to view events that have occurred as inevitable.
  - Individuals assume they could have foreseen the outcomes of uncertain events.

  Hindsight bias can lead to overconfidence in ability to predict outcomes. It may also cause investors to cast aside valid analysis techniques that did not turn out to be correct in favor of poor techniques that turned out well by chance.

Cognitive Errors: Information-Processing Biases

These are related more to the processing of information and less to the decision-making process.

Anchoring and adjustment bias refers to basing expectations on a prior number and overweighting its importance, making adjustments in relation to that number as new information arrives. Examples would be estimating the value of a security relative to its current value or making estimates of earnings per share relative to a previously reported value or relative to a prior estimate. Anchoring leads to underestimating the implications of new information. New data should be considered objectively without regard to any initial anchor point.

Mental accounting bias refers to viewing money in different accounts or from different sources differently when making investment decisions. This conflicts with the idea that security decisions should be made in the context of the investor's overall portfolio of assets based on their financial goals and risk tolerance.

An example of mental accounting is an investor who receives an unexpected bonus at work and chooses to invest it in a very risky biotechnology stock, reasoning that the bonus is "found money" that can acceptably be risked on speculation. In fact, while such a stock may have a place in the investor's portfolio, decisions about whether and how much of it to include should be based on a total portfolio approach. Another example may be an investor who receives an inheritance from a parent and segregates those funds into low-risk bonds out of a desire to "not lose any of the money the parent worked so hard to save." This behavior also conflicts with the principle that investments should be considered in the context of the entire portfolio.

The result of such mental accounting may be that an investor's overall portfolio is not optimal given the investor's circumstances, investment goals, and risk tolerance. It can cause an investor to hold positions that offset each other, rather than considering investments in the context of their correlation of returns. One common form of mental accounting bias is a tendency to view income differently from capital appreciation. This may cause an investor to hold a mix of income-producing and non-income-producing securities that does not match the investor's circumstances.

A study by Camerer et al. (1997)[1] investigated behavior of New York taxi drivers. The study suggests that each new day was a separate account in the minds of the taxi drivers. It appears that New York taxi drivers have a reference rate for daily income and perceive a loss if they fail to meet it (which we may also view as an example of anchoring). On rainy days, demand for taxis is high, while on sunny days, the demand is low. Logically, you would expect taxi drivers to work more hours on rainy days to maximize their incomes. In reality, the opposite was true. Taxi drivers worked longer hours on sunny days as they strove to hit their target income. On rainy days, once the target was achieved, the taxi drivers stopped working.

Framing bias occurs when decisions are affected by the way in which the question or data is "framed." In other words, the way a question is phrased can influence the answer given. Tversky and Kahneman (1980)[2] illustrate framing bias with the following example.

Example: Framing bias (framing as a gain)

The United States is preparing for the outbreak of an unusual disease, which is expected to kill 600 people. Two alternative programs have been proposed. If Program A is adopted, 200 people will be saved. If Program B is adopted, there is a one-third probability that 600 people will be saved and a two-thirds probability that no one will be saved. Which program will people choose?

Answer:

Program A is typically selected. Although the expected value of both Program A and Program B is 200 lives saved, the majority choice is risk averse. The prospect of saving 200 lives with certainty is more attractive than the risky option with the same expected value.

Example: Framing bias (framing as a loss)

A different group of individuals is given the same issue, but the two programs are framed differently. If Program A is adopted, 400 people will die. If Program B is adopted, there is a one-third probability that nobody will die and a two-thirds probability that 600 will die.

Which program will people choose?

Answer:

In this situation, Program B is typically selected. The majority choice is now risk-taking, with the certain death of 400 people being less acceptable than a two-thirds chance that 600 people will die.

In these examples, the two programs presented (A and B) are identical for both groups, but the framing of the information results in different choices. For the first group, the information was presented in the context of a gain, while for the second group, it was in the context of a loss. Loss aversion—the idea that we fear losses more than we value gains—is then demonstrated by the choices made.

An example of framing bias in an investment context is overestimating the significance of short-term price volatility (risk) without weighing it against long-term considerations. This may result in overly conservative portfolios.

Investment managers must take care to avoid framing bias when creating questions to assess an investor's risk tolerance. Failing to properly assess risk tolerance may identify investors as more or less risk averse than they actually are, resulting in portfolios that are inconsistent with the investors' needs.

Availability bias refers to putting undue emphasis on information that is readily available, easy to recall, or based narrowly on personal experience or knowledge. Availability bias occurs when individuals judge the probability of an event occurring by the ease with which examples and instances come to mind. By the very nature of memory, more-recent events are typically easier to recall than events further in the past, which leads to the bias of attaching too much significance to events that have occurred recently and too little to events that occurred further in the past. People also tend to assume that if something is easily remembered, it must occur with a higher probability.

Example: Availability bias

Imagine a word is picked at random from a dictionary. Is it more likely that the word has the letter *r* as the first letter in the word or the third letter?

Answer:

When faced with this problem, most individuals state that it is more likely the letter *r* will be the first letter. In fact, in the English language, there are approximately three times more words with *r* as the third letter than the first. Individuals find it easier to recall words that begin with *r* than words with *r* as the third letter, which distorts their estimation of probabilities.

Availability bias may lead market participants to choose a manager or investment based on advertising or recalling they have heard the name. They may limit their universe of potential investments to familiar firms, resulting in inappropriate asset allocations and lack of diversification. They may also overreact to recent market conditions while ignoring data on historical market performance, or they may place too much emphasis on events that receive a large amount of media attention.

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[1] Colin Camerer, Linda Babcock, George Loewenstein, and Richard Thaler, "Labor Supply of New York City Cab Drivers: One Day at a Time," *Quarterly Journal of Economics* 112 (1997): 407–442.

[2] Amos Tversky and Daniel Kahneman, "The Framing of Decisions and the Rationality of Choice," Stanford University Department of Psychology, Defense Technical Information Center – Technical Report (1980)

Module 87.2: Emotional Biases

Professor's Note

Some of the terms we are about to discuss have already come up in the discussion of cognitive biases. In general, if an investor's view is based on unconscious emotion that the holder is unwilling or unable to change, we should regard it as an emotional bias. If a bias can be overcome with a relatively simple change in thought process or information, we should regard it as a cognitive bias.

While there is no formally accepted definition, these six biases generally arise from emotion and feelings rather than through conscious thought:

1. **Loss-aversion bias** arises from feeling more pain from a loss than pleasure from an equal gain. Kahneman and Tversky (1979)[1] investigated differences between how people feel when they gain and when they lose and how that affects behavior when faced with risk. They found that individuals' willingness to take a gamble (risk) was very different when facing a loss or a gain.

   Consider the following two scenarios:

   Scenario 1: An individual is given $10. The individual is then given the following options:

   - Take an additional $5 with certainty.
   - Flip a coin and win an additional $10 if it lands heads up or nothing if it lands tails up.

   Both options represent a gain relative to the original $10, and the expected value of the gain is $5 for either option. Option 1 creates a guaranteed outcome of $15. Option 2 introduces uncertainty, with equal probabilities of an outcome of $10 or $20. Most individuals chose the riskless Option 1 over the riskier Option 2.

   Scenario 2: An individual is given $20. The individual is then given the following options:

   - Take a $5 loss with certainty.
   - Flip a coin and lose nothing if it lands heads up but lose $10 if it lands tails up.

   Both options represent a potential loss relative to the original $20, and the expected loss is $5 for either option. Most individuals chose risky Option 2 over the riskless certain loss of Option 1.

   In both scenarios, the expected value of the individual's wealth is $15. The options given to the individuals are also identical, with Option 1 resulting in a guaranteed outcome of $15 and Option 2 providing equal probabilities of either $10 or $20. Yet, when faced with gains, people preferred certainty, and when faced with losses, they preferred risk.

The conclusion is that individuals display asymmetrical responses to gains and losses. Kahneman and Tversky suggested that people look at decisions relative to a reference rate. Anything below the reference rate is seen as a loss, and anything above the reference rate is seen as, a gain. The reference rate in Scenario 1 was the $10 initially given to the individual, and in Scenario 2, it was $20. The conflicting responses to the scenarios are explained by attitudes toward gains and losses. Crucially, people fear losses far more than they value gains. Thus, in Scenario 2, they were willing to take the risk in the hope of avoiding a loss.

Professor's Note

Be sure to understand the difference between risk aversion and loss aversion. A risk-averse investor is simply an investor who, given two investments with the same expected returns, would select the investment with the lowest risk. A loss-averse investor is one who feels greater pain (decreases in utility) from losses than satisfaction (increase in utility) from gains. As a result, the individual is more likely to take a risk in the hope of avoiding losses than in the hope of achieving gains.

Consequences of loss-aversion bias may include trading too much by selling for small gains, which increases transaction costs and decreases returns, or incurring too much risk by continuing to hold assets that have deteriorated in quality and lost value. If an initial decline in value occurs, loss-averse investors may take excessive risk in the hope of recovering (investment managers may be particularly susceptible to this behavior). A loss-averse investor might view a position inappropriately as a gain or a loss based on the framing of the reference point.

2. **Overconfidence bias** occurs when market participants overestimate their own intuitive ability or reasoning. It can show up as *illusion of knowledge* when they think they do a better job of predicting than they actually do. Combined with *self-attribution bias*, individuals may give themselves personal credit when things go right (*self-enhancing*) but blame others or circumstances when things go wrong (*self-protecting*). *Prediction overconfidence* leads individuals to underestimate uncertainty and the standard deviation of their predictions, while *certainty overconfidence* occurs when they overstate the probability they will be right.

   While overconfidence is both cognitive and emotional, it is more emotional in nature because it is difficult for most individuals to correct and is rooted in the desire to feel good. Overconfidence bias may cause market participants to underestimate risk, overestimate return, and fail to diversify sufficiently.

3. **Self-control bias** occurs when individuals lack self-discipline and favor short-term satisfaction over long-term goals. Often, individuals are not prepared to make short-term sacrifices to meet their long-term goals. They may favor small payoffs now at the expense of larger payoffs in the future, which is known as *hyperbolic discounting*.

   Self-control bias may result in insufficient savings to fund retirement needs, which in turn may cause an investor to take excessive risk to try to compensate for insufficient savings accumulation. It may also result in overemphasis on income-producing assets to meet short-term needs.

   Self-control bias might be mitigated by establishing an appropriate investment plan (asset allocation) and a budget to achieve sufficient savings. Both should be reviewed on a regular basis.

4. **Status quo bias** occurs when comfort with an existing situation causes an individual to be resistant to change. If investment choices include the option to maintain existing investments or allocations, or if a choice will happen unless the participant opts out, status quo choices become more likely.

Companies have found that automatically enrolling workers in retirement savings schemes, with the option to opt out, increases participation compared with making it necessary for employees to opt in. Thaler and Sunstein (2008)[2] argue for framing choices in this way to achieve better participation rates in retirement plans, as well as other choices, such as whether to register as an organ donor.

Consequences of status quo bias may include holding portfolios with inappropriate risk and not considering other, better investment alternatives.

5. **Endowment bias** occurs when an asset is felt to be special and more valuable simply because it is already owned. For example, a spouse may hold on to securities the deceased spouse purchased, for reasons like sentiment that are unrelated to the current merits of the securities. Endowment bias is common with inherited assets and might be detected or mitigated by asking a question such as "Would you make this same investment with new money today?"

   In studies, individuals have been asked to state their minimum sales price for an asset they own (say $25) and the maximum price they would pay now to buy the same asset (say $23). The price at which they would be willing to sell tended to be higher than the price they would pay. This has been explained as an endowment effect. Once they own an asset, they act as if it is worth more than they would pay.

   Market participants who exhibit endowment bias may be failing to sell assets that are no longer appropriate for their investment needs, or they hold assets with which they are familiar because they provide some intangible sense of comfort.

6. **Regret-aversion bias** occurs when market participants do nothing out of excessive fear that actions could be wrong. They attach undue weight to errors of commission (doing something that turns out wrong) and not enough weight to errors of omission (not doing something that would have turned out right). Their sense of regret and pain is stronger for acts of commission. This is quite similar to status quo bias.

   Herding behavior is a form of regret aversion where participants go with the consensus or popular opinion. Essentially, participants tell themselves they are not to blame if others are wrong too.

   Consequences of regret-aversion bias may include excess conservatism in the portfolio because it is easy to see that riskier assets do at times underperform. Therefore, an investor might not buy riskier assets so as not to experience regret when they decline.

LOS 87.c: Describe how behavioral biases of investors can lead to market characteristics that may not be explained by traditional finance.

In our Equity Investments reading on Market Efficiency, we developed the ideas of anomalies (results that do not fit the prevailing model of securities risks and returns) and market inefficiencies (anomalies that present opportunities to earn positive risk-adjusted returns). Many market anomalies have been explained by small sample size, time period bias, or inadequacies in the specification of prevailing models of returns. In the same way, some anomalies once considered evidence of market inefficiency have been explained by the possible misspecification of risk, leading to inaccurate risk adjustment of returns.

Bubbles and subsequent crashes have been with us as long as trading markets have existed. Some have offered explanations for these extremes of valuation based on rational behavior. Investors who leave markets that they determine are exhibiting characteristics of a bubble may miss out on high returns over extended periods. On the other hand, investors who stay invested based on a belief that they will be able to get out at or near the market top often find that that is not realistic. Either type of mistake, leaving the market too early or staying fully invested too long, can be quite damaging to a fund manager's career. Exploiting the fact that a market is "in a bubble" for an arbitrage profit may not be possible due to the restrictions on and risk of short selling in a rapidly rising market.

While behavioral finance does not provide an overall explanation for bubbles and their aftermath, some have claimed that cognitive and emotional biases exhibited during such periods may have some causal effects. These claims include the following:

- *Overconfidence* may lead to overtrading, underestimation of risk, and lack of diversification.
- Persistently good results combined with *self-attribution* bias can fuel overconfidence, as can *hindsight* bias (as investors give themselves credit for choosing profitable stocks in a bull market).
- *Confirmation bias* may lead investors to ignore or misinterpret new information suggesting that valuations will not continue to rise, or to misinterpret initial decreases in asset values as simply another buying opportunity.
- *Anchoring* may cause investors to believe recent highs are rational prices even after prices begin their eventual decline.
- Fear of *regret* may keep even very skeptical investors in the market.

One anomaly that has been persistent over time is the value/growth anomaly. Value stocks (low market-book, low P/E, high dividend yield stocks) have outperformed growth stocks (high P/E, low dividend yield stocks) over long periods. Fama and French (1992) found that adding risk factors for firm size and book-to-market ratio to a model of stock returns eliminated the evidence of value stock outperformance. These results suggest that the extra returns to value stocks were compensation for additional risks captured by firm size and book-to-market ratios.

Others have suggested behavioral factors as the cause of the seeming outperformance of value stocks. The **halo effect** is a version of representativeness in which a company's good characteristics, such as fast growth and a rising stock price, are extended into a conclusion that it is a good stock to own, leading to overvaluation of growth stocks.

The fact that investors tend to invest heavily in firms in their domestic country in a global portfolio, or more heavily in firms operating in their region of a country, is considered anomalous in that rationality suggests greater diversification. Such **home bias**, it is claimed, may result from a belief that they have better access to information or simply an emotional desire to invest in companies "closer to home." Similarly, investors may underestimate the risk or overestimate the future returns of firms whose products they use or firms for which they are exposed to a great amount of positive marketing messages.

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[1]Daniel Kahneman and Amos Tversky, "Prospect Theory: An Analysis of Decision Under Risk, *Econometrica* 47, no.2 (1979): 263–91.

[2]Richard H. Thaler and Cass R. Sunstein, *Nudge: Improving Decisions About Health, Wealth, and Happiness* (New York: Penguin Group, 2008).

Reading 87: Key Concepts

LOS 87.a

Cognitive errors result from the inability to analyze information or from basing decisions on partial information. Individuals try to process information and make rational decisions, but they may lack the capacity or sufficient information to do so. Cognitive errors can be divided into belief perseverance errors and processing errors.

Emotional biases are caused by the way individuals frame information and decisions, rather than the process used to analyze and interpret information. Emotional bias is more of a spontaneous reaction.

LOS 87.b

Cognitive Errors: Belief Perseverance

- Conservatism bias.
- Confirmation bias.
- Representativeness bias.
- Control bias.
- Hindsight bias.

Cognitive Errors: Information Processing

- Anchoring and adjustment.
- Mental accounting bias.
- Framing bias.
- Availability bias.

Emotional Biases

- Loss-aversion bias.
- Overconfidence bias.
- Self-control bias.
- Status quo bias.
- Endowment bias.
- Regret-aversion bias.

LOS 87.c

Many reported anomalies have been explained by inadequate specification of security risk, but some have attempted to explain the existence of persistent anomalies as the results of cognitive errors and emotional biases.

Behavioral finance has not explained bubbles and crashes, but some cognitive errors and emotional biases may be exhibited during bubbles and crashes.

Reading 88: Introduction to Risk Management

LOS 88.a: Define risk management.

The **risk management** process seeks to (1) identify the risk tolerance of the organization, (2) identify and measure the risks that the organization faces, and (3) modify and monitor these risks.

The process does not seek to minimize or eliminate all of these risks. The organization may increase its exposure to risks it decides to take because it is better able to manage and respond to them. The organization may decrease its exposure to risks that it is less well able to manage and respond to by making organizational changes, purchasing insurance, or entering into hedging transactions. Through these choices the firm aligns the risks it takes with its risk tolerances for these various types of risk.

Risk (uncertainty) is not something to be avoided by an organization or in an investment portfolio. Returns above the risk-free rate are earned by taking on risk. While returns for any period are not under the control of managers, the specific risks and overall level of risk the organization takes are under their control. We can think of risk management as determining organizational risks, determining the optimal bundle of risks for the organization, and implementing risk mitigation strategies to achieve that bundle of risks.

We describe the principles of risk management here in a framework that can be applied broadly, not only to firms or organizations in general, but also to the management of investment portfolios and financial firms, and even to individuals deciding how much risk and which specific risks they will take. Individuals follow a similar approach, selecting a bundle of risks that is optimal for maximizing their expected utility (rather than returns or profit).

LOS 88.b: Describe features of a risk management framework.

An overall **risk management framework** encompasses several activities, including:

- Establishing processes and policies for risk governance.
- Determining the organization's risk tolerance.
- Identifying and measuring existing risks.
- Managing and mitigating risks to achieve the optimal bundle of risks.
- Monitoring risk exposures over time.
- Communicating across the organization.
- Performing strategic risk analysis.

This framework is general, but all of these elements should be addressed in any comprehensive risk management framework. Only by understanding the risks the organization faces, and having the processes and procedures in place to effectively manage and monitor these risks, can an organization align its risk exposures to the goals of the organization.

LOS 88.c: Define risk governance and describe elements of effective risk governance.

Risk governance refers to senior management's determination of the risk tolerance of the organization, the elements of its optimal risk exposure strategy, and the framework for oversight of the risk management function. Risk governance seeks to manage risk in a way that supports the overall goals of the organization so it can achieve the best business outcome consistent with the organization's overall risk tolerance. Risk governance provides organization-wide guidance on the risks that should be pursued in an efficient manner, risks that should be subject to limits, and risks that should be reduced or avoided.

A risk management committee can provide a way for various parts of the organization to bring up issues of risk measurement, integration of risks, and the best ways to mitigate undesirable risks.

LOS 88.d: Explain how risk tolerance affects risk management.

Determining an organization's **risk tolerance** involves setting the overall risk exposure the organization will take by identifying the risks the firm can effectively take and the risks that the organization should reduce or avoid. Some of the factors that determine an organization's risk tolerance are its expertise in its lines of business, its skill at responding to negative outside events, its regulatory environment, and its financial strength and ability to withstand losses.

When analyzing risk tolerance, management should examine risks that may exist within the organization as well as those that may arise from outside. The various risks the firm is exposed to must each be considered and weighted against the expected benefits of bearing those risks and how these fit the overall goals of the organization.

LOS 88.e: Describe risk budgeting and its role in risk governance.

Risk budgeting is the process of allocating firm resources to assets (or investments) by considering their various risk characteristics and how they combine to meet the organization's risk tolerance. The goal is to allocate the overall amount of acceptable risk to the mix of assets or investments that have the greatest expected returns over time.

The risk budget may be a single metric, such as portfolio beta, value at risk, portfolio duration, or returns variance. A risk budget may be constructed based on categories of investments, such as domestic equities, domestic debt securities, international equities, and international debt securities. Another way to allocate a risk budget is to identify specific risk factors that comprise the overall risk of the portfolio or organization. In this case, specific risk factors that affect asset classes to varying degrees, such as interest rate risk, equity market risk, and foreign exchange rate risk, are estimated and aggregated to determine whether they match the overall risk tolerance of the organization.

LOS 88.f: Identify financial and non-financial sources of risk and describe how they may interact.

Financial risks are those that arise from exposure to financial markets. Examples are:

- **Credit risk.** This is the uncertainty about whether the counterparty to a transaction will fulfill its contractual obligations.

- **Liquidity risk.** This is the risk of loss when selling an asset at a time when market conditions make the sales price less than the underlying fair value of the asset.
- **Market risk.** This is the uncertainty about market prices of assets (stocks, commodities, and currencies) and interest rates.

Non-financial risks arise from the operations of the organization and from sources external to the organization. Examples are:

- **Operational risk.** This is the risk that human error, faulty organizational processes, inadequate security, or business interruptions will result in losses. An example of an operational risk is **cyber risk**, which refers to disruptions of an organization's information technology.
- **Solvency risk.** This is the risk that the organization will be unable to continue to operate because it has run out of cash.
- **Regulatory risk.** This is the risk that the regulatory environment will change, imposing costs on the firm or restricting its activities.
- **Governmental or political risk** (including **tax risk).** This is the risk that political actions outside a specific regulatory framework, such as increases in tax rates, will impose significant costs on an organization.
- **Legal risk.** This is the uncertainty about the organization's exposure to future legal action.
- **Model risk.** This is the risk that asset valuations based on the organization's analytical models are incorrect.
- **Tail risk.** This is the risk that extreme events (those in the tails of the distribution of outcomes) are more likely than the organization's analysis indicates, especially from incorrectly concluding that the distribution of outcomes is normal.
- **Accounting risk.** This is the risk that the organization's accounting policies and estimates are judged to be incorrect.

For individuals, risks, such as risk of death (**mortality risk**) prior to providing for their families' future needs and the risk of living longer than anticipated (**longevity risk**) so that assets run out, are very important in financial planning. Mortality risk is most often addressed with life insurance, and longevity risk can be reduced by purchasing a lifetime annuity. Risk of health care expenses is addressed with health insurance. Although the risks for an individual are in some ways different from those of organizations, the overall approach is the same, choosing which risks to bear (self-insure), which risks to prevent or avoid, and which risks to take in order to maximize the expected outcome in terms of personal utility or satisfaction.

The various risks an organization faces are not independent; they interact in many ways. Consider a firm with market risk that it reduces with option contracts. If markets decline significantly, the firm is owed a payment from the firm on the other side of the option trade, so now there is significant counterparty or credit risk. There also may be legal risks if the counterparty seeks to avoid the payment through loopholes in the contract. Credit losses and legal losses may result in greater liquidity risk as positions must be sold. Additional losses from selling in a declining or less liquid market may increase solvency risk because of the negative impact on the firm's cash position.

Interactions among risks must be considered because such interactions are many and frequent. They can be especially important during periods of stress in financial markets, when risk management is most important to the health and possibly the survival of the organization.

LOS 88.g: Describe methods for measuring and modifying risk exposures and factors to consider in choosing among the methods.

Measures of risk for specific asset types include standard deviation, beta, and duration.

- **Standard deviation** is a measure of the volatility of asset prices and interest rates. Standard deviation may not be the appropriate measure of risk for non-normal probability distributions, especially those with negative skew or positive excess kurtosis (fat tails).
- **Beta** measures the market risk of equity securities and portfolios of equity securities. This measure considers the risk reduction benefits of diversification and is appropriate for securities held in a well-diversified portfolio, whereas standard deviation is a measure of risk on a stand-alone basis.

- **Duration** is a measure of the price sensitivity of debt securities to changes in interest rates.

Derivatives risks (sometimes referred to as "the Greeks") include:

- **Delta.** This is the sensitivity of derivatives values to the price of the underlying asset.
- **Gamma.** This is the sensitivity of delta to changes in the price of the underlying asset.
- **Vega.** This is the sensitivity of derivatives values to the volatility of the price of the underlying asset.
- **Rho.** This is the sensitivity of derivatives values to changes in the risk-free rate.

Tail risk is the uncertainty about the probability of extreme (negative) outcomes. Commonly used measures of tail risk (sometimes referred to as **downside risk**) include Value at Risk and Conditional VaR.

Value at risk (VaR) is the minimum loss over a period that will occur with a specific probability. Consider a bank that has a one-month VaR of $1 million with a probability of 5%. That means that a one-month loss of at least $1 million is expected to occur 5% of the time. Note that this is not the maximum one-month loss the bank will experience; it is the minimum loss that will occur 5% of the time. VaR does not provide a maximum loss for a period. VaR has become accepted as a risk measure for banks and is used in establishing minimum capital requirements.

There are various methods of calculating VaR, and both the inputs and models used will affect the calculated value, perhaps significantly. As is always the case with estimates of risk, incorrect inputs or inappropriate distribution assumptions will lead to misleading results. Given these limitations, VaR should be used in conjunction with other risk measures.

Conditional VaR (CVaR) is the expected value of a loss, *given* that the loss exceeds a minimum amount. Relating this to the VaR measure presented previously, the CVaR would be the expected loss, given that the loss was at least $1 million. It is calculated as the probability-weighted average loss for all losses expected to be at least $1 million. CVaR is similar to the measure of loss given default that is used in estimating risk for debt securities.

Subjective and Market-Based Estimates of Risk

Two methods of risk assessment that are used to supplement measures such as VaR and CVaR are stress testing and scenario analysis. **Stress testing** examines the effects of a specific (usually extreme) change in a key variable such as an interest rate or exchange rate. **Scenario analysis** refers to a similar what-if analysis of expected loss but incorporates changes in multiple inputs. A given scenario might combine an interest rate change with a significant change in oil prices or exchange rates.

Quantifying the risk to an organization of very infrequent events is quite difficult. The risk of the bankruptcy of a firm that has never experienced significant financial distress is often a subjective estimate rather than a data-driven estimate. Estimates of risk can also be based on the market prices of insurance, derivatives, or other securities that can be used to hedge those risks. These hedging costs provide information on market participants' aggregate estimate of the expected loss of specific risks.

Operational risks are difficult to quantify for a single organization because they are very difficult to predict and may result in very large costs to the organization. One way to approach this problem is to examine a large sample of firms in order to determine an overall probability of significant losses due to operational risks and the average loss of firms that have experienced such losses.

Unexpected changes in tax laws or the regulatory environment can impose large costs on an organization. The political nature of such changes makes them quite difficult to predict. Subjective estimates, rather than data-driven quantitative estimates, are necessary. As is often the case, even a subjective, non-quantitative estimate of risk probabilities and magnitudes is better than not addressing the risk factor at all.

Modifying Risk Exposures

Risk management does not seek to eliminate all risks. The goal is to retain the optimal mix of risks for the organization. This may mean taking on more of some risks, decreasing others, and eliminating some altogether. Once the risk management team has estimated various risks, management may decide to prevent or avoid a risk, accept a risk, transfer a risk, or shift a risk.

One way to avoid a risk is to not engage in the activity with the uncertain outcome. If political risks in a country are to be avoided, simply not investing in securities of firms based in that country or not expanding a business enterprise to that country would avoid those risks. A decision to avoid certain risks typically would come from top management as a part of establishing the risk tolerance of the organization and would be instituted because the risks are judged to outweigh the potential benefits of specific activities.

Some risks can be prevented. The risk of a data breach can be prevented with a greater level of security for the data and stronger processes. In this case, the benefits of reducing or eliminating the risk are judged to be greater than the cost of doing so.

For risks that management has decided to bear, the organization will seek to bear them efficiently. **Diversification** may offer a way to more efficiently bear a specific risk.

Sometimes the term **self-insurance** is used to describe a situation where an organization has decided to bear a risk. Note, however, that this simply means that it will bear any associated losses from this risk factor. It is possible that this represents inaction rather than the result of analysis and strategic decision-making. In some cases, the firm will establish a reserve account to cover losses as a way of mitigating the impact of losses on the organization.

For a risk an organization has decided not to bear, risk transfer or risk shifting can be employed. With a **risk transfer**, another party takes on the risk. Insurance is a type of risk transfer. The risk of fire destroying a warehouse complex is shifted to an insurance company by buying an insurance policy and paying the policy premiums. Insurance companies diversify across many risks so the premiums of some insured parties pay the losses of others. Ideally, the various risks the insurance company insures are not highly correlated, as that can reduce or eliminate any diversification benefits. An insurance company with highly correlated risks (or a single very large risk) may itself shift some of the resulting risk by buying reinsurance from another company.

With a **surety bond**, an insurance company has agreed to make a payment if a third party fails to perform under the terms of a contract or agreement with the organization. For example, a company may be exposed to losses if a key supplier does not deliver on time, slowing a project and resulting in penalty payments by the company. Insurers also issue **fidelity bonds**, which will pay for losses that result from employee theft or misconduct. Managements that purchase insurance, surety bonds, or fidelity bonds have determined that the benefits of risk reduction are greater than the cost of the insurance.

Risk shifting is a way to change the distribution of possible outcomes and is accomplished primarily with derivative contracts. For example, financial firms that do not want to bear currency risk on some foreign currency denominated debt securities can use forward currency contracts, futures contracts, or swaps to reduce or eliminate that risk. A firm with a large position in a specific stock can buy put options that provide a minimum sale price for the securities, altering the distribution of possible outcomes (in this case providing a floor value for the securities). On the other hand, a firm could sell call options on a specific stock, altering the distribution of possible outcomes by giving up some of the upside potential of the stock but decreasing its downside risk by the amount of the premiums received from the sale of the call options.

Choosing Among Risk Modification Methods

Organizations may use multiple methods of risk modification to reduce a single risk. The criterion is always a comparison of the costs and benefits of risk modification. Some risks may be mitigated by diversification, some shifted by insurance where it is available and economical, some shifted though the use of derivatives, and some simply borne or self-insured. The end result is a risk profile that matches the risk tolerance established for the organization and includes the risks that top management has determined match the goals of the organization in terms of cost versus potential returns.

Reading 88: Key Concepts

LOS 88.a

Risk management is the process of identifying and measuring the risks an organization (or portfolio manager or individual) faces, determining an acceptable level of overall risk (establishing risk tolerance), deciding which risks should be taken and which risks should be reduced or avoided, and putting the structure in place to maintain the bundle of risks that is expected to best achieve the goals of the organization.

LOS 88.b

An overall risk management framework should address the following activities:

- Identifying and measuring existing risks.
- Determining the organization's overall risk tolerance.
- Establishing the processes and policies for risk governance.
- Managing and mitigating risks to achieve the optimal bundle of risks.
- Monitoring risk exposures over time.
- Communicating across the organization.
- Performing strategic risk analysis.

LOS 88.c

Risk governance refers to senior management's determination of the risk tolerance of the organization, the elements of its optimal risk exposure strategy, and the framework for oversight of the risk management function.

LOS 88.d

The risk tolerance for an organization is the overall amount of risk it will take in pursuing its goals and is determined by top management.

LOS 88.e

Risk budgeting is the process of allocating the total risk the firm will take (risk tolerance) to assets or investments by considering the risk characteristics of each and how they can be combined to best meet the organization's goals. The budget can be a single risk measure or the sum of various risk factors.

LOS 88.f

Financial risks are those that arise from exposure to financial markets, including credit risk, liquidity risk, and market risk. Non-financial risks are the risks from the operation of the organization and from sources external to the organization. Individuals face mortality and longevity risk, in addition to financial risks.

Interactions among risks are frequent and can be especially significant during periods of stress in financial markets.

LOS 88.g

Risk of assets is measured by standard deviation, beta, or duration. Derivatives risk measures include delta, gamma, vega, and rho. Tail risk is measured with value at risk (VaR) or conditional VaR. Some risks must be measured subjectively.

An organization may decide to bear a risk (self-insurance), avoid or take steps to prevent a risk, efficiently manage a risk through diversification, transfer a risk with insurance or a surety bond, or shift a risk (change the distribution of uncertain outcomes) with derivatives.

Organizations may use multiple methods of risk modification after considering the costs and benefits of the various methods. The end result is a risk profile that matches the organization's risk tolerance and includes the risks that top management has determined match the organization's goals.

Reading 89: Ethics and Trust in the Investment Profession

Module 89.1: Ethics and Trust

LOS 89.a: Explain ethics.

Ethics can be described as a set of shared beliefs about what is good or acceptable behavior and what is bad or unacceptable behavior. Ethical conduct has been described as behavior that follows moral principles and is consistent with society's ethical expectations.

Ethical conduct has also been described as conduct that improves outcomes for **stakeholders**, who are people directly or indirectly affected by the conduct. Examples of stakeholders in the case of investment professionals include their clients, coworkers, employers, and the investment profession as a whole. Some decisions may bring positive results for you, but negative consequences for a stakeholder, such as a coworker. Ethical conduct is behavior that balances your self-interest with the impact on others.

LOS 89.b: Describe the role of a code of ethics in defining a profession.

A **code of ethics** is a written set of moral principles that can guide behavior by describing what is considered acceptable behavior. Having a code of ethics is a way to communicate the values, principles, and expectations of an organization or other group of people and provides a general guide to what constitutes acceptable behavior. Some codes of ethics include a set of rules or standards that require some minimum level of ethical behavior.

A **profession** refers to a group of people with specialized skills and knowledge who serve others and agree to behave in accordance with a code of ethics. A professional code of ethics is a way for a profession to communicate to the public that its members will use their knowledge and skills to serve their clients in an honest and ethical manner.

LOS 89.c: Describe professions and how they establish trust.

A **profession** is an occupational group (e.g., doctors or lawyers) that has requirements of specialized expert knowledge, and often a focus on ethical behavior and service to the larger community or society. Additionally, a profession may have the following characteristics:

- A code and standards for professional behavior.
- A regulatory body to enforce rules concerning professional behavior and monitor the ethical behavior of members.
- A focus on the needs of their clients (e.g., students, patients).
- A focus on service to society.
- A requirement to put client interests first.
- A focus on or requirement for continuing education.

Ways that professions establish trust include:

- Requiring high standards of expertise, knowledge, and skill.
- Establishing standards of ethical behavior.
- Monitoring professional conduct.
- Encouraging continuing education to maintain and increase competence.
- Being focused on clients' needs.
- Mentoring and inspiring others in the profession.

LOS 89.d: Describe the need for high ethical standards in investment management.

Investment professionals have a special responsibility because they are entrusted with their clients' wealth. The responsibility to use their specialized knowledge and skills to both protect and grow client assets makes high ethical standards all the more important. Investment advice and management are intangible products, making quality and value received more difficult to evaluate than for tangible products such as a laptop computer or a restaurant meal. For this reason, trust in investment professionals takes on an even greater importance than in many other businesses.

Failure to act in a highly ethical manner can damage not only client wealth but also impede the success of investment firms and investment professionals because potential investors will be less likely to use their services.

Unethical behavior by financial services professionals can have negative effects for society as a whole. The financial services industry serves as an intermediary between savers and those seeking financing for their business activities. A lack of trust in financial advisors will reduce the funds entrusted to them and increase the cost of raising capital for business investment and growth. When investors cannot rely on the information they receive from financial services professionals, this adds another layer of risk on top of the investment risks that investors face. Even the perception of additional risk will reduce the amounts invested and increase the returns required to attract investor capital.

In addition to reducing the amount of investment overall, unethical behavior—such as providing incomplete, misleading, or false information to investors—can affect the allocation of the capital that is raised. Misallocation of capital to businesses other than those with the most potential for growth and societal benefit reduces the growth of an economy and the well-being of its people. When the allocation of investment capital is constrained or inefficient, the negative consequences extend to all the participants in an economy.

LOS 89.e: Explain professionalism in investment management.

Because clients of investment professionals rely on their expertise, judgment, and ethical principles, many of the characteristics of a profession we have described apply.

Ethical principles are of great importance because clients often do not have significant knowledge about financial securities, fee structures, or sources of potential bias in investment recommendations. Currently, some financial professionals are held to a **suitability standard**, while others are held to a **fiduciary standard**. *Suitability* refers to the match between client return requirements and risk tolerances and the characteristics of the securities recommended. A fiduciary standard is stronger, requiring professionals to use their knowledge and expertise to act in the best interests of the client.

LOS 89.f: Identify challenges to ethical behavior.

One challenge to ethical behavior is that individuals tend to overrate the ethical quality of their behavior on a relative basis and overemphasize the importance of their own personal traits in determining the ethical quality of their behavior.

It is claimed that external or **situational influences** are a more important determinant of the ethical quality of behavior than internal (personal) traits that influence behavior. One situational influence is social pressure from others. Loyalty to an employer, supervisor, organization, or coworkers can cause individuals to act in unethical ways as they place more importance on their self-interest and short-term results than on longer-term results and the ethical quality of their decisions and behavior. The prospect of acquiring more money or greater prestige can cause individuals to engage in unethical behavior.

Firms with strict rules-based compliance procedures run the risk of fostering a culture that is so focused on adhering to compliance rules that individuals only ask themselves what they *can* do. The question of what behavior they *should* engage in, based on ethical principles and longer-term results, is often not addressed in such situations.

LOS 89.g: Compare and contrast ethical standards with legal standards.

Not all unethical actions are illegal, and not all illegal actions are unethical. In some places it may be illegal to report one's employer's actions against the best interests of clients by sharing what is considered private company information with authorities, but doing so may be considered ethical "whistle-blowing" behavior by some. Acts of civil disobedience that are illegal are also considered by many to be ethical behavior. On the other hand, recommending investment in a relative's firm without disclosure may not be illegal, but would be considered unethical by many.

Ethical principles often set a higher standard of behavior than laws and regulations. New laws and regulations often result from recent instances of what is perceived to be unethical behavior. Just as the Securities Act of 1933, the Glass-Steagall Act, and the Securities Exchange Act of 1934 followed the perceived bad behavior by investment professionals and bankers leading to the 1929 market crash, the Sarbanes-Oxley laws followed the accounting scandals at Enron and Worldcom, and the Dodd-Frank Act followed the 2008 financial crisis. New laws and regulations can create opportunities for different unethical behavior. In general, ethical decisions require more judgment and consideration of the impact of behavior on many stakeholders compared to legal decisions.

LOS 89.h: Describe a framework for ethical decision making.

Ethical decisions will be improved when ethics are integrated into a firm's decision-making process. This will allow decision makers and teams to consider alternative actions as well as shorter- and longer-term consequences from various perspectives, improving the ethical aspects of their decisions. To do this it is first necessary that the firm adopt a code of ethics to guide the process.

Such integration provides an opportunity to teach, practice, and reinforce ethical decision-making. This is an important part of developing an ethical culture. The support of senior management for integrating ethics into the decision-making process is also very important in developing a culture and processes that will result in ethical decision-making.

Using a framework for ethical decision making helps individuals identify the important issues involved, examine these issues from multiple perspectives, develop the necessary judgment and decision-making skills required, and avoid unanticipated ethical consequences.

The following ethical decision-making framework is presented in the Level I CFA curriculum:[1]

- Identify: Relevant facts, stakeholders and duties owed, ethical principles, conflicts of interest.
- Consider: Situational influences, additional guidance, alternative actions.
- Decide and act.
- Reflect: Was the outcome as anticipated? Why or why not?

In the first step, decision makers need to identify the facts they have to work with, and the facts they would like to have, before making a decision. Stakeholders—those affected by the decision—must be identified. These stakeholders may include the employer, clients, coworkers, self, family, and others in the industry, and the duties to each stakeholder should be identified. This part of the process will also help in explicitly identifying potential conflicts of interest among the various stakeholders. At this point the decision makers should be able to identify the ethical principles involved in the decision, although greater clarity about those may also be gained throughout the process.

In the second step, the framework suggests situational factors that may influence decision makers should be identified and considered along with any personal biases that may come into play. At this point, decision makers may seek outside guidance which can come from a mentor, colleagues, or friends who have shown good judgment in the past. Guidance may also be sought from the firm's legal and compliance departments. This guidance from alternative sources will help to provide a variety of perspectives from which the decision under consideration can be viewed, as well as help in developing alternatives that should be considered. Finally, the alternative actions that have been identified are all considered, taking into account both the short-term and long-term effects of each alternative action and any potential but unanticipated ethical implications.

In the final step, decision makers should evaluate the outcomes of the actions that were taken. In particular, they should consider whether the decisions had their intended results and whether appropriate consideration was given to ethical principles, situational influences, and duties to clients and other stakeholders.

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[1]"Ethics and Trust in the Investment Profession," Bidhan L Parmar, PhD, Dorothy C. Kelly, CFA, and David B. Stevens, CFA, in CFA Program 2024 Level I Curriculum, Volume 6 (CFA Institute, 2023).

Reading 89: Key Concepts

LOS 89.a

Ethical behavior is that which conforms to a set of rules and moral principles based on shared beliefs about what behavior is acceptable and what behavior is unacceptable.

LOS 89.b

A professional code of ethics is a way for a profession to communicate to the public that its members will use their knowledge and skills to serve their clients in an honest and ethical manner, and can increase public confidence and trust that members will act ethically.

LOS 89.c

A profession is an occupational group that has requirements of specialized expert knowledge. Professions establish trust by requiring high standards of expertise, setting standards for ethical behavior, and monitoring professional conduct.

LOS 89.d

Investment professionals have a special responsibility to use their specialized knowledge and skills to both protect and grow client assets. The fact that investment management is an intangible product makes high ethical standards all the more important in the financial services profession.

LOS 89.e

Some financial professionals are held to a suitability standard, while others are held to a fiduciary standard. Suitability refers to the match between client return requirements and risk tolerances and the characteristics of the securities recommended. A fiduciary standard requires professionals to act in the best interests of the client.

LOS 89.f

Challenges to ethical behavior include overestimating one's own ethical character, considering only near-term consequences and not longer-term consequences of behavior, and letting situational (external) influences, such as peer pressure, unduly affect one's decisions and behavior.

LOS 89.g

Not all unethical actions are illegal, and not all illegal actions are unethical. Laws are more specific than ethical principles and often address prior unethical behavior. Ethical behavior requires more judgment; acts such as civil disobedience may be considered ethical even when they are illegal.

LOS 89.h

A framework for ethical decision-making is designed to lead to better decisions by identifying the stakeholders affected and the conflicts of interest among them, considering alternative actions and the relevant situational influences on decision makers, seeking out different perspectives, and evaluating decisions to see if they had unintended consequences.

LOS 90.a: Describe the structure of the CFA Institute Professional Conduct Program and the process for the enforcement of the Code and Standards.

The CFA Institute Professional Conduct Program is covered by the CFA Institute Bylaws and the Rules of Procedure for Proceedings Related to Professional Conduct. The Program is based on the principles of fairness of the process to members and candidates and maintaining the confidentiality of the proceedings. The CFA Institute Board of Governors has overall responsibility for the Professional Conduct Program, and its Disciplinary Review Committee is responsible for enforcing the Code and Standards.

The CFA Institute Professional Conduct staff conducts inquiries related to professional conduct. Several circumstances can prompt such an inquiry:

1. Self-disclosure by members or candidates on their annual Professional Conduct Statements of involvement in civil litigation or a criminal investigation, or that the member or candidate is the subject of a written complaint.
2. Written complaints about a member or candidate's professional conduct that are received by the Professional Conduct staff.
3. Evidence of misconduct by a member or candidate that the Professional Conduct staff received through public sources, such as a media article or broadcast.
4. A report by a CFA exam proctor of a possible violation during the examination.
5. Analysis of exam materials and monitoring of social media by CFA Institute.

Once an inquiry has begun, the Professional Conduct staff may request (in writing) an explanation from the subject member or candidate and may: (1) interview the subject member or candidate, (2) interview the complainant or other third parties, and/or (3) collect documents and records relevant to the investigation.

The Professional Conduct staff may decide: (1) that no disciplinary sanctions are appropriate, (2) to issue a cautionary letter, or (3) to discipline the member or candidate. In a case where the Professional Conduct staff finds a violation has occurred and proposes a disciplinary sanction, the member or candidate may accept or reject the sanction. If the member or candidate chooses to reject the sanction, the matter will be referred to a disciplinary review panel of CFA Institute members for a hearing. Sanctions imposed may include condemnation by the member's peers or suspension of candidate's continued participation in the CFA Program.

LOS 90.b: Identify the six components of the Code of Ethics and the seven Standards of Professional Conduct.

Code of Ethics

Members of CFA Institute [including Chartered Financial Analyst® (CFA®) charterholders] and candidates for the CFA designation ("Members and Candidates") must:[1]

- Act with integrity, competence, diligence, respect, and in an ethical manner with the public, clients, prospective clients, employers, employees, colleagues in the investment profession, and other participants in the global capital markets.
- Place the integrity of the investment profession and the interests of clients above their own personal interests.
- Use reasonable care and exercise independent professional judgment when conducting investment analysis, making investment recommendations, taking investment actions, and engaging in other professional activities.
- Practice and encourage others to practice in a professional and ethical manner that will reflect credit on themselves and the profession.
- Promote the integrity and viability of the global capital markets for the ultimate benefit of society.
- Maintain and improve their professional competence and strive to maintain and improve the competence of other investment professionals.

The Standards of Professional Conduct

I. Professionalism
II. Integrity of Capital Markets
III. Duties to Clients
IV. Duties to Employers
V. Investment Analysis, Recommendations, and Actions
VI. Conflicts of Interest
VII. Responsibilities as a CFA Institute Member or CFA Candidate

LOS 90.c: Explain the ethical responsibilities required by the Code and Standards, including the sub-sections of each Standard.

Standards of Professional Conduct[2]

I. **PROFESSIONALISM**
   A. **Knowledge of the Law.** Members and Candidates must understand and comply with all applicable laws, rules, and regulations (including the CFA Institute *Code of Ethics* and *Standards of Professional Conduct*) of any government, regulatory organization, licensing agency, or professional association governing their professional activities. In the event of conflict, Members and Candidates must comply with the more strict law, rule, or regulation. Members and Candidates must not knowingly participate or assist in and must dissociate from any violation of such laws, rules, or regulations.
   B. **Independence and Objectivity.** Members and Candidates must use reasonable care and judgment to achieve and maintain independence and objectivity in their professional activities. Members and Candidates must not offer, solicit, or accept any gift, benefit, compensation, or consideration that reasonably could be expected to compromise their own or another's independence and objectivity.
   C. **Misrepresentation.** Members and Candidates must not knowingly make any misrepresentations relating to investment analysis, recommendations, actions, or other professional activities.
   D. **Misconduct.** Members and Candidates must not engage in any professional conduct involving dishonesty, fraud, or deceit or commit any act that reflects adversely on their professional reputation, integrity, or competence.

II. **INTEGRITY OF CAPITAL MARKETS**

A. **Material Nonpublic Information.** Members and Candidates who possess material nonpublic information that could affect the value of an investment must not act or cause others to act on the information.
B. **Market Manipulation.** Members and Candidates must not engage in practices that distort prices or artificially inflate trading volume with the intent to mislead market participants.

III. **DUTIES TO CLIENTS**

A. **Loyalty, Prudence, and Care.** Members and Candidates have a duty of loyalty to their clients and must act with reasonable care and exercise prudent judgment. Members and Candidates must act for the benefit of their clients and place their clients' interests before their employer's or their own interests.
B. **Fair Dealing.** Members and Candidates must deal fairly and objectively with all clients when providing investment analysis, making investment recommendations, taking investment action, or engaging in other professional activities.
C. **Suitability.**
   1. When Members and Candidates are in an advisory relationship with a client, they must:
      a. Make a reasonable inquiry into a client's or prospective clients' investment experience, risk and return objectives, and financial constraints prior to making any investment recommendation or taking investment action and must reassess and update this information regularly.
      b. Determine that an investment is suitable to the client's financial situation and consistent with the client's written objectives, mandates, and constraints before making an investment recommendation or taking investment action.
      c. Judge the suitability of investments in the context of the client's total portfolio.
   2. When Members and Candidates are responsible for managing a portfolio to a specific mandate, strategy, or style, they must make only investment recommendations or take only investment actions that are consistent with the stated objectives and constraints of the portfolio.
D. **Performance Presentation.** When communicating investment performance information, Members or Candidates must make reasonable efforts to ensure that it is fair, accurate, and complete.
E. **Preservation of Confidentiality.** Members and Candidates must keep information about current, former, and prospective clients confidential unless:
   1. The information concerns illegal activities on the part of the client or prospective client,
   2. Disclosure is required by law, or
   3. The client or prospective client permits disclosure of the information.

IV. **DUTIES TO EMPLOYERS**

A. **Loyalty.** In matters related to their employment, Members and Candidates must act for the benefit of their employer and not deprive their employer of the advantage of their skills and abilities, divulge confidential information, or otherwise cause harm to their employer.
B. **Additional Compensation Arrangements.** Members and Candidates must not accept gifts, benefits, compensation, or consideration that competes with or might reasonably be expected to create a conflict of interest with their employer's interest unless they obtain written consent from all parties involved.
C. **Responsibilities of Supervisors.** Members and Candidates must make reasonable efforts to ensure that anyone subject to their supervision or authority complies with applicable laws, rules, regulations, and the Code and Standards.

V. **INVESTMENT ANALYSIS, RECOMMENDATIONS, AND ACTIONS**

A. **Diligence and Reasonable Basis.** Members and Candidates must:
   1. Exercise diligence, independence, and thoroughness in analyzing investments, making investment recommendations, and taking investment actions.
   2. Have a reasonable and adequate basis, supported by appropriate research and investigation, for any investment analysis, recommendation, or action.
B. **Communication with Clients and Prospective Clients.** Members and Candidates must:

1. Disclose to clients and prospective clients the basic format and general principles of the investment processes they use to analyze investments, select securities, and construct portfolios and must promptly disclose any changes that might materially affect those processes.
2. Disclose to clients and prospective clients significant limitations and risks associated with the investment process.
3. Use reasonable judgment in identifying which factors are important to their investment analyses, recommendations, or actions and include those factors in communications with clients and prospective clients.
4. Distinguish between fact and opinion in the presentation of investment analysis and recommendations.

C. **Record Retention.** Members and Candidates must develop and maintain appropriate records to support their investment analysis, recommendations, actions, and other investment-related communications with clients and prospective clients.

VI. **CONFLICTS OF INTEREST**

A. **Disclosure of Conflicts.** Members and Candidates must make full and fair disclosure of all matters that could reasonably be expected to impair their independence and objectivity or interfere with respective duties to their clients, prospective clients, and employer. Members and Candidates must ensure that such disclosures are prominent, are delivered in plain language, and communicate the relevant information effectively.

B. **Priority of Transactions.** Investment transactions for clients and employers must have priority over investment transactions in which a Member or Candidate is the beneficial owner.

C. **Referral Fees.** Members and Candidates must disclose to their employer, clients, and prospective clients, as appropriate, any compensation, consideration, or benefit received by, or paid to, others for the recommendation of products or services.

VII. **RESPONSIBILITIES AS A CFA INSTITUTE MEMBER OR CFA CANDIDATE**

A. **Conduct as Participants in CFA Institute Programs.** Members and Candidates must not engage in any conduct that compromises the reputation or integrity of CFA Institute or the CFA designation or the integrity, validity, or security of CFA Institute programs.

B. **Reference to CFA Institute, the CFA Designation, and the CFA Program.** When referring to CFA Institute, CFA Institute membership, the CFA designation, or candidacy in the CFA Program, Members and Candidates must not misrepresent or exaggerate the meaning or implications of membership in CFA Institute, holding the CFA designation, or candidacy in the CFA Program.

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[2]Ibid.

Reading 91: Guidance for Standards I–VII

Module 91.1: Guidance for Standards I(A) and I(B)

LOS 91.a: Demonstrate the application of the Code of Ethics and Standards of Professional Conduct to situations involving issues of professional integrity.
LOS 91.b: Recommend practices and procedures designed to prevent violations of the Code of Ethics and Standards of Professional Conduct.
LOS 91.c: Identify conduct that conforms to the Code and Standards and conduct that violates the Code and Standards.

The Standards of Professional Conduct comprise seven Standards (I–VII) and a total of 22 subsections. These Standards and their application are described in the *Standards of Practice Handbook, 11th Edition, 2014,* published by CFA Institute.

The *Standards of Practice Handbook* is included in its entirety in the CFA curriculum in Volume 6. You can also download a PDF of the *Standards of Practice Handbook* at the website for CFA Institute, www.cfainstitute.org. A third alternative is to purchase the *Standards of Practice Handbook* through Amazon (make sure you get the 11th edition) for about $40 or get the Kindle edition for $0.99.

> **Professor's Note**
>
> You should carefully read the *Standards of Practice Handbook* multiple times in preparation for your Level I exam (yes, the entire book, including all examples). Fifteen to twenty percent of your exam questions will be based on this book and the short learning module concerning Global Investment Performance Standards (GIPS). Given that much of this material must simply be memorized, we suggest that your final reading of the *Standards of Practice Handbook* be on the Friday prior to your exam. You probably do not need to read all the examples that day, but if you highlighted some points during an earlier reading, you can revisit those as you go through all the Standards, the guidance, and the recommended best practices.

In our summary of the Standards of Practice, we focus on describing three things: (1) actions that clearly violate the subsection, (2) the behaviors that each subsection is intended to either encourage or discourage, and (3) recommended best practices for members and their firms.

In many cases the actions that members and candidates must not take are explained using terms open to interpretation, such as "reasonable," "adequate," and "token."

Some examples from the Standards themselves are:
…use reasonable care and judgment to achieve…
…accept any gift, that reasonably could be expected to compromise…

…act with reasonable care and exercise prudent judgment…
…deal fairly and objectively with all clients…
...make a reasonable inquiry into…
…make reasonable efforts to ensure…
…might reasonably be expected to create a conflict of interest with…
…Have a reasonable and adequate basis…
…Use reasonable judgment in…
…matters that could be reasonably expected to impair…

The requirement of the LOS is that you know what constitutes a violation, not that you draw a distinction between what is "reasonable" and what is not in a given situation. We believe the exam writers take this into account and that if they intend, for example, to test whether a recommendation has been given without reasonable care and judgment, it will likely be clear either that the care and judgment exhibited by the analyst did not rise to the level of "reasonable," or that it did.

No monetary value for a "token" gift is given in the Standards, although it is recommended that a firm establish such a monetary value for its employees. Here, again, the correct answer to a question will not likely hinge on candidate's determination of what is a token gift and what is not. Questions should be clear in this regard. A business dinner is likely a token gift, but a week at a condominium in Aspen or tickets to the Super Bowl are likely not. Always look for clues in the questions that lead you to the question-writer's preferred answer choice, such as "lavish" entertainment and "luxury" accommodations.

Next, we present a summary of each subsection of the Standards of Professional Conduct. For each one, we first detail actions that violate the Standard and then list actions and behaviors that are recommended within the Standards. We suggest you learn the violations especially well so you understand that the other items are recommended. For the exam, it is not necessary to memorize the Standard number and subsection letter. Knowing that an action violates, for example, Professionalism, rather than Duties to Employers or Duties to Clients, should be sufficient in this regard. Note that some actions may violate more than one Standard.

One way to write questions for this material is to offer a reason that might make one believe a Standard does not apply in a particular situation. In most, if not all, cases the "reason" does not change the requirement of the Standard. If you are prohibited from some action, the motivations for the action or other circumstances simply do not matter. If the Standard says it's a violation, it's a violation. An exception is when intent is key to the Standard, such as intending to mislead clients or market participants in general.

Standard I: Professionalism[1]

Standard I(A) Knowledge of the Law

Members and Candidates must understand and comply with all applicable laws, rules, and regulations (including the CFA Institute Code of Ethics and Standards of Professional Conduct) of any government, regulatory organization, licensing agency, or professional association governing their professional activities. In the event of conflict, Members and Candidates must comply with the more strict law, rule, or regulation. Members and Candidates must not knowingly participate or assist in and must dissociate from any violation of such laws, rules, or regulations.

The Standards begin with a straightforward statement: Don't violate any laws, rules, or regulations that apply to your professional activities. This includes the Code and Standards, so any violation of the Code and Standards will also violate this subsection.

A member may be governed by different rules and regulations among the Standards, the country in which the member resides, and the country where the member is doing business. Follow the most strict of these, or, put another way, do not violate any of the three sets of rules and regulations.

If you know that violations of applicable rules or laws are taking place, either by coworkers or clients, CFA Institute strongly encourages members and candidates to report potential violations.[2] One way to do so is to approach your supervisor or compliance department to remedy the situation. If they will not or cannot, then you must dissociate from the activity (e.g., not working with a trading group you know is not allocating client trades properly according to the Standard on Fair Dealing, or not using marketing materials that you know or should know are misleading or erroneous). If this cannot be accomplished, you may, in an extreme case, have to resign from the firm to be in compliance with this Standard.

Recommendations for Members

- Establish, or encourage employer to establish, procedures to keep employees informed of changes in relevant laws, rules, and regulations.
- Review, or encourage employer to review, the firm's written compliance procedures on a regular basis.
- Maintain, or encourage employer to maintain, copies of current laws, rules, and regulations.
- When in doubt about legality, consult supervisor, compliance personnel, or a lawyer.
- When dissociating from violations, keep records documenting the violations, encourage employer to bring an end to the violations.
- There is no requirement in the Standards to report wrongdoers, but local law may require it; members are "strongly encouraged" to report violations to CFA Institute Professional Conduct Program.

Recommendations for Firms

- Have a code of ethics.
- Provide employees with information on laws, rules, and regulations governing professional activities.
- Have procedures for reporting suspected violations.

Standard I(B) Independence and Objectivity

Members and Candidates must use reasonable care and judgment to achieve and maintain independence and objectivity in their professional activities. Members and Candidates must not offer, solicit, or accept any gift, benefit, compensation, or consideration that reasonably could be expected to compromise their own or another's independence and objectivity.

Analysts may face pressure or receive inducements to give a security a specific rating, to select certain outside managers or vendors, or to produce favorable or unfavorable research and conclusions. Members who allow their investment recommendations or analysis to be influenced by such pressure or inducements will have violated the requirement to use reasonable care and to maintain independence and objectivity in their professional activities. Allocating shares in oversubscribed IPOs to personal accounts is a violation.

Normal business entertainment is permitted. Members who accept, solicit, or offer things of value that could be expected to influence the member's or others' independence or objectivity are violating the Standard. Gifts from clients are considered less likely to compromise independence and objectivity than gifts from other parties. Client gifts must be disclosed to the member's employer prior to acceptance, if possible, but after acceptance, if not.

Members may prepare reports paid for by the subject firm if compensation is a flat rate not tied to the conclusions of the report (and if the fact that the research is issuer-paid is disclosed). Accepting compensation that is dependent on the conclusions, recommendations, or market impact of the report, and failure to disclose that research is issuer-paid, are violations of this Standard.

Recommendations for Members

Members or their firms should pay for their own travel to company events or tours when practicable and limit use of corporate aircraft to trips for which commercial travel is not an alternative.

Recommendations for Firms

- Restrict employee participation in IPOs and private placements; require pre-approval for participation.
- Appoint a compliance officer, have written policies on independence and objectivity and clear procedures for reporting violations.
- Limit gifts, other than from clients, to token items only.

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[2]Ibid.

Standard I(C) Misrepresentation

Members and Candidates must not knowingly make any misrepresentations relating to investment analysis, recommendations, actions, or other professional activities.

Misrepresentation includes knowingly misleading investors, omitting relevant information, presenting selective data to mislead investors, and plagiarism. Plagiarism is using reports, forecasts, models, ideas, charts, graphs, or spreadsheets created by others without crediting the source. Crediting the source is not required when using projections, statistics, and tables from recognized financial and statistical reporting services. When using models developed or research done by other members of the firm, it is permitted to omit the names of those who are no longer with the firm as long as the member does not represent work previously done by others as his alone.

Actions that would violate the Standard include:

- Presenting third-party research as your own, without attribution to the source.
- Guaranteeing a specific return on securities that do not have an explicit guarantee from a government body or financial institution.
- Selecting a valuation service because it puts the highest value on untraded security holdings.
- Selecting a performance benchmark that is not comparable to the investment strategy employed.
- Presenting performance data or attribution analysis that omits accounts or relevant variables.
- Offering false or misleading information about the analyst's or firm's capabilities, expertise, or experience.
- Using marketing materials from a third party (outside advisor) that are misleading.

Recommendations for Members

- Prepare a summary of experience, qualifications, and services a member is able to perform.
- Encourage employers to develop procedures for verifying marketing materials provided by third parties concerning their capabilities, products, and services.
- Cite the source of any summaries of materials provided by others.
- Keep copies of all reports, articles, or other materials used in the preparation of research reports.
- Provide a list, in writing, of the firm's available services and qualifications.
- Periodically review documents and communications of members for any misrepresentation of employee or firm qualifications and capabilities.

Standard I(D) Misconduct

Members and Candidates must not engage in any professional conduct involving dishonesty, fraud, or deceit or commit any act that reflects adversely on their professional reputation, integrity, or competence.

The first part here regarding professional conduct is clear: no dishonesty, fraud, or deceit. The second part, while it applies to all conduct by the member, specifically requires that the act, "reflects adversely on their professional reputation, integrity, or competence." The guidance states, in fact, that members must not try to use enforcement of this Standard against another member to settle personal, political, or other disputes that are not related to professional ethics or competence.

Recommendations for Firms

- Develop and adopt a code of ethics and make clear that unethical behavior will not be tolerated.
- Give employees a list of potential violations and sanctions, including dismissal.
- Check references of potential employees.

Standard II: Integrity of Capital Markets

Standard II(A) Material Nonpublic Information

Members and Candidates who possess material nonpublic information that could affect the value of an investment must not act or cause others to act on the information.

Information is "material" if its disclosure would affect the price of a security or if a reasonable investor would want the information before making an investment decision. Information that is ambiguous as to its likely effect on price may not be considered material.

Information is "nonpublic" until it has been made available to the marketplace. An analyst conference call is not public disclosure. Selective disclosure of information by corporations creates the potential for insider-trading violations.

The prohibition against acting on material nonpublic information extends to mutual funds containing the subject securities as well as related swaps and options contracts. It is the member's responsibility to determine if information she receives has been publicly disseminated prior acting or causing others to act on it.

Some members and candidates may be involved in transactions during which they are provided with material nonpublic information by firms (e.g., investment banking transactions). Members and candidates may use this information for its intended purpose, but must not use the information for any other purpose unless it becomes public information.

Under the so-called **mosaic theory**, reaching an investment conclusion through perceptive analysis of public information combined with non-material nonpublic information is not a violation of the Standard.

Recommendations for Members

- Make reasonable efforts to achieve public dissemination by the firm of information they possess.
- Encourage their firms to adopt procedures to prevent the misuse of material nonpublic information.

Recommendations for Firms

Use a firewall within the firm, with elements including:

- Exercise substantial control of relevant interdepartmental communications through a clearance area, such as the compliance or legal department.
- Review employee trades.
- Maintain "watch," "restricted," and "rumor" lists.

Monitor and restrict proprietary trading while a firm is in possession of material nonpublic information. However, prohibiting all proprietary trading while a firm is in possession of material nonpublic information may be inappropriate because it may send a signal to the market. In these cases, firms should only take the opposite side of unsolicited customer trades.

Standard II(B) Market Manipulation

Members and Candidates must not engage in practices that distort prices or artificially inflate trading volume with the intent to mislead market participants.

Member actions may affect security values and trading volumes without violating this Standard. The key point here is that if there is the *intent to mislead*, then the Standard is violated. Of course, spreading false information to affect prices or volume is a violation of this Standard as is making trades intended to mislead market participants.

Standard III: Duties to Clients

Standard III(A) Loyalty, Prudence, and Care

Members and Candidates have a duty of loyalty to their clients and must act with reasonable care and exercise prudent judgment. Members and Candidates must act for the benefit of their clients and place their clients' interests before their employer's or their own interests.

Client interests always come first. Although this Standard does not impose a fiduciary duty on members or candidates where one did not already exist, it does require members and candidates to act in their clients' best interests and recommend products that are suitable given their clients' investment objectives and risk tolerances. Members and candidates must:

- Exercise the prudence, care, skill, and diligence under the circumstances that a person acting in a like capacity and familiar with such matters would use.
- Manage pools of client assets in accordance with the terms of the governing documents, such as trust documents or investment management agreements.
- Make investment decisions in the context of the total portfolio.
- Inform clients of any limitations in an advisory relationship (e.g., an advisor who may only recommend her own firm's products).
- Vote proxies in an informed and responsible manner. Due to cost-benefit considerations, it may not be necessary to vote all proxies.
- Client brokerage, or "soft dollars" or "soft commissions," must be used to benefit the client.
- The "client" may be the investing public as a whole rather than a specific entity or person.

Recommendations for Members

Submit to clients, at least quarterly, itemized statements showing all securities in custody and all debits, credits, and transactions.

Encourage firms to address these topics when drafting policies and procedures regarding fiduciary duty:

- Follow applicable rules and laws.
- Establish investment objectives of client.
- Consider suitability of a portfolio relative to the client's needs and circumstances, the investment's basic characteristics, or the basic characteristics of the total portfolio.
- Diversify.
- Deal fairly with all clients in regard to investment actions.
- Disclose conflicts.

- Disclose compensation arrangements.
- Vote proxies in the best interest of clients and ultimate beneficiaries.
- Maintain confidentiality.
- Seek best execution.

Standard III(B) Fair Dealing

Members and Candidates must deal fairly and objectively with all clients when providing investment analysis, making investment recommendations, taking investment action, or engaging in other professional activities.

Do not discriminate against any clients when disseminating recommendations or taking investment action. "Fairly" does not mean "equally." In the normal course of business, there will be differences in the time emails, faxes, and other communications are received by different clients.

Different service levels are acceptable, but they must not negatively affect or disadvantage any clients. Disclose the different service levels to all clients and prospects, and make premium levels of service available to all those willing to pay for them.

Give all clients a fair opportunity to act on every recommendation. Clients who are unaware of a change in the recommendation for a security should be advised of the change before an order for the security is accepted.

Treat clients fairly in light of their investment objectives and circumstances. Treat both individual and institutional clients in a fair and impartial manner. Members and candidates should not take advantage of their position in the industry to disadvantage clients (e.g., taking shares of an oversubscribed IPO).

Recommendations for Members

- Encourage firms to establish compliance procedures requiring proper dissemination of investment recommendations and fair treatment of all customers and clients.
- Maintain a list of clients and holdings—use to ensure that all holders are treated fairly.

Recommendations for Firms

- Limit the number of people who are aware that a change in recommendation will be made.
- Shorten the time frame between decision and dissemination.
- Publish personnel guidelines for pre-dissemination—have in place guidelines prohibiting personnel who have prior knowledge of a recommendation from discussing it or taking action on the pending recommendation.
- Disseminate new or changed recommendations simultaneously to all clients who have expressed an interest or for whom an investment is suitable.
- Develop written trade allocation procedures—ensure fairness to clients, timely and efficient order execution, and accuracy of client positions.
- Disclose trade allocation procedures.
- Establish systematic account review—ensure that no client is given preferred treatment and that investment actions are consistent with the account's objectives.
- Disclose available levels of service.

Standard III(C) Suitability

1. When Members and Candidates are in an advisory relationship with a client, they must:

   a. Make a reasonable inquiry into a client's or prospective client's investment experience, risk and return objectives, and financial constraints prior to making any investment recommendation or taking investment action and must reassess and update this information regularly.

   b. Determine that an investment is suitable to the client's financial situation and consistent with the client's written objectives, mandates, and constraints before making an investment recommendation or taking investment action.

   c. Judge the suitability of investments in the context of the client's total portfolio.

2. When Members and Candidates are responsible for managing a portfolio to a specific mandate, strategy, or style, they must make only investment recommendations or take only investment actions that are consistent with the stated objectives and constraints of the portfolio.

In advisory relationships, members must gather client information at the beginning of the relationship, in the form of an investment policy statement (IPS). Consider clients' needs and circumstances and, thus, their risk tolerance. Consider whether or not the use of leverage is suitable for the client.

If a member is responsible for managing a fund to an index or other stated mandate, he must select only investments that are consistent with the stated mandate.

Unsolicited Trade Requests

An investment manager may receive a client request to purchase a security that the manager knows is unsuitable, given the client's investment policy statement. The trade may or may not have a material effect on the risk characteristics of the client's total portfolio and the requirements are different for each case. In either case, however, the manager should not make the trade until he has discussed with the client the reasons (based on the IPS) that the trade is unsuitable for the client's account.

If the manager determines that the *effect on the risk/return profile of the client's total portfolio is minimal*, the manager, after discussing with the client how the trade does not fit the IPS goals and constraints, may follow his firm's policy with regard to unsuitable trades. Regardless of firm policy, the client must acknowledge the discussion and an understanding of why the trade is unsuitable.

If the trade would have a *material impact on the risk/return profile of the client's total portfolio*, one option is to update the IPS so the client accepts a changed risk profile that would permit the trade. If the client will not accept a changed IPS, the manager may follow firm policy, which may allow the trade to be made in a separate client-directed account. In the absence of other options, the manager may need to reconsider whether to maintain the relationship with the client.

Recommendations for Members

- For each client, put the needs, circumstances, and investment objectives into a written IPS.
- Consider the type of client and whether there are separate beneficiaries, investor objectives (return and risk), investor constraints (liquidity needs, expected cash flows, time, tax, and regulatory and legal circumstances), and performance measurement benchmarks.
- Review the investor's objectives and constraints periodically to reflect any changes in client circumstances.

Standard III(D) Performance Presentation

When communicating investment performance information, Members and Candidates must make reasonable efforts to ensure that it is fair, accurate, and complete.

Members must not misstate performance or mislead clients or prospects about their investment performance or their firm's investment performance.

Members must not misrepresent past performance or reasonably expected performance, and must not state or imply the ability to achieve a rate of return similar to that achieved in the past.

For brief presentations, members must make detailed information available on request and indicate that the presentation has offered only limited information.

Recommendations for Members

- Encourage firms to adhere to Global Investment Performance Standards.
- Consider the sophistication of the audience to whom a performance presentation is addressed.
- Present the performance of a weighted composite of similar portfolios rather than the performance of a single account.
- Include terminated accounts as part of historical performance and clearly state when they were terminated.
- Include all appropriate disclosures to fully explain results (e.g., model results included, gross or net of fees, etc.).
- Maintain data and records used to calculate the performance being presented.

Standard III(E) Preservation of Confidentiality

Members and Candidates must keep information about current, former, and prospective clients confidential unless:

1. The information concerns illegal activities on the part of the client;
2. Disclosure is required by law; or
3. The client or prospective client permits disclosure of the information.

If illegal activities by a client are involved, members may have an obligation to report the activities to authorities.

The confidentiality Standard extends to former clients as well.

The requirements of this Standard are not intended to prevent members and candidates from cooperating with a CFA Institute Professional Conduct Program (PCP) investigation.

Recommendations for Members

- Members should avoid disclosing information received from a client except to authorized coworkers who are also working for the client.
- Members should follow firm procedures for storage of electronic data and recommend adoption of such procedures if they are not in place.

Standard IV: Duties to Employers

Standard IV(A) Loyalty

In matters related to their employment, Members and Candidates must act for the benefit of their employer and not deprive their employer of the advantage of their skills and abilities, divulge confidential information, or otherwise cause harm to their employer.

This Standard is applicable to employees. If members are independent contractors, rather than employees, they have a duty to abide by the terms of their agreements.

Members must not engage in any activities that would injure the firm, deprive it of profit, or deprive it of the advantage of employees' skills and abilities.

Members should always place client interests above interests of their employer, but consider the effects of their actions on firm integrity and sustainability.

There is no requirement that the employee put employer interests ahead of family and other personal obligations; it is expected that employers and employees will discuss such matters and balance these obligations with work obligations.

There may be isolated cases where a duty to one's employer may be violated in order to protect clients or the integrity of the market, when the actions are not for personal gain.

Independent practice for compensation is allowed if a notification is provided to the employer fully describing all aspects of the services, including compensation, duration, and the nature of the activities and the employer consents to all terms of the proposed independent practice before it begins.

When leaving an employer, members must continue to act in their employer's best interests until their resignation is effective. Activities that may constitute a violation include:

- Misappropriation of trade secrets.
- Misuse of confidential information.
- Soliciting employer's clients prior to leaving.
- Self-dealing.
- Misappropriation of client lists.

Employer records on any medium (e.g., home computer, tablet, cell phone) are the property of the firm.

When an employee has left a firm, simple knowledge of names and existence of former clients is generally not confidential. There is also no prohibition on the use of experience or knowledge gained while with a former employer. If an agreement exists among employers (e.g., the U.S. "Protocol for Broker Recruiting") that permits brokers to take certain client information when leaving a firm, a member may act within the terms of the agreement without violating the Standard.

Members and candidates must adhere to their employers' policies concerning social media. When planning to leave an employer, members and candidates must ensure that their social media use complies with their employers' policies for notifying clients about employee separations.

Recommendations for Members

Members are encouraged to give their employer a copy of the Code and Standards.

Best practice is to use separate social media accounts for personal and professional communications.

Recommendations for Firms

Employers should not have incentive and compensation systems that encourage unethical behavior.

Standard IV(B) Additional Compensation Arrangements

Members and Candidates must not accept gifts, benefits, compensation, or consideration that competes with or might reasonably be expected to create a conflict of interest with their employer's interest unless they obtain written consent from all parties involved.

Compensation includes direct and indirect compensation from a client and other benefits received from third parties.

Written consent from a member's employer includes email communication.

Understand the difference between an additional compensation arrangement and a gift from a client:

- If a client offers a bonus that depends on the *future performance* of her account, this is an additional compensation arrangement that requires written consent in advance.
- If a client offers a bonus to reward a member for her account's *past performance*, this is a gift that requires disclosure to the member's employer to comply with Standard I(B) Independence and Objectivity.

Recommendations for Members

Make an immediate written report to the employer detailing any proposed compensation and services, if additional to that provided by the employer.

Members and candidates who are hired to work part time should discuss any arrangements that may compete with their employer's interest at the time they are hired and abide by any limitations their employer identifies.

Recommendations for Firms

Details of additional compensation, including any performance incentives, should be verified by the offering party.

Standard IV(C) Responsibilities of Supervisors

Members and Candidates must make reasonable efforts to ensure that anyone subject to their supervision or authority complies with applicable laws, rules, regulations, and the Code and Standards.

Members must make reasonable efforts to prevent employees from violating laws, rules, regulations, or the Code and Standards, as well as make reasonable efforts to detect violations.

An adequate compliance system must meet industry standards, regulatory requirements, and the requirements of the Code and Standards.

Members with supervisory responsibilities have an obligation to bring an inadequate compliance system to the attention of firm's management and recommend corrective action.

A member or candidate faced with no compliance procedures or with procedures he believes are inadequate must decline supervisory responsibility in writing until adequate procedures are adopted by the firm.

If there is a violation, respond promptly and conduct a thorough investigation while increasing supervision or placing limitations on the wrongdoer's activities.

Recommendations for Members

A member should recommend that his employer adopt a code of ethics. Members should encourage employers to provide their codes of ethics to clients.

Once the compliance program is instituted, the supervisor should:

- Distribute it to the proper personnel.
- Update it as needed.
- Continually educate staff regarding procedures.
- Issue reminders as necessary.
- Require professional conduct evaluations.
- Review employee actions to monitor compliance and identify violations.

Recommendations for Firms

Employers should not commingle compliance procedures with the firm's code of ethics—this can dilute the goal of reinforcing one's ethical obligations.

While investigating a possible breach of compliance procedures, it is appropriate to limit the suspected employee's activities.

Adequate compliance procedures should:

- Be clearly written.
- Be easy to understand.
- Designate a compliance officer with authority clearly defined.
- Have a system of checks and balances.
- Outline the scope of procedures.
- Outline what conduct is permitted.
- Contain procedures for reporting violations and sanctions.
- Structure incentives so unethical behavior is not rewarded.

Standard V: Investment Analysis, Recommendations, and Actions

Standard V(A) Diligence and Reasonable Basis

Members and Candidates must:

1. Exercise diligence, independence, and thoroughness in analyzing investments, making investment recommendations, and taking investment actions.
2. Have a reasonable and adequate basis, supported by appropriate research and investigation, for any investment analysis, recommendation, or action.

The application of this Standard depends on the investment philosophy adhered to, members' and candidates' roles in the investment decision-making process, and the resources and support provided by employers. These factors dictate the degree of diligence, thoroughness of research, and the proper level of investigation required.

The level of research needed to satisfy the requirement for due diligence will differ depending on the product or service offered. A list of things that should be considered prior to making a recommendation or taking investment action includes:

- Global and national economic conditions.
- A firm's financial results and operating history, and the business cycle stage.
- Fees and historical results for a mutual fund.
- Limitations of any quantitative models used.
- A determination of whether peer group comparisons for valuation are appropriate.

Recommendations for Members

Members should encourage their firms to adopt a policy for periodic review of the quality of third-party research, if they have not. Examples of criteria to use in judging quality are:

- Review assumptions used.
- Determine how rigorous the analysis was.
- Identify how timely the research is.
- Evaluate objectivity and independence of the recommendations.

Members should *encourage their firms to consider* these policies and procedures supporting this Standard:

- Have a policy requiring that research reports and recommendations have a basis that can be substantiated as reasonable and adequate.
- Have detailed, written guidance for proper research and due diligence.
- Have measurable criteria for judging the quality of research, and base analyst compensation on such criteria.
- Have written procedures that provide a minimum acceptable level of scenario testing for computer-based models and include standards for the range of scenarios, model accuracy over time, and a measure of the sensitivity of cash flows to model assumptions and inputs.
- Have a policy for evaluating outside providers of information that addresses the reasonableness and accuracy of the information provided and establishes how often the evaluations should be repeated.
- Adopt a set of standards that provides criteria for evaluating external advisers and states how often a review of external advisers will be performed.

Standard V(B) Communication With Clients and Prospective Clients

Members and Candidates must:

1. Disclose to clients and prospective clients the basic format and general principles of the investment processes they use to analyze investments, select securities, and construct portfolios and must promptly disclose any changes that might materially affect those processes.
2. Disclose to clients and prospective clients significant limitations and risks associated with the investment process.
3. Use reasonable judgment in identifying which factors are important to their investment analyses, recommendations, or actions and include those factors in communications with clients and prospective clients.
4. Distinguish between fact and opinion in the presentation of investment analyses and recommendations.

All means and types of communication with clients are covered by this Standard, not just research reports or other written communications.

Members must distinguish between opinions and facts and always include the basic characteristics of the security being analyzed in a research report. Expectations based on statistical modeling and analysis are not facts.

Members must explain to clients and prospects the investment decision-making process used.

In preparing recommendations for structured securities, allocation strategies, or any other nontraditional investment, members must communicate those risk factors specific to such investments. In all cases, members should communicate the potential gains and losses on the investment clearly in terms of total returns.

Members must communicate significant changes in the risk characteristics of an investment or investment strategy.

Members must update clients regularly about any changes in the investment process, including any risks and limitations that have been newly identified.

When using projections from quantitative models and analysis, members may violate the Standard by not explaining the limitations of the model and the assumptions it uses, which provides a context for judging the uncertainty regarding the estimated investment result.

Members and candidates must inform clients about limitations inherent to an investment. Two examples of such limitations are liquidity and capacity. Liquidity refers to the ability to exit an investment readily without experiencing a significant extra cost from doing so. Capacity refers to an investment vehicle's ability to absorb additional investment without reducing the returns it is able to achieve.

Recommendations for Members

Selection of relevant factors in a report can be a judgment call so members should maintain records indicating the nature of the research, and be able to supply additional information if it is requested by the client or other users of the report.

Standard V(C) Record Retention

Members and Candidates must develop and maintain appropriate records to support their investment analyses, recommendations, actions, and other investment-related communications with clients and prospective clients.

Members must maintain research records that support the reasons for the analyst's conclusions and any investment actions taken. Such records are the property of the firm. All communications with clients through any medium, including emails and text messages, are records that must be retained.

A member who changes firms must re-create the analysis documentation supporting her recommendation using publicly available information or information obtained from the company and must not rely on memory or materials created at her previous firm.

Recommendations for Members

If no regulatory standards or firm policies are in place, the Standard recommends a seven-year minimum holding period.

Recommendations for Firms

This recordkeeping requirement generally is the firm's responsibility.

Standard VI: Conflicts of Interest

Standard VI(A) Disclosure of Conflicts

Members and Candidates must make full and fair disclosure of all matters that could reasonably be expected to impair their independence and objectivity or interfere with respective duties to their clients, prospective clients, and employer. Members and Candidates must ensure that such disclosures are prominent, are delivered in plain language, and communicate the relevant information effectively.

Members must fully disclose to clients, prospects, and their employers all actual and potential conflicts of interest in order to protect investors and employers. These disclosures must be clearly stated.

The requirement that all potential areas of conflict be disclosed allows clients and prospects to judge motives and potential biases for themselves. Disclosure of broker-dealer market-making activities would be included here. Board service is another area of potential conflict.

The most common conflict that requires disclosure is actual ownership of stock in companies that the member recommends or that clients hold.

Another common source of conflicts of interest is a member's compensation/bonus structure, which can potentially create incentives to take actions that produce immediate gains for the member with little or no concern for longer-term returns for the client. Such conflicts must be disclosed when the member is acting in an advisory capacity and must be updated in the case of significant change in compensation structure.

Members must give their employers enough information to judge the impact of a conflict, take reasonable steps to avoid conflicts, and report them promptly if they occur.

Recommendations for Members

Any special compensation arrangements, bonus programs, commissions, and incentives should be disclosed.

Standard VI(B) Priority of Transactions

Investment transactions for clients and employers must have priority over investment transactions in which a Member or Candidate is the beneficial owner.

Client transactions take priority over personal transactions and over transactions made on behalf of the member's firm. Personal transactions include situations where the member is a beneficial owner.

Personal transactions may be undertaken only after clients and the member's employer have had an adequate opportunity to act on a recommendation. Note that family member accounts that are client accounts should be treated just like any client account; they should not be disadvantaged.

Members must not act on information about pending trades for personal gain. The overriding considerations with respect to personal trades are that they do not disadvantage any clients.

When requested, members must fully disclose to investors their firm's personal trading policies.

Recommendations for Members

Members can avoid conflicts that arise with IPOs by not participating in them.

Members should encourage their firms to adopt the procedures listed in the following recommendations for firms if they have not done so.

Recommendations for Firms

All firms should have basic procedures in place that address conflicts created by personal investing. The following areas should be included:

- Establish limitations on employee participation in equity IPOs.
- Establish restrictions on participation in private placements. Strict limits should be placed on employee acquisition of these securities and proper supervisory procedures should be in place. Participation in these investments raises conflict of interest issues similar to those of IPOs.
- Establish blackout/restricted periods. Employees involved in investment decision-making should have blackout periods prior to trading for clients—no front running (i.e., purchase or sale of securities in advance of anticipated client or employer purchases and sales). The size of the firm and the type of security should help dictate how severe the blackout requirement should be.
- Establish reporting procedures, including duplicate trade confirmations, disclosure of personal holdings and beneficial ownership positions, and preclearance procedures.

Standard VI(C) Referral Fees

Members and Candidates must disclose to their employer, clients, and prospective clients, as appropriate, any compensation, consideration, or benefit received from or paid to others for the recommendation of products or services.

Members must inform employers, clients, and prospects of any benefit received for referrals of customers and clients, allowing them to evaluate the full cost of the service as well as any potential partiality. All types of consideration must be disclosed.

Recommendations for Members

Members should encourage their firms to adopt clear procedures regarding compensation for referrals.

Members should provide their employers with updates at least quarterly.

Recommendations for Firms

Firms that do not prohibit referral fees should have clear procedures for approval and policies regarding the nature and value of referral compensation received.

Standard VII: Responsibilities as a CFA Institute Member or CFA Candidate

Standard VII(A) Conduct as Participants in CFA Institute Programs

Members and Candidates must not engage in any conduct that compromises the reputation or integrity of CFA Institute or the CFA designation or the integrity, validity, or security of CFA Institute programs.

Members must not engage in any activity that undermines the integrity of the CFA charter. This Standard applies to conduct that includes:

- Cheating on the CFA exam or any exam.
- Revealing anything about either broad or specific topics tested, content of exam questions, or formulas required or not required on the exam.
- Not following rules and policies of the CFA Program.
- Giving confidential information on the CFA Program to candidates or the public.
- Improperly using the designation to further personal and professional goals.
- Misrepresenting information on the Professional Conduct Statement (PCS) or the CFA Institute Professional Development Program.

Members and candidates are not precluded from expressing their opinions regarding the exam program or CFA Institute but must not reveal confidential information about the CFA Program.

Candidates who violate any of the CFA exam policies (e.g., calculator, personal belongings, Candidate Pledge) have violated Standard VII(A).

Members who volunteer in the CFA Program may not solicit or reveal information about questions considered for or included on a CFA exam, about the grading process, or about scoring of questions.

Standard VII(B) Reference to CFA Institute, the CFA Designation, and the CFA Program

When referring to CFA Institute, CFA Institute membership, the CFA designation, or candidacy in the CFA Program, Members and Candidates must not misrepresent or exaggerate the meaning or implications of membership in CFA Institute, holding the CFA designation, or candidacy in the CFA Program.

Members must not make promotional promises or guarantees tied to the CFA designation, such as over-promising individual competence or over-promising investment results in the future (i.e., higher performance, less risk, etc.).

Members must satisfy these requirements to maintain membership:

- Sign the PCS annually.
- Pay CFA Institute membership dues annually.

If they fail to do this, they are no longer active members.

Do not misrepresent or exaggerate the meaning of the CFA designation.

There is no partial CFA designation. It is acceptable to state that a candidate successfully completed the program in three years if, in fact, he did, but claiming superior ability because of this is not permitted.

Recommendations for Members

Members should be sure that their firms are aware of the proper references to a member's CFA designation or candidacy, as errors in these references are common.

Reading 92: Introduction to the Global Investment Performance Standards (GIPS)

Module 92.1: Introduction to GIPS

LOS 92.a: Explain why the GIPS standards were created, who can claim compliance, and who benefits from compliance.

When firms choose their own methodologies for reporting investment performance, the results are not comparable across firms, and firms have a tendency to choose a methodology that makes their performance look good. Some choices that tend to bias results are:

- Choosing a top-performing portfolio and claiming it represents a firm's overall results for all assets managed.
- Excluding terminated accounts, which may tend to be accounts closed by investors due to sub-par performance.
- Selecting time periods to report that put firm performance in the best possible light.

GIPS present a standardized methodology for performance reporting that makes comparison of performance across firms meaningful, provides specific information that is useful to current clients, prospective clients, and investors, and avoids misrepresentation of performance. Widespread usage of GIPS can also give oversight bodies a clearer understanding of the returns achieved and the risks taken by the firms they supervise.

GIPS only apply to firms that manage assets for others. Presenting performance information compliant with GIPS is voluntary for such firms, and they may only claim compliance with GIPS if they comply fully and on a firmwide basis. Other firms related to the asset management business, such as software developers, may state that they endorse GIPS but may not claim compliance with GIPS.

LOS 92.b: Describe the key concepts of the GIPS Standards for Firms.

The GIPS standards for firms consist of eight sections:

1. *Fundamentals of Compliance*. The fundamental issues involved in complying with GIPS are (a) defining the firm, (b) providing GIPS-compliant reports to all clients and prospects, (c) complying with applicable regulations and laws, and (d) presenting information that is neither false nor misleading.
2. *Input Data and Calculation Methodology*. Input data should be consistent in order to establish full, fair, and comparable investment performance presentations. Certain methodologies are required for portfolio return calculations, and certain other methodologies are required for composite return calculations. Uniformity in methods across firms is required so that their results are comparable.
3. *Composite and Pooled Fund Maintenance*. Creation of meaningful, asset-weighted composites is important to achieve a fair presentation. Composite performance is based on the performance of one or more portfolios that have the same investment strategy or investment objective. Composite returns are the asset-weighted average (not a simple average) of the returns on the portfolios that are included in each composite. Pooled funds must be included in a composite if they fit its definition.
4. *Composite Time-Weighted Return Report*.

5. *Composite Money-Weighted Return Report.*
6. *Pooled Fund Time-Weighted Return Report.*

7. *Pooled Fund Money-Weighted Return Report.*

   Sections 4 through 7 contain required and recommended procedures for reporting the performance of composites and pooled funds, as well as the necessary disclosures. There are some disclosures that all firms must make, but some disclosures may not apply to all firms. If a disclosure is not applicable to a specific firm, the firm is not required to include any statement regarding it. A firm that has met all the requirements of GIPS may include an appropriate claim of compliance.

8. *GIPS Advertising Guidelines.* If an advertisement includes a claim of compliance with GIPS, the advertisement must comply with these guidelines. The guidelines do not apply to advertisements that do not reference the firm's GIPS compliance.

LOS 92.c: Explain the purpose of composites in performance reporting.

A **composite** is a grouping of individual discretionary portfolios representing a similar investment strategy, objective, or mandate. Examples of possible composites are large capitalization stocks, investment-grade domestic bonds, and accounts managed to match the performance of a specific securities index. Reporting on the performance of composites gives clients and prospects information about the firm's success in managing various types of securities and results for various investment styles.

A composite, such as international equities, must include all fee-paying, discretionary portfolios (current and past) that the firm has managed in accordance with a particular strategy. The firm should identify which composite each managed portfolio is to be included in before the portfolio's performance is known. This prevents firms from choosing portfolios to include in a composite in order to create composites with superior returns. All fee-paying discretionary portfolios managed by the firm must be included in at least one composite.

LOS 92.d: Describe the fundamentals of compliance, including the recommendations of the GIPS standards with respect to the definition of the firm and the firm's definition of discretion.

The definition of the firm, for purposes of GIPS compliance, must be the corporation, subsidiary, or division that is held out to clients as a business entity. If a firm has different geographic locations (e.g., all doing business under the name of Bluestone Advisers), then the definition of the firm should include all the various geographic locations and their clients.

The definition of **discretion** refers to how a firm determines which of the portfolios it manages should be included in a composite. A firm may determine that a portfolio is nondiscretionary—and therefore not include it in a composite—if the client places restrictions on it that prevent the manager from carrying out the intended strategy.

LOS 92.e: Describe the concept of independent verification.

Firms are *encouraged* to pursue independent verification of their compliance with GIPS. Verification applies to the entire firm's performance measurement practices and methods, not a selected composite.

If a firm chooses to pursue verification, it must be performed by a third party, not by the firm itself, on a firmwide basis. This third-party verifier must attest that (1) the firm has complied with all GIPS requirements for composite construction on a firmwide basis and (2) the firm's processes and procedures are established to present performance in accordance with the calculation methodology required by GIPS, the data requirements of GIPS, and in the format required by GIPS.

Verified firms should include the following disclosure language:

> [Insert name of firm] has been verified for the periods [insert dates] by [name of verifier]. A copy of the verification report is available upon request.

LOS 92.a

GIPS are principles that investment firms can voluntarily follow. They are designed to avoid misrepresentations of historical investment results to clients and prospects. GIPS allow clients to more easily compare investment performance among investment firms and have more confidence in reported performance.

LOS 92.b

The eight sections of the GIPS standards for firms are:

1. Fundamentals of Compliance.
2. Input Data and Calculation Methodology.
3. Composite and Pooled Fund Maintenance.
4. Composite Time-Weighted Return Report.
5. Composite Money-Weighted Return Report.
6. Pooled Fund Time-Weighted Return Report.
7. Pooled Fund Money-Weighted Return Report.
8. GIPS Advertising Guidelines.

LOS 92.c

A composite is a grouping of discretionary portfolios representing a similar investment strategy, objective, or mandate. A composite must include all portfolios (current and past) that the firm has managed in accordance with this particular strategy. The firm should identify which composite each managed portfolio is to be included in before the portfolio's performance is known.

LOS 92.d

GIPS compliance must be firmwide, where the firm is the distinct business entity that is held out to clients and prospects.

A firm must establish criteria to determine which of the portfolios it manages are discretionary or nondiscretionary.

LOS 92.e

Firms are encouraged to pursue independent verification of GIPS compliance. If they seek verification, it must be performed by a third party, not by the firm itself. The third-party verifier must attest that (1) the firm has complied with all GIPS requirements for composite construction on a firmwide basis and (2) the firm's processes and procedures are established to present performance in accordance with the calculation methodology required by GIPS, the data requirements of GIPS, and in the format required by GIPS.

Reading 93: Ethics Application

LOS 93.a: Evaluate practices, policies, and conduct relative to the CFA Institute Code of Ethics and Standards of Professional Conduct.
LOS 93.b: Explain how the practices, policies, and conduct do or do not violate the CFA Institute Code of Ethics and Standards of Professional Conduct.

Standard I: Professionalism

Standard I(A) Knowledge of the Law

Case 1:

A member reports to his supervisor that their firm is overcharging clients by seeking to recover the cost of expenses that are actually reimbursable. The actions taken only remedy the situation for some clients but not for others. The member must disassociate from the activity and not work with any clients who are still being overcharged.

Case 2:

A member violates the Standard by failing to investigate transactions in an account that appear to be at high risk of violating money-laundering laws (because it was a long-standing client who has relationships with board members).

Case 3:

A member violates the Standards (and the law) by forging customer signatures for expediency.

Standard I(B) Independence and Objectivity

Case 1:

A member violates the Standard by contributing to a politician's campaign, believing that it may lead to preferential treatment with regard to receiving management contracts for government pension fund money.

Standard I(C) Misrepresentation

Case 1:

A member violates the Standard by assuring a client that returns on a fund will outweigh the penalties incurred from shifting funds from an existing investment, essentially guaranteeing a specific rate of return on an investment that has no actual guarantee of returns.

Case 2:

A proposal to manage pension assets lists all personnel and their qualifications, but while the proposal is under consideration, one of the key personnel leaves the firm. The member is required to inform the potential client of this personnel change (misrepresentation by omission).

Case 3:

A member who is the CEO of an electric car company posts on social media that private financing has been secured to take the company private at $420 per share when that is untrue and the member admits the price "was a joke." This is clearly a violation of the Standard regarding misrepresentation.

Standard I(D) Misconduct

Case 1:

A member is arrested for minor criminal offenses that are part of civil disobedience related to protesting and expressing her personal beliefs. This is not necessarily a violation of the Standard regarding misconduct, which states in part that members are prohibited from "engaging in any professional conduct involving dishonesty, fraud, or deceit" or committing an act that "reflects adversely on their professional reputation, integrity, or competence."

Case 2:

A member violates the Standard by using his firm's error-correction policy to effectively give his own money to a client account in order to make his management of the account look better.

Standard II: Integrity of Capital Markets

Standard II(A) Material Nonpublic Information

Case 1:

A member violates the Standard by using material nonpublic information he acquired by overhearing a friend's phone conversation and inferring that a takeover offer for a specific company was imminent.

Case 2:

A member violates the Standard by sharing information with clients about a regulator's positive response to trials of a company's new drug that he learned of during a meeting of analysts with company management. The information is considered nonpublic because sharing it with a specific group of analysts was "selective sharing" of the information and cannot be considered public disclosure.

Standard II(B) Market Manipulation

Case 1:

A member violates the Standard by fraudulently including the names of people who do not actually own shares to meet the minimum required number of shareholders for a firm to be listed on an exchange, misleading market participants about the potential liquidity of the shares.

Standard III: Duties to Clients

Standard III(A) Loyalty, Prudence, and Care

Case 1:

A member violates the Standard by putting a policy into an investment firm's client agreements that indicates that their representatives are excused in some cases from acting in a client's best interest or failing to adequately investigate suitability of recommendations to clients, and that clients' rights to make claims for violating securities laws are restricted. Members and candidates cannot "opt out" of the Standards.

Case 2:

A member executes trades for a client who self-directs their own account and has received the firm's policies regarding margin requirements. In this case, the member's requirements to act in the client's best interests are limited, compared with a relationship in which the member is providing investment advice. There is no intent to limit the member's responsibilities to act in good faith and not misrepresent the services to be provided according to firm policies. The terms regarding the firm's actions in the event of a margin shortfall are open to negotiation at the inception of the relationship.

Case 3:

A member allocates expenses to a client that arise from actions designed to reduce total expenses (staying over a weekend to reduce transportation costs but increasing lodging costs). This is not a violation, but charging a client for expenses that benefit other clients or that are for personal activities of the member is.

Standard III(B) Fair Dealing

Case 1:

A member treats clients fairly by simultaneously sending emails regarding a change in investment recommendation. Providing individual clients with updates or clarifications regarding the change in recommendation is not a violation. Further, the member's firm offers some clients, for an additional fee, weekly updates about securities that may indicate a change in recommendation may be forthcoming. This is not a violation as long as clients are all informed about the availability of the weekly updates for a fee and as long as no change in recommendation is included in a weekly update that would disadvantage clients who do not subscribe to the additional service.

Standard III(C) Suitability

Case 1:

A member violates the Standard by recommending investments that carry more risk than is suitable for some clients, even though there are tax advantages to the recommended portfolios.

Case 2:

A member violates the Standard by making a client-requested change to their portfolio without investigating the client circumstances adequately to determine whether the requested investment is suitable for the clients.

Standard III(D) Performance Presentation

Case 1:

A member violates the Standard by presenting performance data based on a composite of separately managed accounts the firm managed before creating the fund that is being reported, giving the impression that the fund has been in existence for many years when, in fact, it is relatively new.

Standard III(E) Preservation of Confidentiality

Case 1:

Member A downloads clients' personally identifiable information to his personal server at home to make working from home easier and gets hacked. The member has violated the Standard by not taking adequate steps to protect client information. Member B, who is the firm's head of compliance, likely violated the Standard by not taking the proper steps to protect client information, as evidenced by the fact that Member A was able to download sensitive client information to his personal server.

Standard IV: Duties to Employers

Standard IV(A) Loyalty

Case 1:

A member violates the Standard by making harmful statements about Firm A, her current employer, and promoting the firm that she intends to move to while she is still employed by Firm A.

Case 2:

A member is pressured by his firm to sell the firm's proprietary investment products that are relatively expensive and haven't performed well. He has complained to management, and they have done nothing. He copies client records and he records conversations with his supervisor to document his conduct. He takes this information to securities regulators. His actions are not a violation of the Standard because he is acting in the interest of his clients.

Case 3:

A member violates the Standard by taking client information with her when she leaves her firm; the client list is the property of her current firm. Although she only intends to send her clients thank-you notes, the client list she copied has personal information.

Standard IV(B) Additional Compensation Arrangements

Case 1:

A member works for a firm that produces issuer-paid research reports and is involved with the decision of which companies her firm will cover. Some companies seeking coverage by the firm offer her a bonus payment if their firm is selected. Because this creates a possible conflict between her interests and her firm's interests, she must get approval in writing from her employer to accept the bonus arrangement.

Standard IV(C) Responsibilities of Supervisors

Case 1:

A member with supervisory responsibility over a branch office violates the Standard by not making reasonable efforts to ensure that those under his supervision are not engaging in misconduct. He also violates the Standard by not having clear written compliance policies and procedures in place, along with employee training. If these are not in place, he must decline to take supervisory responsibility.

Case 2:

A member violates the Standard by accepting the title of chief compliance officer even though she has no experience in compliance, is denied permission to contact clients or review client communications, and is not allowed to enforce company policies. She is in violation of the Standard because it requires her to make reasonable efforts to both detect and prevent violations of laws, rules, and regulations by those subject to her supervision. Knowing she could not do this, she should have declined to take on supervisory responsibilities.

Standard V: Investment Analysis, Recommendations, and Actions

Standard V(A) Diligence and Reasonable Basis

Case 1:

A member violates the Standard by recommending the purchase of shares of a company for which he has not performed a diligent, thorough, and independent analysis. A second member bases her recommendation on the analysis of the first member and incorporates part of the first member's research report in her own and is thereby also in violation of the Standards.

Standard V(B) Communication With Clients and Prospective Clients

Case 1:

A member's firm begins calculating client fees using different methods from those spelled out in the disclosures made to them when they opened their accounts. The member is required by the Standard to inform clients that the fee calculation methods have changed, prior to changing the methods, even if fees are not higher overall using the new processes.

Case 2:

A credit rating agency changes its methodology of determining the ratings of commercial mortgage-backed securities without disclosing the change to potential users of the ratings. A member who is responsible for publishing the ratings violates the Standard by publishing ratings without disclosing the change in methodology.

Standard V(C) Record Retention

Case 1:

A member violates the Standard by not updating client records in a timely manner. Although he keeps himself updated on changes in client circumstances and adjusts portfolios accordingly, he does not do a good job of keeping written client profiles up to date.

Standard VI: Conflicts of Interest

Standard VI(A) Disclosure of Conflicts

Case 1:

A member receives payment from third-party subadvisors that she uses to manage client funds invested in some asset classes. To avoid violating the Standard, she must disclose these arrangements to clients because the payments may influence her choice of subadvisors.

Standard VI(B) Priority of Transactions

Case 1:

A member violates the Standard by buying shares and call options in his personal account just prior to purchasing large blocks of the same stocks in client accounts (front-running the trades), in anticipation of an increase in the price of the stocks as a result of the large purchases.

Case 2:

A member violates the Standard by telling friends and relatives about large buy orders he is to execute for clients of his employer prior to executing those orders, allowing them to front-run those trades for quick profits.

Case 3:

A member enters client trades grouped as block orders and allocates them to specific client accounts after the market closes for the day. The member violates the Standard (and likely the Standard on fair dealing as well) by allocating profitable trades to personal accounts and allocating a disproportionate amount of losing trades to his largest client accounts where they will likely go unnoticed.

Standard VI(C) Referral Fees

Case 1:

A member invites existing clients who have referred very profitable accounts to her to lavish parties, rewarding them with discounts on fees and gift cards as well. The member violates the Standard by not disclosing these "referral fees" to all existing clients and prospective clients.

Standard VII: Responsibilities as a CFA Institute Member or CFA Candidate

Standard VII(A) Conduct as Participants in CFA Institute Programs

Case 1:

A member who teaches exam-prep classes hosts a post-exam party for candidates who took the exam. He solicits their general impressions about the difficulty of the exam and their opinions on the most difficult exam questions. He is permitted to share their opinions about the difficulty of the exam with future candidates in his exam-prep classes. However, he is not permitted to solicit or share any information about specific exam questions or about which topics were or were not tested on the exam.

Standard VII(B) Reference to CFA Institute, the CFA Designation, and the CFA Program

Case 1:

A previous member who has not paid her dues to CFA Institute violates the Standard by using the CFA designation in marketing materials and on her business cards. Another member violates the Standard by claiming that all senior employees, which includes the member who has not paid her dues, are CFA charterholders.

Formulas

$$\text{leveraged return} = \frac{r(V_0 + V_B) - r_B V_B}{V_0}$$

rate of return for an investor after fees:

$$r = \frac{V_1 \text{ - } V_0 \text{ - total fees}}{V_0}$$

futures price ≈ spot price × (1 + risk-free rate) + storage costs − convenience yield

$$r \times (V_0 + V_B) \text{ - } (r_B \times V_B)$$

$$\text{total fees} = m\ V_1 + \max\,[0, p(V_1 \text{ - } V_0)]$$

where:

m = management fee

P = performance fee

V_0 = beginning-of-period assets

V_1 = end-of-period assets